Tenebris and Lux

Comprehensive Master Guide

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Table of Contents

1. Executive Summary . . . 1
Volume I: World Canon, Metaphysics & Sacyran Lore . . . 2
    2. Chapter 1: The Primordial Neutrality of Lux and Tenebris . . . 3
    3. Canonical Introduction . . . 4
    4. 1. The Primordial Cosmology: Neutrality of the Apex Forces . . . 5
        4.1. Lux (The Apex of Radiance) . . . 5
        4.2. Tenebris (The Apex of Darkness) . . . 5
        4.3. The Moral Sway of the Wielders . . . 5
    5. 2. The Twin Relics: Shield of Lux & Sword of Tenebris . . . 6
        5.1. Harmonized Equilibrium . . . 6
    6. 3. The Kingdom of Sacyra: Setting & Society . . . 7
        6.1. Governance and Courtly Politics . . . 7
        6.2. The Crisis: The Re-Awakening of Dark Lord Malakor . . . 7
    7. 4. The Dual Protagonists . . . 8
        7.1. 1. Elijah Ro-enheart (The Vanguard of Duty) . . . 8
        7.2. 2. Katrina Jaren (The Free-Spirited Blade) . . . 8
        7.3. Dynamic Interplay . . . 8
    8. 5. The Tactical Strategy RPG (TRPG) Paradigm . . . 9
    9. Chapter 2: Socio-Political Strata of the Kingdom of Sacyra . . . 10
    10. Overview . . . 11
    11. 1. Elijah Ro-enheart (The Aegis of Lineage) . . . 12
        11.1. Biography & Heritage . . . 12
        11.2. Combat & Tactical Doctrine (TRPG) . . . 12
        11.3. Narrative Arc & Internal Conflict . . . 12
    12. 2. Katrina Jaren (The Blade of Freedom) . . . 13
        12.1. Biography & Heritage . . . 13
        12.2. Combat & Tactical Doctrine (TRPG) . . . 13
        12.3. Narrative Arc & Internal Conflict . . . 13
    13. 3. Dual-Protagonist Dynamics: The Wielders of Harmony . . . 14
        13.1. Tactical Synergy on the Grid . . . 14
    14. 4. Antagonist: Dark Lord Malakor (The Void Sovereign) . . . 15
    15. Chapter 3: High School Inception & TRPG Genesis . . . 16
    16. Abstract . . . 17
    17. 1. Historical Provenance: The 2014–2018 High School Genesis . . . 18
    18. 2. Narrative Realignment: The Kingdom of Sacyra . . . 19
        18.1. 1. The Kingdom of Sacyra (/s□□□ki□r□/) . . . 19
        18.2. 2. The Dual Protagonists . . . 19
    19. 3. Ludological Evolution: The Tactical Strategy RPG (TRPG) . . . 20

19.1. 1. The Dual-Layer Combat Loop . . . 20
19.2. 2. Architectural Implications for the C Engine . . . 20
20. 4. Next Milestone Roadmap . . . 21
Volume II: Dramatis Personae & Dual Protagonist Dynamics . . . 22
21. Chapter 4: Dual Protagonist Dynamics & Moral Continuum . . . 23
22. Overview . . . 24
23. 1. Thematic & Social Contrast Matrix . . . 25
24. 2. Interpersonal Chemistry & Banter . . . 26
24.1. Narrative Evolution Across the Campaign . . . 26
25. 3. Tactical Synergy on the TRPG Battlefield . . . 27
26. Chapter 5: Elijah Ro-enheart: Vanguard of Lux . . . 28
27. Overview . . . 29
28. 1. Identity & Heritage . . . 30
29. 2. Psychological Profile & Demeanor . . . 31
29.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin) . . . 31
30. 3. Martial Doctrine & TRPG Grid Mechanics . . . 32
30.1. Tactical Synergies . . . 32
31. Chapter 6: Lady Katrina Jaren: The Outrider Blade . . . 33
32. Overview . . . 34
33. 1. Identity & Heritage . . . 35
34. 2. Psychological Profile & Demeanor . . . 36
34.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin) . . . 36
35. 3. Martial Doctrine & TRPG Grid Mechanics . . . 37
35.1. Tactical Synergies . . . 37
36. Chapter 7: Grandma Magolia: Matriarch of Dio Village . . . 38
37. Overview . . . 39
38. 1. Identity & Family Ties . . . 40
39. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin) . . . 41
40. 3. The Heirloom Timepiece: Grandma’s Watch . . . 42
41. Chapter 8: Monica the Mystic & Goddess Lyri . . . 43
42. Overview . . . 44
43. 1. Identity & Mystical Lineage . . . 45
44. 2. Canonical Voice & Demeanor (2017–2018 RPG Maker Origin) . . . 46
45. 3. Arcane Role in the Campaign . . . 47
46. Chapter 9: Eric the Lumberjack & The Moral Litmus . . . 48
47. Overview . . . 49
48. 1. Identity & Background . . . 50
49. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin) . . . 51
50. 3. The Test of Honesty & Moral Continuum . . . 52
51. Chapter 10: John the Thief & Outlaw Resistance . . . 53
52. Overview . . . 54

53. 1. Identity & Underworld Standing . . . 55
54. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin) . . . 56
55. 3. Quest & Mechanics Role . . . 57
56. Chapter 11: The St. Baron & Feudal Exploitation . . . 58
57. Overview . . . 59
58. 1. Identity & Standing . . . 60
59. 2. Socio-Economic Conflict in Dio Lower Ward . . . 61
60. 3. Narrative & Quest Role . . . 62
61. Chapter 12: High Priest Roté & The Royal Omens . . . 63
62. Overview . . . 64
63. 1. Identity & Standing . . . 65
64. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin) . . . 66
65. 3. Narrative Role & Dynamic . . . 67
66. Chapter 13: King Jihad & The Treasonous Ambition . . . 68
67. Overview . . . 69
68. 1. Identity & Sovereign Office . . . 70
69. 2. Psychological Profile & Secret Ambition . . . 71
69.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin) . . . 71
70. 3. Socio-Political Role in the Sacyra Campaign . . . 72
71. Chapter 14: Commander Gareth Ro-enheart & The Vanished Guard . . . 73
72. Overview . . . 74
73. 1. Identity & Lineage . . . 75
74. 2. Legend & Martial Legacy . . . 76
75. 3. The 10th Birthday Vanishing & Royal Intrigue . . . 77
Volume III: Ludology & Tactical TRPG Combat Mechanics . . . 78
76. Chapter 15: Dual-Layer TRPG Combat Architecture . . . 79
77. Overview . . . 80
78. 1. Dual-Layer Battle Architecture . . . 81
79. 2. Phase 1: Grid-Based Tactical Battlefield . . . 82
79.1. 1. Grid Topography & Spatial Invariants . . . 82
79.2. 2. Movement & Zone of Control (ZOC) . . . 82
79.3. 3. Turn Sequencing . . . 82
80. 3. Phase 2: Cinematic Combat Resolution (Side-View Action) . . . 83
80.1. 1. Visual & Staging Presentation . . . 83
80.2. 2. Combat Mechanics During Cinematic Resolution . . . 83
81. 4. Dual-Protagonist Synergy: Elijah & Katrina . . . 84
81.1. Elijah Ro-enheart (Frontline Guardian) . . . 84
81.2. Katrina Jaren (Tactical Skirmisher) . . . 84
81.3. Tandem Linked Combat (The Duo Duel) . . . 84
82. 5. Persistent State & Moral Sway Feedback . . . 85
83. Chapter 16: The Twin Relics, Combat Stances & Moral Sway . . . 86

84. Overview . . . 87
85. 1. Primordial Neutrality & Relic Mechanics . . . 88
86. 2. The Combat Stance State Machine . . . 89
87. 3. The Moral Sway Continuum . . . 90
88. Chapter 17: World Topography, Provinces & Battlefield Terrain . . . 91
89. Overview . . . 92
90. 1. Spatial Structure: The Provinces of Sacyra . . . 93
90.1. 1. The Garrison Altar (LOC_GARRISON_ALTAR) . . . 93
90.2. 2. Sunken Citadel Ramparts (LOC_CITADEL_GATES) . . . 93
90.3. 3. The Fractured Marches (LOC_DIVIDED_MARCHES) . . . 94
90.4. 4. The Abyssal Threshold (LOC_DARK_LORD_THRESHOLD) . . . 94
91. Chapter 18: Continuous Invariant Testing Philosophy . . . 95
92. Abstract . . . 96
93. 1. The Fundamental Divergence . . . 97
94. 2. Core Pillars of Game Development Testing . . . 98
94.1. Pillar 1: Dynamic State Invariants & Bounded Continuity . . . 98
94.2. Pillar 2: Deterministic Replay ("Golden Master Replay") . . . 98
94.3. Pillar 3: Softlock Immunity & Degenerate Play Detection . . . 98
94.4. Pillar 4: Economy & Combat Balance Invariance . . . 99
94.5. Pillar 5: Frame-Budget Latency Headroom . . . 99
95. Chapter 19: Game Design Curation & Triaging Matrix . . . 100
96. Overview . . . 101
97. 1. Triage Categorization Taxonomy . . . 102
98. 2. Tier 1: RPG Maker RTP Defaults (To Discard or Re-Architect) . . . 103
99. 3. Tier 2: Prototype & Testing Sketches (Concepts to Evolve) . . . 104
100. 4. Tier 3: Fleshed-Out Canonical Pillars (Core Invariants to Reconstruct) . . . 105
100.1. 1. Dual-Protagonist Dynamics & Narrative Branching . . . 105
100.2. 2. The Moral Sway Continuum Engine . . . 105
100.3. 3. Lecode Tactical Battle System (LeTBS) Mechanics . . . 105
100.4. 4. The Monotonic 24-Hour Time Engine . . . 106
Volume IV: The 2017–2018 Demo Baseline & Historical Archive . . . 107
101. Chapter 20: Historical Baseline Overview & Provenance . . . 108
102. Overview . . . 109
103. 1. Historical Provenance & Dataset Strata . . . 110
104. 2. Documentation Taxonomy . . . 111
105. 3. Architectural Extraction Script . . . 112
106. Chapter 21: 7-Act Narrative Script & Dialogue Walkthrough . . . 113
107. Overview . . . 114
108. Act I: The Castle Prophecy & Dark Ambition . . . 115
108.1. Start-Up / Walk of Fear . . . 115
108.1.1. Map 002: Start-Up . . . 115

108.2. The Castle (Outside) . . . 115
108.2.1. Map 003: The Castle (Outside) . . . 115
108.2.2. Map 008: Battle Tutorial . . . 115
108.3. The Castle (Inside) - The Throne Room . . . 117
108.3.1. Map 004: The Castle (Inside) . . . 117
108.3.2. Map 009: Secret Map . . . 117
109. Act II: The Dream Realm & The Ancient Archives . . . 119
109.1. Intro / The Void . . . 119
109.1.1. Map 002: Start-Up . . . 119
109.2. Character Select & The Room of Sages . . . 119
109.2.1. Map 006: Character Select . . . 119
110. Act III: The Trial of Mechanisms & The Dream Eaters . . . 124
110.1. The Switches Puzzle Chamber . . . 124
110.1.1. Map 007: Switches . . . 124
110.1.2. Map 040: Switches . . . 125
110.2. The Secret Crystal Chamber . . . 127
110.2.1. Map 009: Secret Map . . . 127
110.2.2. Map 039: Secret Map . . . 128
110.3. The Battle Tutorial & Dream Eater Ambush . . . 130
110.3.1. Map 008: Battle Tutorial . . . 130
110.3.2. Map 041: Battle Tutorial . . . 131
111. Act IV: Awakening in Dio Village & Family Roots . . . 133
111.1. Grandma’s House Upstairs . . . 133
111.1.1. Map 020: Grandma’s House Upstairs . . . 133
111.1.2. Map 004: The Castle (Inside) . . . 134
111.2. Grandma’s House Downstairs . . . 134
111.2.1. Map 021: Grandma’s House Downstairs . . . 134
111.2.2. Map 007: Switches . . . 136
112. Act V: Dio Village Upper & The Outskirts Crisis . . . 139
112.1. Dio Village Upper / Main Square . . . 139
112.1.1. Map 023: Main Square . . . 139
112.2. Monica’s House & The Mystic’s Brew . . . 141
112.2.1. Map 027: Monica’s House . . . 141
112.3. The Secret Sword Cache . . . 144
112.3.1. Map 024: The Sword Map . . . 144
112.3.2. Map 025: The Sword Map . . . 145
113. Act VI: Moral Crossroads of Dio Lower Ward . . . 146
113.1. Eric the Lumberjack’s House . . . 146
113.1.1. Map 028: Eric’s House . . . 146
113.2. Shopping District & Lower Village . . . 150
113.2.1. Map 030: Shopping District . . . 150

113.3. John the Thief’s Hideout . . . 151
113.3.1. Map 031: John the Thief (Sidequest) . . . 151
114. Act VII: The Frontier, Wilderness & Final Dungeon . . . 152
114.1. World Map Frontier . . . 152
114.2. Wilderness Battle Zones (Demo Areas 3 & 4) . . . 152
114.2.1. Map 020: Grandma’s House Upstairs . . . 152
114.2.2. Map 021: Grandma’s House Downstairs . . . 153
114.3. The Final Dungeon & Boss Threshold . . . 155
114.3.1. Map 023: Main Square . . . 155
114.3.2. Map 024: The Sword Map . . . 158
114.3.3. Map 025: The Sword Map . . . 158
114.3.4. Map 026: Start Up Area . . . 158
114.3.5. Map 027: Monica’s House . . . 158
114.3.6. Map 028: Eric’s House . . . 161
115. Chapter 22: Complete Cast & Tactical Party Compendium . . . 166
116. Overview . . . 167
117. 1. Dual Protagonists . . . 168
117.1. Elijah Ro-enheart (The Steadfast Vanguard) . . . 168
117.2. Lady Katrina Jaren (The Outrider Blade) . . . 168
118. 2. Extended Tactical Party Companions (Folder 2 Additions) . . . 170
118.1. Sebastien (The Infiltrator Outlaw) . . . 170
118.2. Angela (The Evoker Elementalist) . . . 170
118.3. Lucius (The Battle Priest) . . . 170
119. 3. Summoned Celestial Entities . . . 171
119.1. Wind Spirit (Actor 11) . . . 171
119.2. Pixie (Actor 12) . . . 171
119.3. Fire Chicken (Actor 13) . . . 171
120. 4. Key Lore Figures & NPCs . . . 172
121. Chapter 23: LeTBS Tactical Grid System Specification . . . 173
122. Overview . . . 174
123. 1. Battle Field Architecture & Action Economy . . . 175
124. 2. Scope & Area of Effect (AOE) Geometries . . . 176
125. 3. Terrain State Modifications & Dynamic Hazards . . . 177
126. 4. Turn Timeline Manipulation . . . 178
127. Chapter 24: Classes, Growth Curves & Learning Trees . . . 179
128. Overview . . . 180
129. Class 1: Warrior . . . 181
129.1. Skill Learnings for Warrior . . . 181
130. Class 2: Mage . . . 182
130.1. Skill Learnings for Mage . . . 182
131. Class 3: Priest . . . 183

131.1. Skill Learnings for Priest . . . 183
132. Class 4: Summoner . . . 184
132.1. Skill Learnings for Summoner . . . 184
133. Class 5: Hero . . . 185
133.1. Skill Learnings for Hero . . . 185
134. Class 6: Thief . . . 186
135. Class 7: Start . . . 187
136. Class 10: ---Special--- . . . 188
137. Class 11: Guard . . . 189
137.1. Skill Learnings for Guard . . . 189
138. Class 12: Wind Spirit . . . 190
138.1. Skill Learnings for Wind Spirit . . . 190
139. Class 13: Pixie . . . 191
139.1. Skill Learnings for Pixie . . . 191
140. Class 14: Fire Chicken . . . 192
140.1. Skill Learnings for Fire Chicken . . . 192
141. Chapter 25: The Complete Grimoire (75+ Skills & Formulas) . . . 193
142. Overview . . . 194
143. Basic Martial Maneuvers . . . 195
144. Knight Martial Techniques . . . 196
145. Mage Elemental Spells . . . 197
146. Priest Holy Arts & Turn Manipulation . . . 199
147. Summoner Totems & Rites . . . 202
148. Summon Beast Techniques . . . 203
149. Enemy Bestial & Void Techniques . . . 204
150. Chapter 26: Bestiary, Destructible Props & Troop Formations . . . 205
151. Overview . . . 206
152. 1. Monster Bestiary . . . 207
153. 2. Neutral & Interactive Grid Entities . . . 208
154. 3. Troop Encounter Formations . . . 209
155. Chapter 27: Armory, Inventory & Key Lore Relics . . . 210
156. Overview . . . 211
157. 1. Weapons & Relic Armaments . . . 212
158. 2. Defensive Armor & Relic Wards . . . 213
159. 3. Consumable Items & Quest Relics . . . 214
160. Chapter 28: Engine Systems, Time Clock & Switch/Variable Index . . . 215
161. Overview . . . 216
162. 1. The Monotonic Time Engine . . . 217
163. 2. The Moral Sway Continuum Engine . . . 218
164. 3. Global Switches Directory . . . 219
165. 4. Global Variables Directory . . . 221

166. Chapter 29: Historical Dataset Evolution & Delta Analysis . . . 222
167. Overview . . . 223
168. 1. Dataset Characteristics & Metrics . . . 224
169. 2. Core Differences & Evolution Strategy . . . 225
Volume V: Software Architecture & The 5-Block Invariant Protocol . . . 226
170. Chapter 30: OmniCode 5-Block Architectural Framework . . . 227
171. Architecture Overview . . . 228
172. 1. The Universal 5-Block Code Structure . . . 229
173. 2. Core Mechanics & State Invariants . . . 230
173.1. The Twin Relics: Shield and Sword . . . 230
173.2. Invariant Axioms . . . 230
174. 3. Multi-Phase Engine Evolution . . . 231
175. Chapter 31: Dual-Frontend Presentation Layer (CLI + Native X11) . . . 232
176. Abstract . . . 233
177. 1. The Core Architectural Philosophy . . . 234
177.1. Universal Action Dispatcher & Intent Emitter . . . 234
178. 2. Native Desktop GUI Architecture: Why Not Web? . . . 236
178.1. Double-Buffering & Render Synchronization . . . 236
178.2. Dynamic Visual Palette: The Twin Relics . . . 237
179. 3. The 5-Block Implementation Mapping . . . 238
179.1. Presentation Architecture Breakdown (`include/gui/gui.h` & `src/gui/gui.c`) . . . 238
180. 4. Headless CI & Deterministic Replay . . . 239
181. 5. Custom Engine Evolution Roadmap . . . 240
182. Chapter 32: Central Manifest & Pre-Flight Diagnostic Integrity . . . 241
183. Specification Abstract & Purpose . . . 242
184. 1. Expected Architecture: The Central Manifest Pattern . . . 243
184.1. 1.1 Structural Hierarchy . . . 243
184.2. 1.2 The Diagnostic Severity Spectrum . . . 243
184.3. 1.3 Expected Pre-Flight Engine States . . . 244
185. 2. Architectural Metadata Dimensions & Pedagogical Balance . . . 245
185.1. 2.1 Striking the Balance: Beginner-Accessible yet Senior-Rigorous . . . 245
185.2. 2.2 Explicit Temporal Stamping & Version History . . . 245
185.3. 2.3 The 5-Dimension Compiled Schema (`module_metadata_t`) . . . 245
185.4. 2.4 Active Dependency Graph Enforcement . . . 246
186. 3. Expected Pre-Flight "Loose Bolts" System Checks . . . 247
187. 4. Target Verification Criteria ("Expected" Baseline) . . . 248
188. 5. Actual Results vs. Expected Gap Analysis . . . 249
188.1. 5.1 Empirical Benchmark Telemetry . . . 249
188.2. 5.2 How to Reproduce and Verify Every Result . . . 250
188.3. 5.3 Gap Analysis & Next Architectural Horizons . . . 251
189. Chapter 33: Layer 0 Core Utilities Code Tour . . . 252

190. Overview . . . 253
191. 1. Mathematical Invariant Utilities (`math_utils.h` & `math_utils.c`) . . . 254
191.1. Architectural Roles: . . . 254
192. 2. String Manipulation Utilities (`string_utils.h` & `string_utils.c`) . . . 255
192.1. Architectural Roles: . . . 255
193. 3. 2D Geometry Utilities (`geom_utils.h` & `geom_utils.c`) . . . 256
193.1. Architectural Roles: . . . 256
194. 4. Monotonic Timing Utilities (`time_utils.h` & `time_utils.c`) . . . 257
194.1. Architectural Roles: . . . 257
195. 5. Central Manifest & Pre-Flight System (`manifest.h` & `manifest.c`) . . . 258
195.1. Architectural Roles: . . . 259
196. Chapter 34: Layer 1 Simulation Engine Code Tour . . . 260
197. Overview . . . 261
198. 1. State Structure & Memory Layout (`player_state_t`) . . . 262
199. 2. Universal Action Dispatcher (`game_process_command`) . . . 263
199.1. Execution Pipeline: . . . 263
200. 3. Diagnostic Self-Auditing (`game_audit_state`) . . . 264
201. Chapter 35: Layer 2 Presentation Viewport Code Tour . . . 265
202. Overview . . . 266
203. 1. The Native X11 Architecture . . . 267
203.1. Key Specifications: . . . 267
204. 2. TrueColor Palette & Alpha Opacity . . . 268
205. 3. Event Loop & Action Routing . . . 269
206. 4. Viewport Layout Auditing (`gui_audit_layout`) . . . 270
207. Chapter 36: Layer 4 Verification & Test Suite Code Tour . . . 271
208. Overview . . . 272
209. 1. CMake & CTest Test Targets . . . 273
210. 2. State Invariant Suite (`tests/test_game_invariants.c`) . . . 274
211. 3. Gameplay Scenarios & Replay Suite (`tests/test_gameplay_scenarios.c`) . . . 275
Volume VI: Engineering Chronology & Development Logs . . . 276
212. Chapter 37: DevLog 001: Genesis, Paradigm Pivot & Invariants . . . 277
213. Abstract . . . 278
214. 1. Project Genesis: Clean Foundations and the 5-Block Standard . . . 279
215. 2. The Early Prototype: The Survival-Battery Fallacy . . . 280
216. 3. The Paradigm Pivot: One Clarification That Changed Everything . . . 281
216.1. 1. From Passive Attrition to Active Agency . . . 281
216.2. 2. Disentangling Darkness from the Villain . . . 281
216.3. 3. The Moral Sway Continuum . . . 281
217. 4. Dual-Frontend Architecture: Native X11 & Terminal CLI . . . 282
218. 5. Why Game Development Testing Differs from Software Testing . . . 283
218.1. 1. Game State Invariant Testing (`tests/test_game_invariants.c`) . . . 283

218.2. 2. Gameplay Scenarios & Replay Testing (tests/test_gameplay_scenarios.c) . . . 283
219. 6. What Lies Ahead . . . 284
220. Chapter 38: DevLog 003: Central Manifest & Pre-Flight Spec . . . 285
221. Abstract & Engineering Intent . . . 286
222. 1. The Core Problem: Loose Bolts & Silent Corruption . . . 287
223. 2. Architectural Implementation . . . 288
223.1. 2.1 Compiled Metadata Structures . . . 288
223.2. 2.2 Subsystem Integrations . . . 288
224. 3. "Expected" vs. "Actual" Gap Analysis . . . 289
225. 4. Summary & Next Steps . . . 290
226. Chapter 39: DevLog 004: Pedagogical Metadata & Active DAG Contracts . . . 291
227. Abstract & Engineering Intent . . . 292
228. 1. The Pedagogical Balancing Act: Beginner Mission vs. Senior Contracts . . . 293
228.1. 1.1 The Dual-Audience Protocol . . . 293
229. 2. Temporal Precision: Mandating Full ISO 8601 Calendar Dates . . . 295
230. 3. Active DAG Dependency Contracts: Turning Comments into Machine Tripwires . . . 296
230.1. 3.1 Compiled 5-Dimension Metadata Schema . . . 296
230.2. 3.2 Pre-Flight DAG Enforcement Algorithm . . . 297
231. 4. Subsystem Dependency Manifest . . . 298
232. 5. Empirical Verification & Test Results . . . 299
232.1. 5.1 CTest Verification Suite . . . 299
232.2. 5.2 Pre-Flight System Audit Telemetry . . . 299
232.3. 5.3 Active DAG Dependency Tripwire Test . . . 300
233. 6. Conclusion & Next Horizons . . . 301
234. Chapter 40: DevLog 005: Ontological Sub-Assemblies & 80-Column Discipline . . . 302
235. Abstract & Engineering Intent . . . 303
236. 1. The Need for IDE-First Hover Documentation . . . 304
236.1. 1.1 The Clangd Hover Contract . . . 304
237. 2. The Machine Metaphor: From Cosmos to Fasteners . . . 305
238. 3. Visual Border & Delimiter Standards . . . 306
239. 4. Chronological Reorganization of Developer Logs . . . 308
240. 5. Empirical Verification Telemetry . . . 309
241. 6. Summary . . . 310
242. Chapter 41: DevLog 006: Source Implementation & Root Pedagogical Commenting . . . 311
243. Executive Summary . . . 312
244. 1. Architectural Intent: End-to-End Documentation Consistency . . . 313
245. 2. Source Code Implementation Matrix . . . 314
246. 3. Root Configuration Pedagogical Enhancement . . . 315
246.1. 3.1 CMakeLists.txt . . . 315
246.2. 3.2 .gitignore . . . 315
246.3. 3.3 README.md . . . 315

247. 4. Strict 80-Column Verification Audit . . . 316
248. 5. Verification & Telemetry Results . . . 317
249. Chapter 42: DevLog 007: Archival Extraction & The 2017–2018 Demo Baseline . . . 318
250. Abstract . . . 319
251. 1. Historical Context: Rediscovery of the Genesis Datasets . . . 320
252. 2. The Extraction Toolchain: `docs/tools/extract_demo_data.py` . . . 321
252.1. Core Functions of the Extraction Engine . . . 321
253. 3. The 10-Volume Baseline Documentation Compendium . . . 322
254. 4. C11 Engine Rebuilding Roadmap . . . 323

Chapter 1. Executive Summary

This Master Guide serves as the definitive single source of truth for **Tenebris and Lux**. It unites world canon, character dossiers, ludology, the 2017–2018 demo baseline, software architecture, and engineering dev logs into a comprehensive compendium.

Volume I: World Canon, Metaphysics & Sacyran Lore

Cosmic neutrality of Lux and Tenebris, chivalric Kingdom of Sacyra, and the existential menace of Dark Lord Malakor.

Chapter 2. Chapter 1: The Primordial Neutrality of Lux and Tenebris

Source File: `/home/professorseanex/Tenebris and Lux/docs/lore/world_and_lore.adoc`

Chapter 3. Canonical Introduction

This document establishes the living world-building canon, cosmological ontology, and historical epic of **Tenebris and Lux** ("Darkness and Light"). Conceived in 2014–2018 as the foundational game project of **CreativeWorkzStudio**, this is the epic of two primordial relics—**The Shield of Lux** and **The Sword of Tenebris**—set within the storied **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**). The story is driven by two central protagonists: **Elijah Ro-enheart**, son of a former head of the Royal Guard, and **Katrina Jaren**, a noble-born daughter navigating royal court expectations while yearning for the Adventurers' Guild, as the re-awakening **Dark Lord Malakor** rises from the abyss.

Chapter 4. 1. The Primordial Cosmology: Neutrality of the Apex Forces

> "Light without shadow is a blinding desert that calcifies the spirit. Shadow without light is a suffocating void where purpose drowns. The Shield preserves and confines; the Sword severs and liberates. The relics know neither virtue nor malice—only the soul that wields them."
>
> — The Sacyran Chronicle of Antiquity

In the cosmology of **Tenebris and Lux**, the cosmos is governed by two apex forces that transcend human morality. Neither is inherently good nor evil:

4.1. Lux (The Apex of Radiance)

- **Ontological Essence**: Manifestation, illumination, cohesion, preservation, and structure.
- **The Dual Nature of Light**:
  - **Virtuous Manifestation**: Wards the vulnerable with impenetrable golden barriers, unmasks dark deceits, preserves cultural knowledge, and inspires steadfast courage.
  - **Tyrannical Manifestation**: Scorches the earth in unyielding glare, blinds those seeking mercy, enforces inflexible order, and petrifies living will into stone.

4.2. Tenebris (The Apex of Darkness)

- **Ontological Essence**: Mystery, dissolution, rest, severance, and the fertile void.
- **The Dual Nature of Darkness**:
  - **Virtuous Manifestation**: Veils the oppressed from predators, cuts through unholy bonds and tyrant armor, grants restful oblivion to the weary, and pierces through arrogant illusions.
  - **Malevolent Manifestation**: Spreads creeping rot, feeds paranoia, dissolves conscience into nihilism, and serves the destructive malice of the Dark Lord.

4.3. The Moral Sway of the Wielders

Neither Lux nor Tenebris dictates a character's morality. Good and evil belong entirely to the **intent, conscience, and actions of the wielder**. In the Kingdom of Sacyra, a knight channeling the blinding radiance of Lux can act as a cold oppressor, while a warrior wielding the void-quenched blade of Tenebris can stand as the people's ultimate protector.

Chapter 5. 2. The Twin Relics: Shield of Lux & Sword of Tenebris

Before the crowning of Sacyra's first monarch, primordial celestial smiths forged two sovereign armaments to anchor the realm against cosmic oblivion:

Relic	The Shield of Lux	The Sword of Tenebris
Aspect	The Sovereign Aegis of Radiance	The Sovereign Blade of the Abyss
Physical Form	A towering kite-shield of burnished solar silver, centered with a living celestial sunstone that pulses with warm golden resonance.	A two-handed greatsword forged from star-quenched obsidian, its edge rippling with deep twilight violet mist and void-tempered steel.
Primal Function	Warding, kinetic deflection, barrier projection, illumination, and stasis shielding.	Severance, armor piercing, shadow manipulation, reality-edge slicing, and mist cloaking.
Virtuous Wielding	Wards allies, shields innocents from siege engines, and unmasks lurking infiltrators.	Slices through corrupted war machines, cuts captive chains, and strikes from the dark to liberate.
Corrupted Wielding	Blinds subjects into terrified compliance; traps souls in eternal petrified stasis.	Indiscriminate butchery; feeding the blade's hunger on mortal dread; total subjugation.

5.1. Harmonized Equilibrium

The Shield and Sword were never intended to war against one another; they are complementary halves of a cosmic balance: - When wielded in tandem or in synchronized coordination by trusted companions (**The Stance of Equilibrium**), the Shield deflects kinetic onslaught while the Sword opens devastating counter-attack avenues. - If one relic is wielded without the tempering influence of the other, its bearer risks succumbing to either calcifying stasis or devouring void-sickness.

Chapter 6. 3. The Kingdom of Sacyra: Setting & Society

The **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**, /s□□□ki□r□/) is a grand, historic realm defined by stone bastions, sweeping borderlands, and deeply rooted chivalric traditions.

6.1. Governance and Courtly Politics

- **The Crown & The Royal Guard**: Sacyra's royal palace commands the central citadel. The **Royal Guard** represents the pinnacle of martial honor and personal devotion to the throne, sworn to protect the realm against external ruin and internal treachery.
- **The Noble Houses**: Hereditary aristocratic families hold vast regional fiefs, supply knights to the royal army, and compete for influence and appointed seats on the prestigious **Royal Court**.
- **The Adventurers' Guild**: An independent, meritocratic brotherhood operating throughout Sacyra's frontier outposts and merchant crossroads. Unlike the rigid court or the ceremonial nobility, the Guild takes on perilous monster contracts, frontier patrols, and ancient ruin expeditions, valuing practical skill and personal freedom above noble pedigree.

6.2. The Crisis: The Re-Awakening of Dark Lord Malakor

After centuries of dormancy sealed beneath the Abyssal Vaults along Sacyra's furthest borders, the ancient terror **Dark Lord Malakor** has broken free.

- Malakor is not a personification of Tenebris, but a corrupt mortal arch-sorcerer who twisted dark magic into a tool of catastrophic conquest.
- Seeking to conquer Sacyra, Malakor commands legions of void-blighted fiends, corrupted constructs, and dread vanguard captains.
- His vanguard has already shattered the outer border redoubts, threatening to overrun Sacyra unless heroes arise who can awaken and master the legendary Shield and Sword.

Chapter 7. 4. The Dual Protagonists

The heart of **Tenebris and Lux** is the intertwined journey of two young figures from opposite ends of Sacyra's social fabric:

7.1. 1. Elijah Ro-enheart (The Vanguard of Duty)

- **Lineage**: Born into a proud line of soldiers who have served in Sacyra's royal army for generations. His father was a legendary former head of the **Royal Guard**.
- **Role & Archetype**: Frontline combatant, tactician, and stalwart protector. Raised under a strict code of chivalric duty, martial conditioning, and sacrificial defense.
- **Character Dilemma**: Strives to live up to the immense shadow of his father's legacy while questioning whether blind obedience to royal orders truly protects the people of Sacyra when court politics paralyze the realm.
- **Relic Attunement**: Naturally drawn to the **Shield of Lux**, anchoring defensive formations, absorbing enemy assaults, and safeguarding his squad.

7.2. 2. Katrina Jaren (The Free-Spirited Blade)

- **Lineage**: A daughter of **House Jaren**, one of the most prominent noble families in the Kingdom of Sacyra.
- **Role & Archetype**: Agile skirmisher, perceptive scout, and tactical striker. Possesses natural martial brilliance and a fiercely independent spirit.
- **Character Dilemma**: Raised and groomed since childhood to take a ceremonial seat upon the Royal Court, Katrina suffocates under high-society diplomacy. She secretly longs to break free of courtly expectations and join the **local Adventurers' Guild**, seeking genuine adventure and frontline camaraderie.
- **Relic Attunement**: Naturally resonates with the **Sword of Tenebris**, channeling its fluid, decisive severance to bypass armored juggernauts and cut through enemy flanks.

7.3. Dynamic Interplay

Together, Elijah and Katrina represent the living synthesis of Sacyra: military discipline paired with adventurer initiative; steadfast defense matched with decisive offense; the golden starlight of Lux intertwined with the twilight edge of Tenebris.

Chapter 8. 5. The Tactical Strategy RPG (TRPG) Paradigm

Unlike standard linear dungeon crawlers, **Tenebris and Lux** is built as a **Tactical Strategy RPG**:

1. **Strategic Grid Battlefield**:
    - Battles take place across multi-elevation grid maps (provinces, castle courtyards, bridges, and mountain passes).
    - Units position according to terrain bonuses, line-of-sight, directional flanking, and movement budgets.
2. **Cinematic Combat Resolution**:
    - When a unit initiates combat against an enemy on the grid, the game transitions into a dynamic **traditional side-scroll / action-based cinematic duel**.
    - Attacks, parries, spell wards, and severance slashes play out with visual weight and player agency before returning to the grid.
3. **Moral Sway Influence**:
    - Dialogue choices, tactical decisions (mercy vs. ruthless execution), and relic usage alter the party's **Moral Sway** (`[-100, +100]`), influencing recruitable companions, kingdom politics, and the story's ultimate conclusion.

Chapter 9. Chapter 2: Socio-Political Strata of the Kingdom of Sacyra

Source File: `/home/professorseanex/Tenebris and Lux/docs/lore/characters.adoc`

Chapter 10. Overview

This document provides complete biographical dossiers, martial doctrines, and thematic arcs for the central protagonists and key figures of **Tenebris and Lux** in the **Kingdom of Sacyra**.

Chapter 11. 1. Elijah Ro-enheart (The Aegis of Lineage)

> "My father taught me that a shield is not a wall to hide behind, but a promise made to the people standing behind you."
>
> — Elijah Ro-enheart

11.1. Biography & Heritage

- **Full Name**: Elijah Ro-enheart
- **Affiliation**: Kingdom of Sacyra Royal Army (formerly Royal Cadet Corps)
- **Family Lineage**: The Ro-enhearts have served the Sacyran crown for five unbroken generations. Elijah's father, **Commander Gareth Ro-enheart**, was the esteemed former Head of the Royal Guard, famed for holding the Sunken Citadel against an allied barbarian siege decades prior.
- **Upbringing**: Raised in the barracks and training quadrangles of the capital citadel. Trained rigorously in shield craft, formation tactics, and military law from childhood.
- **Visual & Demeanor**: Broad-shouldered, armored in fitted silver-steel infantry plate over deep navy gambeson. Earnest, disciplined, observant, and burdened by the immense expectations of his family name.

11.2. Combat & Tactical Doctrine (TRPG)

- **Battlefield Role (Grid)**: Heavy Frontline Anchor / Guardian Vanguard.
- **Grid Attributes**: High defense, heavy physical mitigation, zone-of-control denial, ally interception.
- **Cinematic Action Style (Side-View)**:
  - Solid, grounded sword-and-shield techniques.
  - Heavy shield-bashes that stagger incoming charges and break enemy poise.
  - Barrier projection: Raises the **Shield of Lux** to cast wide golden auric domes that deflect projectile swarms.

11.3. Narrative Arc & Internal Conflict

Elijah enters the war believing unquestioningly in military discipline and hierarchy. However, as the Dark Lord Malakor's invasion accelerates and noble houses in the royal court squabble over political favor rather than sending reinforcements to border villages, Elijah is forced to confront what true honor demands: does a soldier serve the written decree of a hesitant court, or the living people of Sacyra?

Chapter 12. 2. Katrina Jaren (The Blade of Freedom)

> "Silk dresses and court etiquette won’t turn back a vanguard blade. Let my cousins argue over tax registries; my place is on the frontier."
>
> — Katrina Jaren

12.1. Biography & Heritage

- **Full Name**: Lady Katrina Jaren
- **Affiliation**: House Jaren (Aristocracy of Sacyra); Aspirant of the Adventurers' Guild
- **Family Lineage**: **House Jaren** is an old, wealthy noble house holding lucrative trade vineyards and silver mines in Sacyra’s eastern province. They command significant political clout on the Royal Court.
- **Upbringing**: Groomed by private tutors, diplomats, and court heralds to occupy a powerful hereditary seat on the Royal Court to advance family interests. Katrina despised the stifling, hypocritical court etiquette, sneaking away to train with Guild sellswords and ranger outriders.
- **Visual & Demeanor**: Quick-footed, clad in tailored lightweight leather armor over noble-weave tunic, with rapier and long-dagger sheaths. Quick-witted, fiercely autonomous, skeptical of bureaucratic protocol, and fiercely loyal to companions who earn her respect.

12.2. Combat & Tactical Doctrine (TRPG)

- **Battlefield Role (Grid)**: High-Mobility Skirmisher / Flanking Striker.
- **Grid Attributes**: High movement tiles, evasive evasion against counter-attacks, armor-piercing critical strikes, hit-and-run tactics.
- **Cinematic Action Style (Side-View)**:
  - Fluid, dynamic acrobatics, quick dashes, and precise target severance.
  - Channels the **Sword of Tenebris** to cloak in mist, closing distances instantly.
  - Unleashes crescent twilight blades that slice clean through heavy automaton plating and void-cursed armor.

12.3. Narrative Arc & Internal Conflict

Katrina longs for the unvarnished honesty of the **Adventurers' Guild**, where skill and character matter more than noble pedigree. Yet her noble education gave her a sharp analytical mind and an understanding of political subterfuge that simple soldiers lack. As she takes up the Sword of Tenebris alongside Elijah, she must reconcile her thirst for personal freedom with her inescapable responsibility to the survival of Sacyra.

Chapter 13. 3. Dual-Protagonist Dynamics: The Wielders of Harmony

The heart of the gameplay and story is the contrast and evolving partnership between Elijah and Katrina:

Dimension	Elijah Ro-enheart	Katrina Jaren
Social Caste	Military Soldier / Son of the Royal Guard Head	High Aristocracy / House Jaren
Institution	Sacyran Royal Army (Structure, Law, Duty)	Adventurers' Guild Aspirant (Freedom, Merit, Exploration)
Relic Affinity	**The Shield of Lux** (Defense, Preservation, Barrier)	**The Sword of Tenebris** (Offense, Severance, Velocity)
Combat Rhythm	Anchor, Stagger, Absorb, Hold the Line	Flank, Pierce, Evade, Sever Key Threats
Moral Perspective	Starts with Lawful Duty; learns the necessity of independent conscience	Starts with Rebellious Independence; learns the weight of sacrifice and leadership

13.1. Tactical Synergy on the Grid

- **The Warding Wall**: Elijah positions on a bottleneck grid tile, activating Shield of Lux aura to increase Katrina’s evasion and protect her from ranged artillery.
- **The Twilight Riposte**: When an enemy assaults Elijah’s shield, Katrina triggers an interception attack from an adjacent grid square, executing a devastating tandem strike.
- **Relic Sharing**: While each hero naturally attunes to one relic, critical narrative crises and high-level TRPG skills allow them to pass or dual-wield the twin armaments, unlocking the supreme **Stance of Equilibrium**.

Chapter 14. 4. Antagonist: Dark Lord Malakor (The Void Sovereign)

- **Identity**: The ancient necromantic lord of the deep Abyssal Vaults bordering Sacyra.
- **Nature**: An immortal warlord who long ago submerged his mortal body in forbidden void mists, Malakor embodies tyrannical ambition.
- **Objective**: Overrun the Kingdom of Sacyra, capture both the Shield of Lux and the Sword of Tenebris, and use their primordial power to permanently extinguish free will across the continent.
- **The Vanguard**:
  - **Umbral Centurions**: Corrupted soldiers encased in blackened, spiked plate armor.
  - **Blighted Automata**: Ancient war-golems reanimated by dark soul-engines.
  - **Void Stalkers**: Ethereal shadow beasts that bypass heavy armor to strike at unit rear lines.

Chapter 15. Chapter 3: High School Inception & TRPG Genesis

Source File: /home/professorseanex/Tenebris and Lux/docs/dev_logs/2026/10/lore_and_design/devlog_002_the_sacyra_chronicles_and_trpg_vision.adoc

Chapter 16. Abstract

This developer log chronicles the foundational provenance and overarching creative vision of **Tenebris and Lux**, rooted in its 2014–2018 high school inception as the premier game project of CreativeWorkzStudio. It documents the elimination of placeholder factions in favor of the canonical **Kingdom of Sacyra**, the introduction of dual protagonists **Elijah Ro-enheart** and **Katrina Jaren**, and the technical specification of the game's true genre: a **Tactical Strategy RPG (TRPG)** fusing grid-based tactical maneuvering with cinematic traditional side-scroll / action-based combat resolution.

Chapter 17. 1. Historical Provenance: The 2014–2018 High School Genesis

Before CreativeWorkzStudio evolved into a multifaceted publishing house, **Tenebris and Lux** was its very first major game in active development.

- **The RPG Maker Era (2014–2018)**: Conceived while the author was in high school learning game design and programming. The initial iterations were built using the default offerings of RPG Maker, exploring community plugins, and crafting characters through its Character Generator.
- **The Lost Assets & The Blank Canvas**: With no legacy digital assets surviving from that era, the project stands not as a nostalgic port, but as a complete ground-up rebirth—realizing the original creative vision with the systems engineering maturity, mathematical rigor, and architectural discipline of modern custom engine design.

Chapter 18. 2. Narrative Realignment: The Kingdom of Sacyra

During early prototype scaffolding, generic placeholder factions ("The Solar Order" and "The Umbral Covenant") were inserted to test basic moral sway logic. On October 26, 2026, the creator firmly realigned the canon, stripping away these synthetic factions to reveal the true world:

18.1. 1. The Kingdom of Sacyra (/s□□□ki□r□/)

- The setting is the historic **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**).
- Sacyra is a structured, chivalric realm anchored by three distinct socio-political pillars:
  1. **The Crown & Royal Guard**: Guardians of legal order, national defense, and the ancestral throne.
  2. **The Noble Houses & Royal Court**: High aristocracy governing regional fiefs and vying for political dominance on the royal council.
  3. **The Adventurers' Guild**: A pragmatic, frontier-based brotherhood operating beyond courtly vanity, answering commoner monster distress calls and exploring untamed ruins.
- **The Existential Threat**: The ancient necromantic sovereign **Dark Lord Malakor** breaking through the Abyssal Vaults, threatening to extinguish all mortal agency.

18.2. 2. The Dual Protagonists

Rather than a generic lone wanderer, the narrative is anchored by two contrasting protagonists whose fates, lineages, and martial philosophies intertwine:

Character	Elijah Ro-enheart	Lady Katrina Jaren
Social Origin	Military Lineage (Common/Barracks Nobility)	High Aristocracy (House Jaren)
Family Heritage	Son of the former Head of the Royal Guard	Daughter of an influential noble house on the Royal Court
Internal Desire	Living up to his father’s legendary military legacy while discovering what true honor demands	Escaping stifling aristocratic court expectations to join the local **Adventurers' Guild**
Relic Synergy	**The Shield of Lux** (Sanctuary, Warding, Formation Hold)	**The Sword of Tenebris** (Severance, Mist-Veil, Critical Strike)
TRPG Doctrine	Frontline heavy vanguard; zone-of-control denial; ally interception	Agile outrider skirmisher; bypassing enemy lines; flanking assassin

Chapter 19. 3. Ludological Evolution: The Tactical Strategy RPG (TRPG)

The game's true mechanical identity is a **Tactical Strategy RPG (TRPG)**. This fundamentally informs how the simulation engine and presentation viewports are being constructed:

19.1. 1. The Dual-Layer Combat Loop

The combat loop operates across two synchronized phases:

1. **Strategic Grid Phase**:
    - The battle takes place on a discrete 2D grid representing castles, ramparts, thickets, and void-blighted frontier passes.
    - Units position based on movement ranges, terrain defense bonuses, directional facing (Front, Flank, Rear), and Zone-of-Control (ZOC) restrictions.
2. **Cinematic Combat Resolution (Side-View Action)**:
    - When a unit initiates combat against an enemy target, the game transitions into a dynamic **2D traditional side-scroll / action-based clash**.
    - Attacks play out with timed player agency: precision blocks with the Shield of Lux, poise-breaking shield bashes, and devastating severance slashes with the Sword of Tenebris.
    - Upon duel conclusion, remaining unit health, stagger, and relic resonance seamlessly persist back to the tactical grid.

19.2. 2. Architectural Implications for the C Engine

Knowing this was originally an RPG Maker project gives us clear architectural directives: * **Decoupled Data Architecture**: Static definitions (`Units`, `Skills`, `Equipment`, `TerrainTypes`) must exist as clean data tables distinct from session state (`player_state_t`, `grid_t`). * **Deterministic Event Pipeline**: Interactive map features (gates, switches, dialog triggers) will follow a clean, reentrant command interpreter model. * **Zero Engine Bloat**: By building this custom in C11 rather than relying on bloated runtimes, our TRPG will maintain sub-millisecond turn updates, instantaneous scene transitions, and absolute softlock immunity.

Chapter 20. 4. Next Milestone Roadmap

1. **Phase 1 (Complete)**: Established OmniCode 5-block layering, Layer 0 utilities, CTest verification suite, and canonical documentation.
2. **Phase 2 (Active)**: Expand simulation state to model dual protagonists (Elijah & Katrina), party data structures, and the foundational 2D tactical grid.
3. **Phase 3**: Implement the cinematic side-scroll combat transition in the presentation layer.
4. **Phase 4**: Expand the tactical campaign across the provinces of Sacyra to confront the vanguard of Dark Lord Malakor.

Volume II: Dramatis Personae & Dual Protagonist Dynamics

Comprehensive character dossiers, dual protagonist synergies, and regional faction leaders.

Chapter 21. Chapter 4: Dual Protagonist Dynamics & Moral Continuum

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/protagonists/dual_protagonist_dynamics.adoc

Chapter 22. Overview

The narrative core and mechanical heartbeat of **Tenebris and Lux** is the interplay between **Elijah Ro-enheart** and **Katrina Jaren**. Contrasting in social caste, philosophical outlook, and combat rhythm, their evolving partnership mirrors the cosmic relationship between Light and Darkness.

Chapter 23. 1. Thematic & Social Contrast Matrix

Dimension	Elijah Ro-enheart	Katrina Jaren
Social Caste	Military Lineage / Son of the Guard Head	High Aristocracy / House Jaren
Institutional Allegiance	Royal Army (Structure, Law, Duty)	Adventurers' Guild Aspirant (Freedom, Merit)
Relic Affinity	**The Shield of Lux** (Defense, Preservation)	**The Sword of Tenebris** (Offense, Severance)
Temperament	Courteous, methodical, law-abiding	Irreverent, sharp-tongued, opportunistic
Moral Trajectory	From rigid obedience to moral conscience	From cynical detachment to noble sacrifice

Chapter 24. 2. Interpersonal Chemistry & Banter

Elijah and Katrina embody the classic dramatic friction between an earnest, rule-following guardian and an irreverent, sharp-witted rogue. Where Elijah pauses to consult protocol or verify authority, Katrina acts impulsively, cutting through bureaucratic red tape with sarcastic practicality.

24.1. Narrative Evolution Across the Campaign

1. **Act I (The Provincial Divide)**: Initial friction in Dio Village. Katrina mocks Elijah's military formality; Elijah cautions against Katrina's cynical disregard for property and propriety.
2. **Act II (The Capital Conspiracy)**: Arrival at Sacyra Castle. Katrina's insight into aristocratic deception exposes King Jihad's corruption, forcing Elijah to choose between his oaths to the throne and his duty to the kingdom.
3. **Act III (The Sovereign Eclipse)**: Total metaphysical synchronization. Facing Dark Lord Malakor, the two heroes overcome their individual biases, passing relics fluidly between each other to unlock the Stance of Equilibrium.

Chapter 25. 3. Tactical Synergy on the TRPG Battlefield

On the tactical grid, Elijah and Katrina form a mutually dependent combat pair:

1. **The Phalanx Pivot**: Elijah pins enemy vanguards with his Shield of Lux ZOC aura. Katrina uses high mobility to circle the battle line and strike flanked enemies.
2. **Warded Severance**: Katrina lures high-threat enemies into charging toward her, stepping back onto Elijah's tile perimeter as Elijah executes a staggering shield deflection.
3. **The Stance of Equilibrium**: In moments of ultimate synchronization, both relics resonance counters combine, unlocking unified abilities where radiant starlight and abyssal void strike in a single coordinated turn.

Chapter 26. Chapter 5: Elijah Ro-enheart: Vanguard of Lux

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/protagonists/elijah_ro_enheart.adoc

Chapter 27. Overview

Elijah Ro-enheart is the frontline heavy vanguard anchor of **Tenebris and Lux** and the initial bearer of the **Shield of Lux**. Steeped in the military traditions of the Royal Guard yet haunted by the unexplained disappearance of his father, Elijah represents the struggle between rigid institutional duty and personal honor.

Chapter 28. 1. Identity & Heritage

- **Full Name**: Elijah Ro-enheart
- **Title / Rank**: Cadet Vanguard, Sacyran Royal Defense Corps
- **Affiliation**: Kingdom of Sacyra Royal Army; House Ro-enheart
- **Family Lineage**:
  - **Father**: Commander Gareth Ro-enheart, legendary former Head of the Royal Guard, celebrated throughout Sacyra for holding the Sunken Citadel. Gareth vanished under mysterious crown orders following Elijah's **10th birthday**, leaving behind unresolved questions and immense expectations.
  - **Grandmother**: **Magolia Ro-enheart** ("Grandma Magolia"), a respected matriarch residing in the Upper Ward of Dio Village, guardian of the sacred family timepiece (**Grandma's Watch**).
- **Visual Profile**: Broad-shouldered, athletic build, steel-blue eyes, and cropped dark hair. Wears fitted silver-steel infantry plate over deep navy quilted gambeson, marked with the defensive crest of the Ro-enhearts.

Chapter 29. 2. Psychological Profile & Demeanor

Elijah is earnest, polite, disciplined, and deeply considerate. While raised within the hierarchy of military discipline, he possesses an intrinsic moral compass that values the safety of innocent civilians above abstract royal decrees.

29.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

- **Respect for Elders & Prudence**: [quote, Elijah, Dio Village - Grandma’s House] "I shouldn’t go in this chest. I need to talk to Grandma first. She’s very particular about her things."
- **The Father Mystery**: [quote, Elijah, Opening Quest] "Maybe it is something about dad! I haven’t seen him since my 10th birthday. This might be my chance to get answers!"
- **Civic Duty & Moral Conscience**: [quote, Elijah, Lower Village Confrontation] "I can’t believe that I am doing this… the baron is the richest man in the village… but he’s been exploiting the town! This is for the good!"

Chapter 30. 3. Martial Doctrine & TRPG Grid Mechanics

Elijah functions as the defensive bulwark of the tactical party, controlling chokepoints and mitigating damage for allied skirmishers.

Dimension	Specification
TRPG Grid Role	Frontline Heavy Vanguard / Fortress Anchor
Movement Budget	3 Cells (Medium-Heavy Infantry)
Zone of Control (ZOC)	Projects heavy ZOC tiles, halting enemy sprint moves
Primary Relic	**The Shield of Lux** (Resonance `[0, 100]`)
Tactical Signature	`Warding Aura`: Projects a 2-cell radius barrier dome
Poise & Knockback	Immune to 1-cell forced displacement; absorbs shock

30.1. Tactical Synergies

1. **Bottleneck Holding**: Elijah anchors narrow corridors, absorbing melee punishment while building Lux resonance via successful deflections.
2. **Aegis Interception**: Steps into adjacent cells to intercept lethal strikes aimed at fragile skirmishers or casters (e.g., Katrina Jaren).

Chapter 31. Chapter 6: Lady Katrina Jaren: The Outrider Blade

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/protagonists/katrina_jaren.adoc

Chapter 32. Overview

Lady Katrina Jaren is the high-mobility skirmisher of **Tenebris and Lux** and the initial wielder of the **Sword of Tenebris**. Born to high aristocracy yet repulsed by the deceit and stagnation of the Royal Court, Katrina seeks autonomy within the meritocratic ranks of the Adventurers' Guild.

Chapter 33. 1. Identity & Heritage

- **Full Name**: Lady Katrina Jaren
- **Title / Standing**: Noble Court Aspirant (Estranged); Guild Novice
- **Affiliation**: House Jaren (Eastern Aristocracy); Adventurers' Guild
- **Family Lineage**:
  - **House Jaren**: One of the wealthiest noble lineages in Sacyra, controlling lucrative vineyard estates and mountain silver mines. They hold significant patronage leverage on the Royal Court.
  - **Kinship with Dio Village**: Despite her family's high status in the capital, Katrina spent formative rebellious seasons under the watchful eye of Grandma Magolia in Dio Village, seeking respite from court etiquette.
- **Visual Profile**: Agile, athletic build, sharp amber eyes, and unbound chestnut hair tied in a practical combat braid. Clad in tailored lightweight leather arming doublets over noble-weave tunic, with rapier and long-dagger scabbards balanced for rapid unsheathing.

Chapter 34. 2. Psychological Profile & Demeanor

Katrina is quick-witted, irreverent, opportunistic, and intensely independent. She detests empty aristocratic pretension and speaks with an unvarnished, sarcastic candor that often unnerves formal soldiers like Elijah. Beneath her cynical exterior lies a fierce loyalty to those she deems genuine.

34.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

- **Impatience with Stagnation & Cynicism**: [quote, Katrina, Dio Village - Grandma’s House] "The will. I love you gran, but a girl gotta get out of this dead-end town. Batty old woman seems to mind that chest… Crazy old woman, that will is mine!"
- **Pragmatic Opportunism**: [quote, Katrina, The Sword Shrine] "Free sword for your girl~!"
- **Fiery Independence & Temper**: [quote, Katrina, Monica’s Grove] "LISTEN HERE YOU OLD-! What is wrong with you… Gran, I’m not your errand girl! This is YOUR friend."
- **Reluctant Heroism**: [quote, Katrina, Lower Village Heist] "Ok… here we go… let’s do this. I have to do at least one good thing today."

Chapter 35. 3. Martial Doctrine & TRPG Grid Mechanics

Katrina excels at rapid battlefield repositioning, flanking maneuvers, and armor-piercing critical strikes.

Dimension	Specification
TRPG Grid Role	High-Mobility Skirmisher / Flanking Striker
Movement Budget	4–5 Cells (Light Scout / Infiltrator)
Zone of Control (ZOC)	Evasive Infiltration: Ignores basic enemy ZOC delays
Primary Relic	**The Sword of Tenebris** (Edge Sharpness `[0, 100]`)
Tactical Signature	`Phantom Slash`: Teleports behind target cell to strike
Evasion & Stealth	Gains 25% evasion when ending turn on obscured tiles

35.1. Tactical Synergies

1. **Flanking Tandem**: Moves across enemy flanks while Elijah holds attention, landing backstab criticals with the abyssal edge of Tenebris.
2. **Hit-and-Run Decoupling**: Uses remaining move points to disengage after striking, retreating behind Elijah's protective barrier dome.

Chapter 36. Chapter 7: Grandma Magolia: Matriarch of Dio Village

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/dio_village/grandma_magolia.adoc

Chapter 37. Overview

Grandma Magolia (Magolia Ro-enheart) is the beloved matriarch of Dio Village, mother to Commander Gareth Ro-enheart, and grandmother to Elijah Ro-enheart. Providing warmth, historical perspective, and the family heirloom (**Grandma's Watch**), Grandma Magolia anchors the heroes to the human heart of Sacyra.

Chapter 38. 1. Identity & Family Ties

- **Full Name**: Magolia Ro-enheart ("Grandma Magolia")
- **Residence**: Grandma’s House (Two-story timber cottage, Dio Upper Ward)
- **Family Lineage**:
  - **Son**: Commander Gareth Ro-enheart (Vanished Head of the Royal Guard)
  - **Grandson**: Elijah Ro-enheart (Bearer of the Shield of Lux)
  - **Surrogate Charge**: Lady Katrina Jaren (who sought haven with Magolia during her rebellion against House Jaren court tutors)
  - **Lifelong Friend**: Monica the Mystic (companion since youth)

Chapter 39. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

Grandma Magolia speaks with gentle affection, nostalgic wisdom, and quiet sadness regarding the decline of provincial Sacyra.

- **The Heirloom Quest**: [quote, Grandma Magolia, Map 21 / Map 7] "Hello sweetie, could you please go upstairs and search in the chest and grab what’s inside? It’s something that has great significance."
- **Reflections on the Kingdom’s Neglect**: [quote, Grandma Magolia, Map 21 / Map 7] "This town used to be full of life. **sighs** The kingdom doesn’t pay much attention to this place anymore…"
- **Passing Down the Heirloom Watch**: [quote, Grandma Magolia, Map 21 / Map 7] "Thank you honey. This watch has been in the family for a long time. Oh, how it brings back memories of better days. I think you should keep it. It’s my gift to you. And here’s a bit of money. I know that you’ll need it more than little ol' me."
- **A Lifelong Bond with Monica**: [quote, Grandma Magolia, Map 21 / Map 7] "Oh me and Monica go way back. She was such a wild child, always getting into mischief. You’ll love her."

Chapter 40. 3. The Heirloom Timepiece: Grandma’s Watch

Grandma Magolia bequeaths **Grandma’s Watch** to the party. In the original 2017–2018 RPG Maker project, this item activated the global calendar and monotonic time engine (Seconds, Minutes, Hours, Days, Weeks, Months, and Seasons). In the modern C11 architecture, this lore item corresponds directly to the Layer 0 `time_utils` subsystem, tracking monotonic game ticks and frame budgets.

Chapter 41. Chapter 8: Monica the Mystic & Goddess Lyri

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/dio_village/monica_and_lyri.adoc

Chapter 42. Overview

Monica the Mystic (known as **The Sage** in village folklore) and her elemental familiar **Lyri** reside in an arcane grove on the outskirts of Upper Dio Village. Sarcastic, sharp-tongued, yet deeply attuned to the metaphysical currents of Sacyra, Monica is the first local authority to confirm that the forces of Darkness have re-awakened.

Chapter 43. 1. Identity & Mystical Lineage

- **Name**: Monica (**The Mystic / The Sage**)
- **Companion Familiar**: **Lyri**, an ethereal nature spirit who perceives metaphysical fluctuations and whispers warnings into Monica's ear.
- **Residence**: Monica's House & Mystic Grove (Dio Upper Ward)
- **Alliances**: Lifelong confidante of Grandma Magolia; contact of Chrono the Sage.

Chapter 44. 2. Canonical Voice & Demeanor (2017–2018 RPG Maker Origin)

Monica possesses a biting, irreverent wit, greeting newcomers with playful insults before demonstrating profound arcane knowledge.

- **Blunt First Impressions**: [quote, Monica to Elijah, Map 27 / Map 29] "And who in the blaze are YOU! … So you are Magolia's grandson. She made you sound more handsome than you are."
- **Clashing with Katrina**: [quote, Monica to Katrina, Map 27 / Map 29] "So you are Magolia's granddaughter. She made you sound prettier than you are."
  Katrina: "LISTEN HERE YOU OLD-!"
  Monica: "Tis only a joke. Now why have you come here? Do you have some milk?"
- **Lyri's Whispers & The Darkness**: [quote, Monica, Map 27 / Map 29] "A wise woman she is. Lyri has been whispering in me ear about the forces of darkness. I'll help, but first, get that hooligan from the city limits. Not only is he preventing folks from leaving, he's also messing with me business."

Chapter 45. 3. Arcane Role in the Campaign

1. **Attunement to the Void**: Through Lyri, Monica senses the breaking of the ancient seals on the Abyssal Threshold long before the Royal Court admits it.
2. **The City Limits Obstacle**: Directs the heroes to clear the town boundary (confronting the "Fighter Kid" blocking the pass) to prove their tactical readiness.
3. **Bridge to the Sages**: Connects Elijah and Katrina with the ancient scholar **Chrono**, guiding them toward their destiny with the twin relics.

Chapter 46. Chapter 9: Eric the Lumberjack & The Moral Litmus

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/dio_village/eric_the_lumberjack.adoc

Chapter 47. Overview

Eric is a former lumberjack residing in the Upper Ward of Dio Village. His livelihood was shattered when a mysterious abyssal portal manifested in the timber forests, driving him into poverty. He introduces the heroes to the **Ring of Fire** and anchors the early **Test of Honesty** moral dilemma.

Chapter 48. 1. Identity & Background

- **Name**: Eric
- **Occupation**: Former Master Woodman / Timberjack (Unemployed due to monsters)
- **Residence**: Eric's Hearth (Upper Dio Village)
- **Economic Situation**: Deeply impoverished, struggling to afford food as monster activity blocks access to the logging roads.

Chapter 49. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

Eric represents the vulnerable provincial working class bearing the brunt of unnatural cosmic incursions.

- **The Struggle of Provincial Labor**: [quote, Eric, Map 28] "We've been seeing a lot of monsters appear near the woods recently. I wonder what's that all about. And with jobs being tight, money is hard to come by."
- **The Story of the Portal & The Ring of Fire**: [quote, Eric, Map 28] "Thank you. Thank you so much, here I believe that this will be more useful to you than me. The name is Eric and that ring has a story. I used to be a lumberjack until that weird portal appeared. I found this ring and held on to it ever since. I'm sure that ring's fire power will come in handy if you plan on leaving the village."

Chapter 50. 3. The Test of Honesty & Moral Continuum

Eric’s home contains a famous interactive vignette from the 2017–2018 project: the **Self-Aware Pot**.

- If the player attempts to loot Eric’s shelf pot, the pot speaks directly to the player: [quote, Self-Aware Pot, Map 28] "What kind of game do you think this is! You can’t just steal whatever you want! How rude!"
- Taking the 10 Gold increments the internal `Moral (Evil)` counter, while gifting charity to Eric increments `Moral (Good)`. This directly fed the original RPG Maker `Morality Meter` and served as the design prototype for the modern C11 engine’s `moral_sway` continuum (`[-100, +100]`).

Chapter 51. Chapter 10: John the Thief & Outlaw Resistance

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/dio_village/john_the_thief.adoc

Chapter 52. Overview

John the Thief is an underworld operative and smuggler residing in the back alleys of Dio Village's Lower Ward. Cynical and gruff, John nonetheless acts as an unofficial protector of the poor, redistributing goods and offering clandestine sidequests to heroes willing to bypass the law.

Chapter 53. 1. Identity & Underworld Standing

- **Name**: John (**"John the Thief"**)
- **Occupation**: Smuggler, Fence, and Independent Rogue Operative
- **Territory**: Lower Ward Alleys & Shopping District
- **Philosophy**: Pragmatic survivalism; contempt for aristocratic landlords.

Chapter 54. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

John initially meets outsiders with hostility, softening only when heroes demonstrate that they care about the common people.

- **Gruff Rebuff**: [quote, John, Map 31] "What do you want kid! Can’t you see that I’m busy!"
- **Recognizing Civic Virtue**: [quote, John, Map 31] "So you’re the one that’s helping the little people of this village out. I have a little task for you…"

Chapter 55. 3. Quest & Mechanics Role

John the Thief introduces the **Stolen Goods** tactical retrieval mechanics. In the 2017–2018 LeTBS combat engine, recovered bandit contraband appeared as interactive grid items with dedicated `Pick Up` and `Push` commands. John serves as the quest receiver who turns recovered goods into vital supplies and gold.

Chapter 56. Chapter 11: The St. Baron & Feudal Exploitation

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/dio_village/st_baron.adoc

Chapter 57. Overview

The St. Baron is the wealthiest aristocrat and dominant landowner of Dio Village. Residing in a secluded manor in the Lower Ward, the St. Baron uses exorbitant rent, grain hoarding, and monopolies to subjugate the impoverished citizens, serving as a primary symbol of noble corruption outside the capital.

Chapter 58. 1. Identity & Standing

- **Title**: The St. Baron of Dio Province
- **Residence**: The St. Baron's Manor (Lower Dio Village)
- **Status**: High provincial gentry; tax collector for the eastern march
- **Reputation**: Ruthless, miserly, and widely despised by the working poor.

Chapter 59. 2. Socio-Economic Conflict in Dio Lower Ward

While the Upper Ward remains pastoral and self-sufficient, the Lower Ward suffers under the St. Baron’s heavy-handed economic extortion. By controlling grain warehouses and money-lending stalls, the Baron has forced shopkeepers and laborers into debt bondage.

- **The Hero’s Moral Justification (Elijah)**: [quote, Elijah, Lower Village Infiltration] "I can’t believe that I am doing this… the baron is the richest man in the village… but he’s been using the town! This is for the good!"
- **The Rogue’s Determination (Katrina)**: [quote, Katrina, Lower Village Infiltration] "Ok… here we go… let’s do this. I have to do at least one good thing today."

Chapter 60. 3. Narrative & Quest Role

The St. Baron represents the mundane human corruption that flourishes when the Crown neglects its borders. Infiltrating the St. Baron’s estate to recover stolen resources for the village provides the player with an early non-combat tactical scenario, forcing Elijah to break legal statutes for the higher moral good.

Chapter 61. Chapter 12: High Priest Roté & The Royal Omens

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/kingdom_and_court/sir_rote.adoc

Chapter 62. Overview

Sir Roté is a veteran scout, senior knight, and herald of the Sacyran Royal Guard. His urgent, breathless arrival at Sacyra Castle serves as the opening narrative incitement of the game, delivering news of the rising island, the unnatural storm, and the return of the primordial Darkness.

Chapter 63. 1. Identity & Standing

- **Full Name**: Sir Roté of the Royal Guard
- **Rank / Office**: Senior Outrider Captain / First Scout of the Realm
- **Affiliation**: Kingdom of Sacyra Royal Army; The Royal Guard
- **Visual Demeanor**: Weather-worn plate mail spattered with mud from frantic galloping, exhausted, earnest, and deeply devoted to the Crown.

Chapter 64. 2. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

Sir Roté represents the perspective of the loyal rank-and-file soldier who believes unquestioningly in his monarch's ability to save the realm.

- **Observation at the Frontier (Start-Up / Castle Gates)**: [quote, Roté, Map 2 & Map 3 / Map 8] "The king… huff… huff… huff… he must know immediately! And to think… The storm… The rising island… All of these events before the Festival of Heroes. The king will know how to stop this, he must know…"
- **The Urgent Report to King Jihad**: [quote, Roté, Throne Room - Map 4 / Map 9] "MY KING! It's here! The darkness… it has returned!"
- **The Uneasy Hesitation**: [quote, Roté, Hearing King Jihad's Gag Order] "Y… Y… Yes my king…"

Chapter 65. 3. Narrative Role & Dynamic

Sir Roté serves as a tragic mirror to Elijah Ro-enheart. Both men revere the oath of the Royal Guard. However, while Roté suppresses his growing unease when King Jihad issues a gag order preventing the people from knowing about the approaching Darkness, Elijah ultimately chooses conscience over blind obedience. Later in the campaign, Roté's internal conflict plays a pivotal role in enabling Elijah and Katrina to infiltrate the castle archives.

Chapter 66. Chapter 13: King Jihad & The Treasonous Ambition

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/kingdom_and_court/king_jihad.adoc

Chapter 67. Overview

King Jihad (referenced in ancient annals as **D. Jihad**) is the reigning monarch of the **Kingdom of Sacyra**. Outwardly a dignified sovereign presiding over the chivalric traditions and the **Festival of Heroes**, King Jihad secretly harbors a perilous ambition: to seize and harness the forbidden abyssal power of the returning Dark Lord Malakor to establish absolute dominion.

Chapter 68. 1. Identity & Sovereign Office

- **Full Name**: King Jihad of Sacyra (Monarchical Title: **D. Jihad**)
- **Office**: Crown Sovereign, King of Sacyra, High Commander of the Realm
- **Seat of Power**: The High Citadel of Sacyra Castle
- **Political Standing**: Balances the aristocratic Noble Houses against the meritocratic Adventurers' Guild while commanding the elite Royal Guard.

Chapter 69. 2. Psychological Profile & Secret Ambition

King Jihad is calculating, authoritative, and outwardly composed. While his subjects view him as a protector against chaos, he views the resurgence of the ancient Darkness not as a tragedy to be repelled, but as an apex weapon to be claimed. Believing himself superior to both holy creeds and dark lords, he seeks to subvert the cosmic void to overthrow all limits on his crown.

69.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

- **The Secret Soliloquy**: [quote, D. Jihad, High Throne Room] "The fools… that power… that wicked power! Soon I will rule over this land, and not even the Dark Lord will have power over me!"
- **Suppressing the Truth**: [quote, King Jihad to Sir Roté] "Thank you Roté, what would I do without your careful watch… Gather up the guard! We need to prepare! Darkness will soon return to this land. And don’t tell the people of this discovery… You all are dismissed."

Chapter 70. 3. Socio-Political Role in the Sacyra Campaign

1. **State Censorship & Denial**: When the mysterious storm and rising island manifest along the frontier, King Jihad silences border outposts and forbids the public from learning the truth before the Festival of Heroes.
2. **The Disappearance of Commander Gareth**: Historical records suggest Gareth Ro-enheart's sudden departure under sealed royal orders was connected to King Jihad's clandestine investigations into the Abyssal Vaults.
3. **The Catalyst for Rebellion**: When Elijah and Katrina uncover that the Crown is actively pursuing void corruption rather than safeguarding the realm, Elijah is forced into open defiance against the very throne his family swore to protect.

Chapter 71. Chapter 14: Commander Gareth Ro-enheart & The Vanished Guard

Source File: /home/professorseanex/Tenebris and Lux/docs/lore/characters/kingdom_and_court/commander_gareth_ro_enheart.adoc

Chapter 72. Overview

Commander Gareth Ro-enheart was the renowned Head of the Royal Guard, hero of the Sunken Citadel, and father to Elijah Ro-enheart. His unexplained disappearance under sealed royal orders following Elijah's 10th birthday forms the primary personal catalyst driving Elijah's journey across Sacyra.

Chapter 73. 1. Identity & Lineage

- **Full Name**: Commander Gareth Ro-enheart
- **Title**: Former Head of the Sacyran Royal Guard; Knight-Commander of Sacyra
- **Kinship**:
  - **Son**: Elijah Ro-enheart (Cadet Vanguard, bearer of the Shield of Lux)
  - **Mother**: Magolia Ro-enheart ("Grandma Magolia", matriarch of Dio Village)
- **Status**: Missing in Action / Sealed Royal Dispatch (Presumed Alive)

Chapter 74. 2. Legend & Martial Legacy

Gareth Ro-enheart is remembered throughout Sacyra as an incorruptible paragon of chivalric virtue. Decades prior, when an allied barbarian host breached the frontier palisades, Commander Gareth anchored the Sunken Citadel for seven days without relief, personally deflecting siege bolts with his reinforced tower shield.

He passed this philosophy to his young son Elijah:

> "A shield is not a wall to hide behind, but a promise made to the people standing behind you."
>
> — Gareth Ro-enheart

Chapter 75. 3. The 10th Birthday Vanishing & Royal Intrigue

On the morning following Elijah's **10th birthday**, Commander Gareth received a direct royal summons from King Jihad and departed the capital with an elite retinue for the southern frontier. Neither he nor his men ever returned.

- **Official Crown Statement**: Gareth was lost during a classified border survey.
- **The Reality**: Clues uncovered by Elijah and Katrina reveal that Gareth discovered King Jihad's clandestine expeditions to unearth the Abyssal Vaults. Refusing to participate in forbidden dark alchemy, Gareth was either imprisoned in a subterranean redoubt or went into deep exile to protect the realm.

Volume III: Ludology & Tactical TRPG Combat Mechanics

Grid-based tactical warfare, relic stance machines, terrain modifiers, continuous invariant testing, and game design curation.

Chapter 76. Chapter 15: Dual-Layer TRPG Combat Architecture

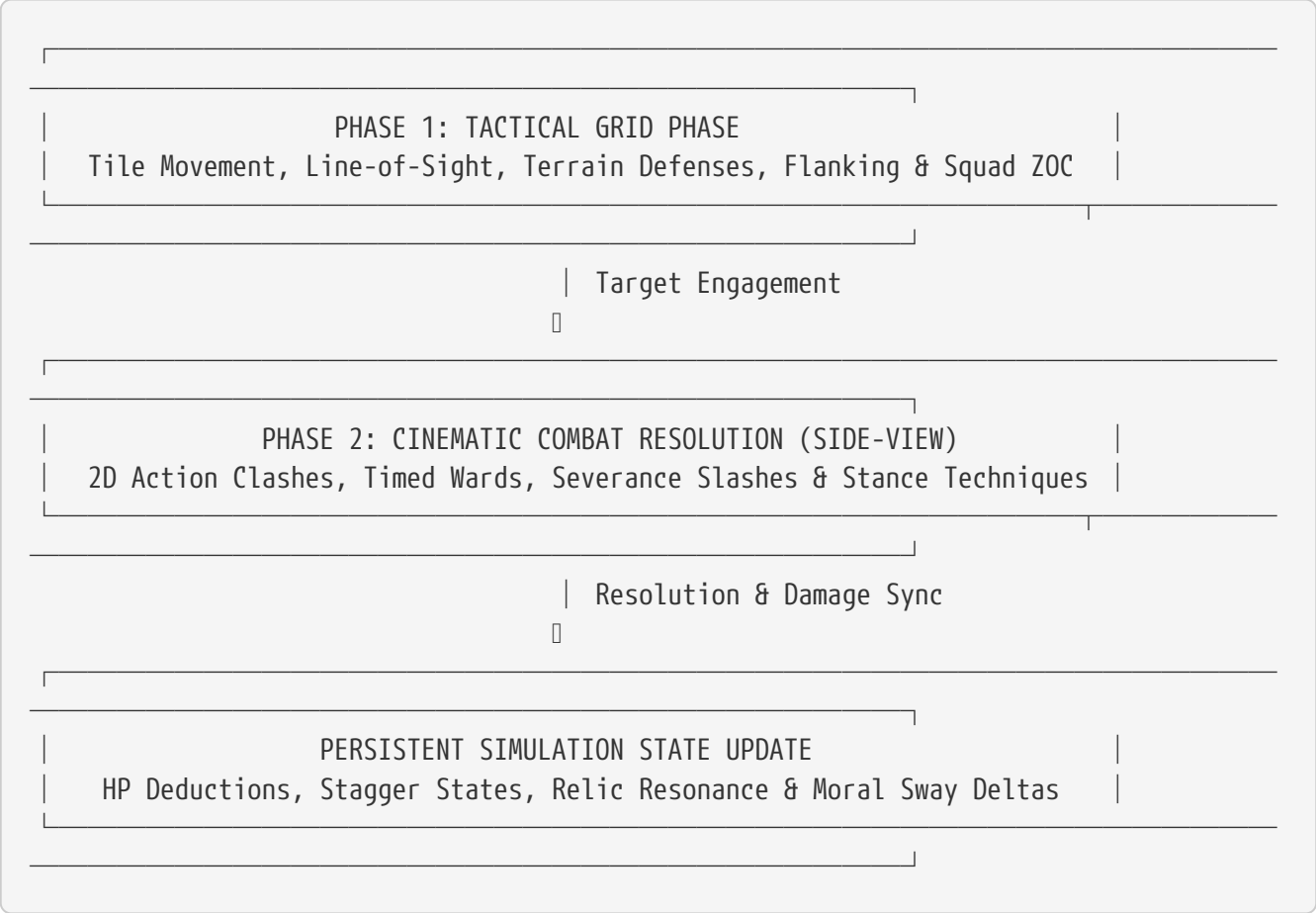
Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/tactical_trpg_system.adoc

Chapter 77. Overview

This document specifies the dual-layer combat architecture of **Tenebris and Lux**: a **Tactical Strategy RPG (TRPG)** featuring a **grid-based tactical battlefield** coupled with **cinematic traditional side-scroll / action-based combat resolution**. Conceived during the 2014–2018 RPG Maker genesis of CreativeWorkzStudio, this design bridges grand tactical positioning with visceral, action-driven 2D engagement.

Chapter 78. 1. Dual-Layer Battle Architecture

The combat engine operates across two seamless, interconnected operational loops:



Chapter 79. 2. Phase 1: Grid-Based Tactical Battlefield

The overarching strategic layer takes place upon a discrete 2D grid map representing the castles, frontier passes, and battlegrounds of the **Kingdom of Sacyra**.

79.1. 1. Grid Topography & Spatial Invariants

- **Coordinate Space**: Discrete integer grid `(grid_x, grid_y)`.
- **Unit Footprint**: Standard units occupy a 1×1 grid tile; major boss constructs (such as Malakor's Siege Automata) occupy 2×2 or 3×3 footprints.
- **Terrain Types & Modifiers**:
  - **Stone Flagstones / Road**: $1.0\times$ movement cost, $+0\%$ evasion.
  - **Ramparts / Elevated Battlement**: $1.5\times$ movement cost, $+20\%$ physical defense, $+1$ attack range for ranged skills.
  - **Frontier Thicket / Rubble**: $2.0\times$ movement cost, $+15\%$ evasion, blocks cavalry charge.
  - **Void Mire (Malakor's Blight)**: $2.5\times$ movement cost, inflicts -5 HP decay per turn on non-void units.

79.2. 2. Movement & Zone of Control (ZOC)

- **Movement Budget**: Units possess a base `movement_tiles` statistic (e.g. Katrina: 6 tiles; Elijah: 4 tiles in heavy armor).
- **Zone of Control**: Moving into an adjacent tile directly forward of a hostile soldier halts further movement for that turn, anchoring frontline formations.
- **Directional Facing**: Units finish their turn facing one of four cardinal directions (`NORTH`, `SOUTH`, `EAST`, `WEST`). Attacking an enemy from their flank grants $+15\%$ critical chance; attacking from the rear guarantees an unguardable strike.

79.3. 3. Turn Sequencing

- **Dynamic Initiative Order**: Turn execution is governed by unit Agility. Fast skirmishers (like Katrina) act earlier in the round, while heavy vanguard armorers act later with higher defensive resilience.

Chapter 80. 3. Phase 2: Cinematic Combat Resolution (Side-View Action)

When an attacker initiates an assault or when two opposing units clash on the grid, the game seamlessly transitions into a dedicated **2D Side-View Cinematic Battle Scene**.

80.1. 1. Visual & Staging Presentation

- The attacker approaches from the left (player units) or right (enemy units) across a dynamic 2D battlefield backdrop matching the current terrain (e.g., Sunken Citadel battlements, ash-strewn frontier).
- Character sprites render with expressive combat animations: Idle combat stance, forward dash, heavy strike, shield block, recoil, stagger, and victory flourishes.

80.2. 2. Combat Mechanics During Cinematic Resolution

1. **Active Input & Timed Agency**:
    - Rather than passive numbers scrolling past, combat resolution incorporates timed player inputs:
    - **Timed Ward (Lux Guard)**: Pressing the ward button precisely upon enemy weapon contact triggers a **Perfect Deflection**, absorbing 100% of incoming damage and staggering the foe.
    - **Twilight Severance (Tenebris Strike)**: Pressing strike during the forward dash activates a **Deep Cleave**, bypassing a percentage of enemy armor.
2. **Stagger & Poise Meter**:
    - Every battler possesses a **Poise / Stagger Gauge** ($0 \dots 100$).
    - Repeated heavy strikes or shield bashes deplete enemy poise. When poise hits zero, the target enters **Staggered Vulnerability**, leaving them defenseless against critical finishers.
3. **Stance Animation Morphing**:
    - The character's active stance directly dictates their cinematic combat posture and move set:
    - `STANCE_WARDING`: Elijah locks the Shield of Lux into the earth; golden starlight forms a kinetic buckler.
    - `STANCE_STRIKING`: Katrina or Elijah lowers their center of gravity, dragging the Sword of Tenebris as twilight embers trail the blade.
    - `STANCE_EQUILIBRIUM`: Fluid parry-and-counter sequencing utilizing both armaments in rapid harmony.

Chapter 81. 4. Dual-Protagonist Synergy: Elijah & Katrina

The tactical depth of **Tenebris and Lux** emerges from combining the distinct capabilities of Elijah and Katrina:

81.1. Elijah Ro-enheart (Frontline Guardian)

Feature	Tactical Grid & Cinematic Specification
Unit Class	Royal Vanguard / Aegis Knight
Grid Role	Anchors the defensive perimeter; draws enemy agro; intercepts projectiles.
Special Trait: Royal Intercept	If an adjacent ally (within 1 tile) is targeted by an attack, Elijah leaps in front to receive the blow, initiating the cinematic defense in their stead.
Relic Mastery	**Shield of Lux**: Projects a 2-tile radius **Sanctuary Aura** on the grid that grants allies $+25\%$ physical defense and immunity to void blight.
Cinematic Moves	**Aegis Slam**, **Radiant Shockwave**, **Bastion Guard**, **Solar Flare Riposte**.

81.2. Katrina Jaren (Tactical Skirmisher)

Feature	Tactical Grid & Cinematic Specification
Unit Class	Guild Outrider / Twilight Blade
Grid Role	Flanks heavy units; assassinates backline spellcasters; disarms siege engines.
Special Trait: Guild Acrobatics	Can move through tiles occupied by enemy infantry without triggering Zone-of-Control lock.
Relic Mastery	**Sword of Tenebris**: Can activate **Shadow Walk**, cloaking in mist on the grid to become untargetable by ranged attacks until her next strike.
Cinematic Moves	**Abyssal Dash**, **Crescent Sever**, **Shadow Dance**, **Mist-Veil Feint**.

81.3. Tandem Linked Combat (The Duo Duel)

When Elijah and Katrina are positioned on adjacent grid tiles and either initiates combat, both heroes enter the cinematic side-view battle scene together:

- **Shield & Sword Flank**: Elijah executes an **Aegis Slam** that knocks the enemy backward, launching them into Katrina's airborne **Crescent Sever**.
- **Harmonized Equilibrium**: Both heroes combine the radiance of Lux and the edge of Tenebris, unleashing a dual finishing art that sweeps across multiple targets.

Chapter 82. 5. Persistent State & Moral Sway Feedback

Upon completion of the cinematic resolution:

1. **HP & Wound Conservation**: Remaining unit health updates onto the tactical grid. Fallen units are removed or enter down-but-not-dead states.
2. **Relic Resonance Tracking**: Energy expended (`shield_power`, `sword_edge`) updates in persistent memory (`player_state_t`).
3. **Moral Sway Evaluation**:
    - Choosing to spare routed enemy captains, warding retreating militia, or using non-lethal shield stagger tilts sway toward **Virtue (+)**.
    - Executing fleeing enemies with abyssal severance or utilizing forbidden void arts tilts sway toward **Malice (-)**.
    - Moral sway unlocks distinct strategic faction options across the Kingdom of Sacyra.

Chapter 83. Chapter 16: The Twin Relics, Combat Stances & Moral Sway

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/relics_and_combat.adoc

Chapter 84. Overview

This document specifies the core ludological mechanics of **Tenebris and Lux** in the **Kingdom of Sacyra**: the twin primordial relics (**The Shield of Lux** and **The Sword of Tenebris**), the dynamic combat stance state machine, the moral sway continuum (`[-100, +100]`), and their integration into the Tactical Strategy RPG system featuring grid maneuvering and cinematic side-view combat resolution.

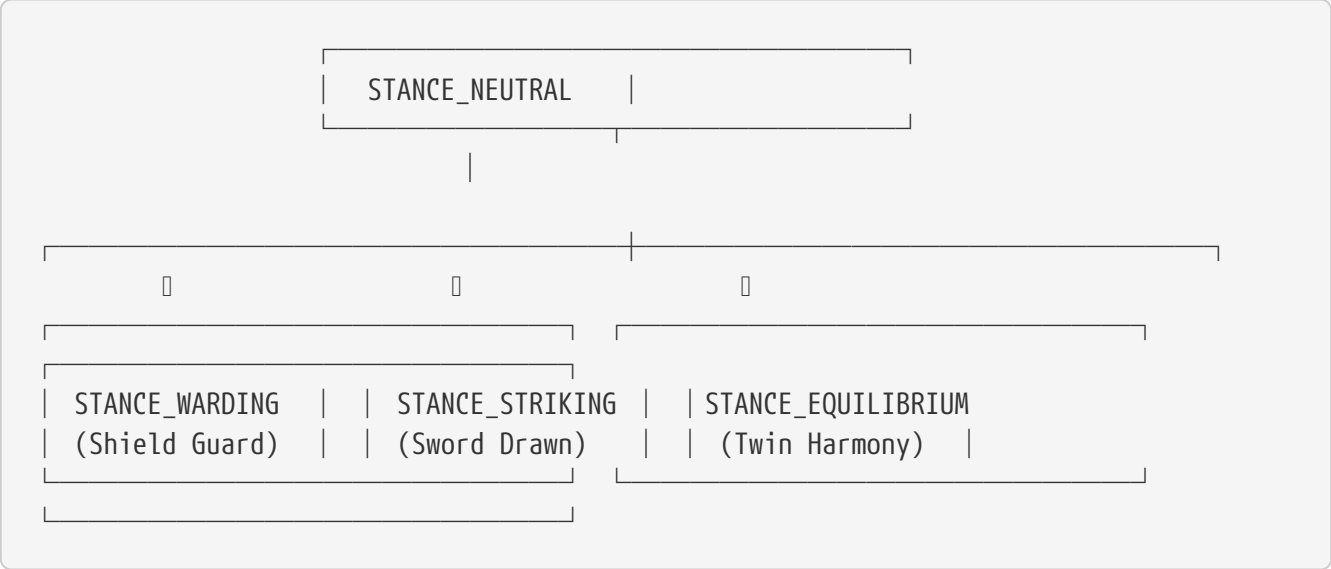
Chapter 85. 1. Primordial Neutrality & Relic Mechanics

Neither relic is inherently good nor evil. They are manifestations of cosmic apex forces whose morality is dictated solely by the intentions and choices of their wielders—**Elijah Ro-enheart** and **Katrina Jaren**:

Relic	The Shield of Lux	The Sword of Tenebris
Aspect	Apex Radiance & Cohesion	Apex Entropy & Dissolution
Energy Gauge	`shield_power` ($0 \dots 100$)	`sword_edge` ($0 \dots 100$)
Primary Bearer	**Elijah Ro-enheart** (Vanguard Defense)	**Katrina Jaren** (Agile Severance)
Grid Role	Zone-of-Control denial, sanctuary aura, ally protection	High-mobility flanking, armor penetration, mist evasion
Cinematic Action	Timed kinetic blocks, shield-bashes, radiant shockwaves	Rapid dash strikes, armor severance, twilight blade waves
Resource Dynamics	Regenerates $+20$ on ward; absorbs incoming assault	Consumes edge on strikes; hones $+15$ outside combat

Chapter 86. 2. The Combat Stance State Machine

The battle posture shifts dynamically across four discrete combat stances:



1. `STANCE_NEUTRAL`: Relics held at waist. Balanced baseline.
    - Standard movement on the tactical grid; baseline attack parameters.
2. `STANCE_WARDING`: Shield raised in protective vanguard high-guard.
    - Grid Effect: Increases allied defense within adjacent tiles by $+25\%$.
    - Cinematic Duel: Boosts Shield power recovery ($+15$), increases stagger damage against attacking foes (25 HP), and wards against incoming status curses.
3. `STANCE_STRIKING`: Greatsword held in aggressive two-handed assault posture.
    - Grid Effect: Increases unit movement by $+1$ tile and grants armor-piercing flank bonuses.
    - Cinematic Duel: Increases strike severance from 30 to 45 damage; reduces edge cost from 12 to 8 points.
4. `STANCE_EQUILIBRIUM`: Harmonized twin-relic synchronization.
    - Grid Effect: Immunity to positioning debuffs and void mire terrain penalties.
    - Cinematic Duel: Simultaneously restores $+15$ Shield power and $+15$ Sword edge; parries incoming strikes while counter-attacking for 20 damage; nudges moral sway toward center (0).

Chapter 87. 3. The Moral Sway Continuum

The moral trajectory of Sacyra's champions is tracked along a continuous bounded integer axis:

```
[ -100: Sovereign of Ruin ] ◀──────── [ 0: Equilibrium ] ────────▶ [ +100: Saint of Mercy ]
```

- **Virtue & Mercy ($+1 \dots +100$)**:
  - Gained by shielding fleeing militia, sparing defeated enemy officers, resolving noble-court corruption through justice, and protecting the innocent with the Shield of Lux.
  - Unlocks royal alliances, Knight Commander endorsements, and sanctuary miracles.
- **Tyranny & Ruin ($-1 \dots -100$)**:
  - Accumulated through ruthless executions, collateral damage in villages, using scorched-earth radiance to blind subjects, or giving in to the void's bloodlust.
  - Unlocks forbidden void arts, mercenary enlistments, and intimidation tactics.
- **Equilibrium (0)**:
  - Maintained by balancing the blade and the aegis; recognizing that light without shadow is a calcified wasteland, and shadow without light is an endless void.
  - Unlocks the ultimate secret synthesis arts of ancient Sacyra.

Chapter 88. Chapter 17: World Topography, Provinces & Battlefield Terrain

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/world_topography.adoc

Chapter 89. Overview

This document specifies the spatial geography, province layouts, environmental encounters, and tactical grid battlefields governing the **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**) as Elijah Ro-enheart and Katrina Jaren defend the realm from Dark Lord Malakor.

Chapter 90. 1. Spatial Structure: The Provinces of Sacyra

The Tactical Strategy RPG campaign progresses across the major defensive tiers of the Kingdom:



90.1. 1. The Garrison Altar (LOC_GARRISON_ALTAR)

- **Role**: The ancestral redoubt of Sacyra's Royal Guard and the starting staging ground.
- **Environment**: High-vaulted halls of carved silver-white granite and polished obsidian pedestals where Elijah's father once held council.
- **Significance**: Here, Elijah Ro-enheart and Katrina Jaren first awaken the dormant Shield of Lux and Sword of Tenebris as reports of Malakor's vanguard arrive.
- **Grid Battlefield**: Smooth stone flags, defensive pillars providing $+15\%$ cover, training dummies, weapon racks.
- **Threat Vigor**: 0 (Sanctified safe zone; tactical deployment and stance tutorial).
- **Connections**: North $\to$ Sunken Citadel Ramparts.

90.2. 2. Sunken Citadel Ramparts (LOC_CITADEL_GATES)

- **Role**: The primary fortified bastion guarding the central valleys of Sacyra.
- **Environment**: Colossal stone ramparts scorched by void-fire; fractured portcullises and smoking ballista batteries.
- **Encounter**: **Malakor's Void Vanguard Captain** (50 Threat HP) commanding shock troops

and blighted footmen.

- **Tactical Grid Elements**:
  - Narrow stone choke-points where Elijah's **Royal Intercept** and Shield of Lux can hold off numerical superiority.
  - Rampart parapets offering Katrina elevated flanking routes and line-of-sight plunging strikes.
- **Connections**: South $\to$ Garrison Altar; North $\to$ Fractured Marches.

90.3. 3. The Fractured Marches (LOC_DIVIDED_MARCHES)

- **Role**: The windswept highlands along the Kingdom's frontier, home to forward outposts of the **Adventurers' Guild** and noble house estates.
- **Environment**: Rolling rocky hills, abandoned watchtowers, and misty ravines where royal army patrols and guild outriders clash with vanguard skirmishers.
- **Encounter**: **Corrupted Golem Vanguard & Shadow Stalkers** (70 Threat HP).
- **Tactical Grid Elements**:
  - Uneven terrain, forest thickets granting $+20\%$ evasion, and muddy ditches that slow heavy armor.
  - Katrina's knowledge of local guild ranger paths allows the party to surprise enemy artillery from the rear.
- **Connections**: South $\to$ Sunken Citadel Ramparts; West $\to$ Abyssal Threshold.

90.4. 4. The Abyssal Threshold (LOC_DARK_LORD_THRESHOLD)

- **Role**: The dark frontier where the fabric of reality bends into the subterranean vaults of Dark Lord Malakor.
- **Environment**: Jagged obsidian monoliths, boiling void-mist chasms, and anti-radiant storms.
- **Encounter**: **Dread Lord Vanguard Commander** ($90 \dots 100$ Threat HP).
- **Tactical Grid Elements**:
  - Void Mire tiles causing -5 HP per turn, demanding strict stance management and the Shield of Lux's purifying aura to cross safely.
  - Multi-tile boss presence demanding coordinated tandem strikes between Elijah and Katrina to shatter dread shields.

Chapter 91. Chapter 18: Continuous Invariant Testing Philosophy

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/game_testing.adoc

Chapter 92. Abstract

In classical software engineering, testing focuses primarily on stateless input/output validation, API schemas, and unit function assertions (`assert(f(x) == y)`). In game development, testing requires an entirely distinct paradigm: verifying continuous state continuity, economy balance, softlock immunity, deterministic replay parity, and strict 60 FPS frame-budget headroom. This document outlines the testing philosophy governing **Tenebris and Lux**.

Chapter 93. 1. The Fundamental Divergence

Dimension	Traditional Software Testing	Game Development Testing
State Nature	Ephemeral, localized, often stateless	Continuous, cumulative, multi-turn state machine
Failure Modes	Null pointer exceptions, HTTP 500s, syntax errors	Softlocks, economy collapse, degenerate gameplay loops, visual desync
Correctness Metric	Does output match expected schema?	Does the game feel right? Is the state bounded, balanced, and unexploitable?
Input Cadence	Controlled, structured API payloads	Unbounded, chaotic human keystrokes, rapid button mash, simultaneous events
Time Budget	Milliseconds/seconds per request	16.6 milliseconds total frame budget (60 FPS) with microseconds for simulation

Chapter 94. 2. Core Pillars of Game Development Testing

94.1. Pillar 1: Dynamic State Invariants & Bounded Continuity

In a game like **Tenebris and Lux**, state variables (`shield_power`, `sword_edge`, `moral_sway`, `threat_hp`) are constantly mutated across hundreds of player turns. A single off-by-one error or uncapped addition will cause state drift, resulting in overflow, negative health zombies, or invulnerability exploits.

- **Invariant Axioms**:
  - `0 ⇐ state.shield_power ⇐ 100` must hold after every single action turn.
  - `0 ⇐ state.sword_edge ⇐ 100` must hold even when spamming strikes at zero energy.
  - `-100 ⇐ state.moral_sway ⇐ +100` must never overflow into undefined integers.
- **Diagnostic Tripwires**: The game engine provides `game_audit_state()`, a self-healing diagnostic routine that asserts all invariants simultaneously and outputs a detailed diagnostic manifest without crashing the game process.

94.2. Pillar 2: Deterministic Replay ("Golden Master Replay")

Games are emergent systems. If two players—or an automated CI runner—replay the exact same sequence of commands from the same seed, the end state must be **100% bitwise identical**:

```
// Golden Master Replay Assertion
TEST_ASSERT(run1.shield_power == run2.shield_power, "Deterministic shield");
TEST_ASSERT(run1.sword_edge   == run2.sword_edge,   "Deterministic sword");
TEST_ASSERT(run1.moral_sway   == run2.moral_sway,   "Deterministic sway");
TEST_ASSERT(run1.threat_hp    == run2.threat_hp,    "Deterministic threat");
```

This ensures that gameplay balance, combat resolution, and narrative branching do not drift across compiler optimizations, operating systems, or refactored engine code.

94.3. Pillar 3: Softlock Immunity & Degenerate Play Detection

A **softlock** occurs when a player is technically still playing, but has no valid action forward: - Running out of relic energy during an un-fleeable battle. - Getting trapped behind a one-way gate with no exit command. - Inputting malformed or rapid chaotic strings that cause the parser to lock up.

Our automated stress suite executes **1,000 rapid chaotic commands** (combining invalid strings, rapid stance transitions, and empty tokens), verifying that the game engine rejects nonsense safely, maintains continuous memory integrity, and never enters a deadlocked state.

94.4. Pillar 4: Economy & Combat Balance Invariance

Game testing must verify that no single player action is universally dominant (the "dominant strategy" flaw). If a player can simply spam `ward` forever without consequence, or if `strike` renders all other actions useless, the game design collapses: - Warding increases shield power and moral virtue, but deals limited stagger damage. - Striking inflicts decisive severance damage, but consumes finite sword edge. - Balancing recovers both relics and parries attacks, but requires timing and resets extreme moral sway.

94.5. Pillar 5: Frame-Budget Latency Headroom

In standard software, a backend database query taking 50 ms is acceptable. In game development, a game running at 60 FPS has **only 16.6 milliseconds** to: 1. Poll hardware events and window queues. 2. Execute game simulation logic. 3. Compute spatial audio and particle animations. 4. Render the scene graph to the backbuffer and blit to the display.

Simulation logic must therefore execute in **microseconds**. In our automated benchmark (`test_gameplay_scenarios`), 10,000 action turns execute in **160 ms** (**16.02 μs per turn**), consuming less than **0.1%** of a 60 FPS frame.

Chapter 95. Chapter 19: Game Design Curation & Triaging Matrix

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/game_design_curation.adoc

Chapter 96. Overview

This document formalizes the architectural triage of the 2017–2018 RPG Maker MV demo datasets. It separates generic RPG Maker Runtime Package (RTP) boilerplate from temporary testing experiments and identifies the authentic, fleshed-out canonical systems to be reconstructed in the native C11 game engine.

Chapter 97. 1. Triage Categorization Taxonomy

To prevent technical debt from legacy game engine defaults from contaminating our custom C11 engine, all extracted game data is triaged into three strict analytical categories:

Category	Definition	Architectural Policy
Tier 1: RTP Defaults & Boilerplate	Stock RPG Maker MV assets, generic items, placeholder sounds, and default formulas.	**Discard / Re-architect**. Never port verbatim into C11.
Tier 2: Prototype & Testing Sketches	Ad-hoc test maps, sandbox battlers, debug switches, and experimental scripts.	**Review & Evolve**. Extract mechanical concepts but redesign within clean strata.
Tier 3: Fleshed-Out Canonical Pillars	Core story scripts, dual-protagonist dialogue dynamics, moral dilemma quests, LeTBS grid tactics, and key relics.	**Preserve & Reconstruct**. Faithfully implement as axiomatic invariants in C11.

Chapter 98. 2. Tier 1: RPG Maker RTP Defaults (To Discard or Re-Architect)

Element	Historical Manifest	Why It Must Be Discarded / Replaced
Basic Consumables	Potion (50G), Magic Water (100G), Dispel Herb (20G), Stimulant (50G)	Generic RTP recovery items. In C11, consumables must tie directly to Sacyran herbalism, relic alchemy, and stance restoration.
Generic Armaments	Standard Sword (100G), Axe (100G), Cane (100G), Bow (100G), Shield (100G), Cloth (100G)	Default placeholder inventory without distinct tactical affinities or stance interactions.
Stock Sound Effects	`Attack2`, `Buzzer1`, `Damage4`, `Collapse1-4`, `Shop1`	Generic MV sound library. C11 audio subsystem will support custom soundscapes.
Placeholder Bestiary	Bat (`HP: 80`), Slime (`HP: 250`), Rabbit (`HP: 55`), Orc (`HP: 300`)	Standard MV demo monsters used for basic stat checks. Replaced by Sacyran fauna and Malakor void aberrations.
Default Title Theme	`Theme6` / `CrossedSwords` title screen graphic	Default MV asset packaging. Replaced by bespoke Sacyran coat of arms and twin relic artwork.

Chapter 99. 3. Tier 2: Prototype & Testing Sketches (Concepts to Evolve)

Element	Historical Location	Tactical / Design Value to Evolve
Sandbox Test Maps	`Map006 Test Areas`, `Map043 Event Map`	Used to test event triggers, movement paths, and sprite coordinates. Demonstrates intent for environmental puzzle mechanics.
Debug Enemies & Troops	`Enemy 14 Test`, `Troop 10 Test`, `Enemy 32 Invincible Neutral`	Proved that neutral, invincible obstacles and destructible props function well on the tactical grid.
Explosive Chickens	Fire Chicken (`Actor 13`), Bomb (`Skill 136`)	Humorous prototype mechanic demonstrating combat summons and dynamic decoy units on the tactical field.
Plugin Experiments	`LeTBS/firestorm.json`, `LeTBS/test.json`, `QMaps/`	Initial experiments with Lecode's Tactical Battle System and Quasi Movement plugins. Proves grid-based TRPG was the true target genre.

Chapter 100. 4. Tier 3: Fleshed-Out Canonical Pillars (Core Invariants to Reconstruct)

The following systems represent the authentic creative vision of **Tenebris and Lux** and must be preserved and reconstructed in C11:

100.1. 1. Dual-Protagonist Dynamics & Narrative Branching

- **Elijah Ro-enheart (`\n[1]`)**: Son of vanished Commander Gareth Ro-enheart. Dutiful, protective vanguard bearer of the Shield of Lux. Dialogue reflects reverence for duty, family loyalty, and hesitation to violate laws.
- **Lady Katrina Jaren (`\n[2]`)**: Daughter of noble House Jaren escaping courtly politics to join the Adventurers' Guild. Agile outrider wielder of the Sword of Tenebris. Dialogue features sardonic wit, impatience with provincial stagnation, and fierce defense of the vulnerable.

100.2. 2. The Moral Sway Continuum Engine

- **Ethical Mechanics**: Discrete moral choices tracking across `Moral (Good)` and `Moral (Evil)` within the invariant continuum `[-100, +100]`.
- **The Test of Honesty (Eric's House, Map 28)**:
  - Stealing Eric's life savings (10G) rewards quick cash but plunges morality.
  - Donating 500G to cover Eric's rent saves a desperate worker and earns the **Ring of Fire**.
- **The Struggle of Dio Village (John the Thief vs. St. Baron, Map 30/31)**:
  - Feudal exploitation vs. outrider rebellion, forcing the heroes to choose between legal compliance and moral justice.

100.3. 3. Lecode Tactical Battle System (LeTBS) Mechanics

- **Grid Movement & Range Geometries**:
  - Movement Points (4 MP baseline), modified by skills like **Light Steps** (+1) and **Heavy Hit** (-3).
  - Scope shapes: `circle(r)`, `circle(max, min)` (annular), `line(n)`, `square(r)`.
- **Terrain Hazards & Obstacles**:
  - **Ice Wall** and **Ice Prison** creating physical destructible barriers (`HP: 20`).
  - **Volcano Ground** and **Burning Tile Spread** (`spread: 100% circle(1)`).
- **Turn Timeline Manipulation**:
  - **Accelerate**, **Decelerate**, and **Emergency Turn** altering the turn queue.

100.4. 4. The Monotonic 24-Hour Time Engine

- **Grandma's Watch**: Family heirloom activating real-time calendar and clock tracking (Seconds, Minutes, Hours, Days, Weeks, Months, Years), corresponding directly to Layer 0 `time_utils`.

Volume IV: The 2017–2018 Demo Baseline & Historical Archive

Verified historical baseline extracted from the original RPG Maker MV datasets, detailing the complete scenario script, LeTBS mechanics, grimoire, and systems.

Chapter 101. Chapter 20: Historical Baseline Overview & Provenance

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/README.adoc

Chapter 102. Overview

This documentation module establishes the comprehensive, verified baseline of the original **2017–2018 RPG Maker MV demo** for **Tenebris and Lux** ("Darkness and Light"). Extracted directly from the archival data directories (`Old Game Data` and `Old_Game_Data_2/data`), this repository anchors our modern C11 reimplementation to its authentic gameplay mechanics, narrative scripts, character arcs, and tactical systems.

Chapter 103. 1. Historical Provenance & Dataset Strata

Between 2014 and 2018, **Tenebris and Lux** was conceived as an ambitious Tactical Strategy RPG (TRPG) combining top-down exploration, puzzle mechanics, and grid-based turn strategy powered by Lecode's Tactical Battle System (LeTBS).

The historical game data is preserved across two distinct repository strata:

Dataset	Directory Anchor	Architectural Role & Characteristics
Folder 1: Primary Narrative & Morality Build	`Old Game Data/`	Contains 37 maps, complete dual-protagonist dialogue branches, the morality engine (Compass of Morality, moral sway continuum, honesty tests), and the core Dio Village quest line (`versionId: 40730002`).
Folder 2: Advanced LeTBS Tactical Build	`Old_Game_Data_2/data/`	Contains 43 maps, extended party members (Sebastien the Thief, Angela the Mage, Lucius the Priest), advanced LeTBS action parameters, full tactical state routines, and character generator assets (`versionId: 2078310`).

Chapter 104. 2. Documentation Taxonomy

This baseline compendium is partitioned into specialized manuals covering every facet of the historical demo:

Document	Focus & Coverage
scenario_and_walkthrough.adoc	Chronological narrative walkthrough, scene-by-scene script, branching dialogue for Elijah vs Katrina, Map 001 through Map 043, puzzle mechanics, and transfer topologies.
cast_and_characters.adoc	Comprehensive character dossiers for protagonists (Elijah & Katrina), extended party members (Sebastien, Angela, Lucius), village elders (Grandma Magolia, Monica, Eric), court figures (King Jihad, Roté), and summons.
combat_and_letbs_mechanics.adoc	Grid-based battle architecture, movement point costs, directional facing, line-of-sight scopes, AOE shapes (`circle`, `line`, `square`), LeTBS action sequences, and tactical states.
classes_and_growth.adoc	7 combat classes (Warrior/Knight, Mage, Priest, Summoner, Hero, Thief, Guard) plus Summons; complete stat growth matrices (Lv 1–99) and level-by-level skill learnings.
grimoire_and_skills.adoc	Complete grimoire of all 75+ historical spells, martial techniques, movement abilities, MP/TP costs, damage formulas, and LeTBS sequence tags.
bestiary_and_troops.adoc	Bestiary of all 22+ enemies and neutral entities (Dream Eaters, Magos, Bandits, Chests, Ice Blocks), stat parameters, AI behavior routines, and troop encounter formations.
armory_and_inventory.adoc	Weapons (Medusa's Sword), Armors (Heart Shield, Ring of Fire), Items (Potions, Magic Water), and Key Lore Relics (Grandma's Watch, Compass of Morality, Book of Monsters, Book of Items).
systems_switches_variables.adoc	Global game engines: Monotonic Time System / Day-Night Cycle, Morality System, and an exhaustive directory of all 100+ switches and variables.
dataset_evolution_delta.adoc	Side-by-side comparative delta analysis between `Old Game Data` and `Old_Game_Data_2/data`, detailing version discrepancies and modern C11 reconciliation strategies.

Chapter 105. 3. Architectural Extraction Script

The entire documentation suite is fully reproducible from source data using our automated Python extraction pipeline:

```
python3 docs/tools/extract_demo_data.py
```

This ensures absolute fidelity to the original 2017–2018 records as we rebuild the game in native C11.

Chapter 106. Chapter 21: 7-Act Narrative Script & Dialogue Walkthrough

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/scenario_and_walkthrough.adoc

Chapter 107. Overview

This document preserves the complete chronological story script, scene transitions, puzzle structures, and verbatim dialogues of the 2017–2018 demo. It documents both the shared story progression and the distinct dual-protagonist narrative branches between **Elijah Ro-enheart** (`\n[1]`) and **Lady Katrina Jaren** (`\n[2]`).

Chapter 108. Act I: The Castle Prophecy & Dark Ambition

108.1. Start-Up / Walk of Fear

Roté's urgent arrival as dark omens rise.

Associated Maps: Map 002

108.1.1. Map 002: Start-Up

- **Speaker/Source: `The Walk of Fear `**

```
\n<Roté>It has been years since I was needed here.<br>The king...<br>The king must
know of these strange events!
```

108.2. The Castle (Outside)

Roté observes the rising island and eerie storm prior to the Festival.

Associated Maps: Map 003, Map 008

108.2.1. Map 003: The Castle (Outside)

- **Speaker/Source:** `Roté Observation`

```
\n<Roté>The storm... The rising island...
All of these events...
All before the Festival of Heroes.
The king will know how to stop this, he must know...
```

108.2.2. Map 008: Battle Tutorial

- **Speaker/Source:** `Dream Eaters`

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
```

```
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
```

```
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

108.3. The Castle (Inside) - The Throne Room

Roté warns King Jihad; King Jihad's treasonous conspiracy against the Dark Lord.

Associated Maps: Map 004, Map 009

108.3.1. Map 004: The Castle (Inside)

- **Speaker/Source:** `The King`

```
\n<D.Jihad>The fools...that power...that wicked power!
Soon I will rule over this land, and not even the Dark Lord
will have power over me!
```

- **Speaker/Source:** `The Meeting`

```
\n<Roté>King Jihad, I am here to inform you today...
It's here...
The darkness... it has returned
\n<Jihad>...Thank you Roté, what would I do without your
careful watch...
\n<Jihad>Gather up the guard! We need to prepare!
Darkness will soon return to this land. And don't tell the people of
this discovery...
\n<Roté>Y...Y...Yes my king...
\n<Jihad>You all are dismissed.
Yes my king.
```

108.3.2. Map 009: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
```

```
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

Chapter 109. Act II: The Dream Realm & The Ancient Archives

109.1. Intro / The Void

The cosmic darkness and threshold.

Associated Maps: Map 001, Map 002

109.1.1. Map 002: Start-Up

- **Speaker/Source: `The Walk of Fear `**

```
\n<Roté>It has been years since I was needed here.<br>The king...<br>The king must
know of these strange events!
```

109.2. Character Select & The Room of Sages

Meeting Chrono; selecting archetype crystals; reviewing ancient archives.

Associated Maps: Map 006, Map 011

109.2.1. Map 006: Character Select

- **Speaker/Source:** `Chrono`

```
\n<???>So...a new hero is to be chosen.
I have seen this when the sages spoke of the darkness returning.
But...who did the sage's say again?
\n<???>A young man. The sages spoke
of a young man that would help us. Now, if
I just look in the archives...
\n<???>His name, his name, what was his
name?
\n<???>Ah yes! His name was [Elijah]. The
sages said that they will decide their
fate, what will they become I wonder...
Click on the objects in the room to select a class.
When you pick a class, talk to the old man once again.
\n<???>A young woman. The sages
spoke of a young woman that would help us.
Now, if I just look in the archives...
\n<???>Her name, her name, what was her
name?
\n<???>Ah yes! Her name was [Katrina]. The
sages said that they will decide their
```

```
fate, what will they become I wonder...
Click on the objects in the room to select a class.
When you pick a class, talk to the old man once again.
```

- **Speaker/Source:** Chrono

```
\n<???>[Elijah], did you pick your fate?
\n<???>[Katrina], did you pick your fate?
```

- **Speaker/Source:** Chrono

```
\n<???>That is perfect! Now, we need to
prepare...[Elijah], follow me.
\n<???>That is perfect! Now, we need to
prepare...[Katrina], follow me.
```

- **Speaker/Source:** Warrior□s Stone

```
The crystal began to glow
The Warrior. Tactical and strong, the warrior is known
to lead troops into victory. With various command
formation and tactics, they can turn the tide of a
battle!
Do you wish to be a Warrior?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Warrior□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Mage□s Stone

```
The crystal began to glow
The Mage. Wise and creative, they rely on the powers
of the magical arts to handle problems and to aid
others Wielding a staff and wearing cloaks and robes,
they assist in battle with magic of various elements
and keep the battle flowing.
Do you wish to be a Mage?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Mage□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal began to glow
The Hero. The one who's fate is to save the world from
darkness. The hero fights with valor and truth, using
various skills and magic to restore balance to the
world!
Do you wish to be the Hero?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Thief□s Stone

```
The crystal began to glow
The Thief. Sly and lurking in the shadows, the thief
takes what they want in order to survive. They are
great at setting up traps and gathering information
that everyone can't find on their own.
Do you wish to be a Thief?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Thief□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Priest□s Stone

```
The crystal began to glow
The Priest. With the power of the church and holy
intervention, the priest using holy magic to purify
lands, obejcts, and even people. They also use their
magic to heal and protect others.
Do you wish to be a Priest?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Priest□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal began to glow
The Summoner. A being of many mystery. With the great
ablilty to summon beings from other worlds, the
Summoner does his best to maintain a balance between
the worlds and often knows that world's culture
Do you wish to be the Hero?
Speak to the old man once more.
```

```
Take your time. This is an important choice.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

Chapter 110. Act III: The Trial of Mechanisms & The Dream Eaters

110.1. The Switches Puzzle Chamber

Boulders, pressure plates, jumping ledges with 'Z', and knowledge gates.

Associated Maps: Map 007, Map 040

110.1.1. Map 007: Switches

- **Speaker/Source:** `Special Room Switch`

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** `Bonus Gold`

```
100\G were found!
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Stone`

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Come, we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
```

```
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** Aura


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** Aura


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** Aura

```
By clicking Z, you can jump across certain ledges.
```

110.1.2. Map 040: Switches

- **Speaker/Source:** Special Room Switch

```
Press the button?
```

```
The strange stones to the big room was removed!
```

- **Speaker/Source:** Bonus Gold

```
100\G were found!
```

- **Speaker/Source:** Restart Switch

```
Restart puzzle?
```

- **Speaker/Source:** Moving Stone

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** Chrono

```
\n<???>Come [Elijah], we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Chrono

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
```

```
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`

```
By clicking Z, you can jump across certain ledges.
```

110.2. The Secret Crystal Chamber

Elemental alignment (Blue, Yellow, Green, Red) to unlock the Heart Shield.

Associated Maps: Map 009, Map 039

110.2.1. Map 009: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
```

```
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

110.2.2. Map 039: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

110.3. The Battle Tutorial & Dream Eater Ambush

Dream illusion shatters; Dream Eaters assault; Chrono passes out; 'AWAKEN!'

Associated Maps: Map 008, Map 041, Map 012, Map 042

110.3.1. Map 008: Battle Tutorial

- **Speaker/Source:** `Dream Eaters`

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
```

```
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

110.3.2. Map 041: Battle Tutorial

- **Speaker/Source:** `Dream Eaters`

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
```

```
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

Chapter 111. Act IV: Awakening in Dio Village & Family Roots

111.1. Grandma's House Upstairs

Awakening from the dream; opening Grandma's chest for Grandma's Watch & Compass of Morality.

Associated Maps: Map 020, Map 004

111.1.1. Map 020: Grandma's House Upstairs

- **Speaker/Source:** `Grandma□s Chest`

```
\n<[Elijah]>I shouldn't go in this chest. I
need to talk to Grandma first.
She's very particular about her things.
\n<[Katrina]>I shouldn't go in this chest. I
need to talk to Grandma first.
Batty old woman seems to remind that chest.
```

- **Speaker/Source:** `Grandma□s Chest`

```
An old watch was in the chest.
\n<[Elijah]>Oh.. it's an old watch.
What's so special about it anyway?
\n<[Katrina]>She made me....grab...an
OLD WATCH! I thought it was
something important like her will!
```

- **Speaker/Source:** `Grandma□s Chest`

```
There is still something inside of the chest
OBTAINED THE COMPASS OF MORALITY
```

- **Speaker/Source:** `Grandma□s Chest`

```
It's all empty now.
```

- **Speaker/Source:** `Wierd Dream`

```
\n<[Elijah]>That dream....what did it mean...
It could just be my imagination but...It
felt like...something more. A hero
```

```
... Me...?
\n<[Katrina]>WHAT! WHAT THE...ME. Man, my mind
needs to get out of lala land and settle
into reality. I can't be a hero,
I won't be a hero...
```

111.1.2. Map 004: The Castle (Inside)

- **Speaker/Source:** `The King`

```
\n<D.Jihad>The fools...that power...that wicked power!
Soon I will rule over this land, and not even the Dark Lord
will have power over me!
```

- **Speaker/Source:** `The Meeting`

```
\n<Roté>King Jihad, I am here to inform you today...
It's here...
The darkness... it has returned
\n<Jihad>...Thank you Roté, what would I do without your
careful watch...
\n<Jihad>Gather up the guard! We need to prepare!
Darkness will soon return to this land. And don't tell the people of
this discovery...
\n<Roté>Y...Y...Yes my king...
\n<Jihad>You all are dismissed.
Yes my king.
```

111.2. Grandma's House Downstairs

Conversation with Grandma Magolia; memories of Commander Gareth Ro-enheart; errand to Monica.

Associated Maps: Map 021, Map 007

111.2.1. Map 021: Grandma's House Downstairs

- **Speaker/Source:** `Grandma`

```
\n<Grandma>Hello sweetie, could you please
go upstairs and search in the chest
and grab what's inside? It's something that
has great signifinance.
\n<[Katrina]>The will. I love you gran,
but a girl gotta get out of this dead-end
town.
```

```
\n<[Elijah]>Maybe is is something
about dad! I haven't seen him since my
10th birthday.
\n<[Elijah]>Sure thing. This might
be my chance to get answers!
\n<[Katrina]>Why not! Crazy old woman...
That will is mine!
\n<[Elijah]>Well...uhh...but...
But if it is about him...
\n<[Katrina]>Gran, why can't you do it?
Why doesn't she do it?
If she wants it she can get it.
```

- **Speaker/Source:** Grandma

```
It's been a while since we've talk.
What do you want to talk about?
\n<Grandma>Oh me and Monica go way back.
She was such a wild child, always getting
into mischief. You'll love her.
\n<Grandma>This town used to be full of
life. sighs The kingdom doesn't
pay much attention to this place anymore...
\c[5]Not avalible in the demo
\n<Grandma>Have you been upstairs yet?
The chest should still be there.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>Thank you honey. This watch has
been in the family for a long time. Oh,
how it brings back memories of better days.
\n<Grandma>I think you should keep it. It's
my gift to you. And here's a bit of money.
I know that you'll need it more than little
ol' me.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>[Katrina] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>[Elijah] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>My friend, Monica,
would know what is causing these strange
```

```
rumors. Can you go to her house and see
what she has to say about all of this?
\n<[Elijah]>Sure thing. It will be nice to
finally leave the house. I might even
get to leave town!
\n<[Katrina]>Why not! I would love to~
I'm out of this poor slum of a town!
\n<[Elijah]>Well..are you sure you can't?
I don't even know her...
\n<[Katrina]>Gran, I'm not your errand girl!
 This is YOUR friend.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

111.2.2. Map 007: Switches

- **Speaker/Source:** Special Room Switch

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** Bonus Gold

```
100\G were found!
```

- **Speaker/Source:** Restart Switch

```
Restart puzzle?
```

- **Speaker/Source:** Moving Stone

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** Chrono

```
\n<???>Come, we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Chrono

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`

```
By clicking Z, you can jump across certain ledges.
```

Chapter 112. Act V: Dio Village Upper & The Outskirts Crisis

112.1. Dio Village Upper / Main Square

The town in disrepair; Fighter Kid blocking the lower gate; visiting Monica.

Associated Maps: Map 022, Map 023, Map 010, Map 012, Map 013

112.1.1. Map 023: Main Square

- **Speaker/Source:** `Fighter Kid`

```
... He's just ignoring you
```

- **Speaker/Source:** `Fighter Kid`

```
You want me to move? I'm blocking the way for others?
Big woop! No one can stop me so I'm staying!
```

- **Speaker/Source:** `Fighter Kid`

```
You're no fun! I'm going home!
```

- **Speaker/Source:** `Mystic□s Home`

```
The door is locked.
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Monica!`

```
\n<[Elijah]>This is great! The town is still
sad with the storm but it's my home. I
need to stay focus and find Monica's house.
Grandma said her house is in front of ours.
\n<[Katrina]>Ugghh. I can't believe we still
live in this dump. Doesn't help that storm
destoryed everything. Now Gran said Monica
live in the house in front of us. Peachy...
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Leave Town`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

112.2. Monica's House & The Mystic's Brew

Meeting Monica; the fallen soldiers totem; offering for Goddess Lyri; errand for a Potion.

Associated Maps: Map 027, Map 029

112.2.1. Map 027: Monica's House

- **Speaker/Source:** `Monica`

```
\n<Monica>And who in the blaze are YOU!
```

```
\n<Monica>...
...
...
\n<Monica>So you are Magolia's grandson.
She made you sound more handsome that
you are.
\n<[Elijah]>HEY! That's not a nice thing
to say!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Elijah]>Nn-...I'm here because my grandma
thinks that evil has returned.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>So you are Magolia's
granddaughter. She made you sound prettier
that you are.
\n<[Katrina]>LISTEN HER YOU OLD-!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Katrina]>What is wrong with you... I'm
just here because my grandma is crazy.
She's babbling about evil forces and all
that nonsense.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
```

```
business. Can you handle that?
```

- **Speaker/Source:** Monica

```
\n<Monica>You have to stop that kid! Our
village will be more devastated than
even with him blocking the road!
```

- **Speaker/Source:** Monica

```
\n<Monica>Good job! I think I have an idea.
Now that the people can, residences and
newcomers have arrived. I notice that a
shopkeeper has came into our town as well.
\n<Monica>I need you to go a buy a potion.
I will use this as a base for my offering
to Lyri. She'll be able to help us in this
dark hour.
Obtained the Book of Monsters
```

- **Speaker/Source:** Monica

```
\n<Monica>Oh yes. This is exactly what I
need for my offering.
\n<Monica>There is one more thing that I
need in order to make this brew suitable
for Lyri. Go out of the town to the west
path. A man has what I need. Get it please.
\n<Monica>If you want to get the potion,
you'll need to go to the lower parts of the
town.
```

- **Speaker/Source:** Monica

```
\n<Monica>Do you need something?
\n<Monica>Me? Why, I'm one of the original
mages for the King. The kingdom was not as
peaceful as is it now. But then, the
Dark Lord tried to take the kingdom.
\n<Monica>He was born from the darkness
that this world had. He planned to take
over this world and fill it with darkness.
The Hero saved this world and brought us
\n<Monica>our peace. The shield and sword.
Oh don't mind this old hag's ranmbling.
Not avalible in demo
```

```
Not avalible in demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `Lyri`

```
It is the deity Lyri, the Goddess of Peace.
```

- **Speaker/Source:** `Totem`

```
The totem has the names of fallen soldiers. One
name has a heart by it and it reads "Miss you".
\n<[Elijah]>Did Monica have a husband?
\n<[Katrina]>Someone married that crazy bat!
```

- **Speaker/Source:** `Totem`

```
The totem is full of offerings.
```

112.3. The Secret Sword Cache

Discovering Medusa's Sword in the hidden glade.

Associated Maps: Map 024, Map 025, Map 017

112.3.1. Map 024: The Sword Map

- **Speaker/Source:** `Medusa□s Sword`

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

112.3.2. Map 025: The Sword Map

- **Speaker/Source:** `Medusa□s Sword`

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

Chapter 113. Act VI: Moral Crossroads of Dio Lower Ward

113.1. Eric the Lumberjack's House

Self-Aware Pot; Test of Honesty: 10G life savings vs 500G rent donation; Ring of Fire reward.

Associated Maps: Map 028

113.1.1. Map 028: Eric's House

- **Speaker/Source:** `Eric`

```
We've been seeing a lot of monster appear near the
recently. I wonder what's that all about. And with
jobs being tight, money is hard to come by.
```

- **Speaker/Source:** `Eric`

```
\n<[Elijah]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Elijah]>Thank you sir, but I...
\n<Eric>Don't mention it, and by the way, the name
is Eric. I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[1]>Thank you. Thank you so much!
\n<[Katrina]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Katrina]>Woah! This is a cool ring dude!
\n<Eric>The name is Eric and that ring has a story.
I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[2]>Thanks dude.
```

- **Speaker/Source:** `Eric`

```
\n<Eric>I'm sure that ring's fire power will come in
handy if you plan on leaving the village
```

- **Speaker/Source:** `Self-Aware Pot`

```
What kind of game do you think this is! You can't
just steal whatever you want! How rude!
```

- **Speaker/Source:** `Test of Honesty`

```
Something is glowing in the pot. Take the pots content?
You recieve 10G
THAT'S ALL MY MONEY! GET OUT NOW!
\n<\n1>I am so sorry! Please, I am not
trying to...
\n<n[2]>AH!
\n<n[2]>Hey, where did you even...
\n<n[2]>AH!
\n<[Elijah]>..steal all your money.
\n<n[1]>I'm sorry...
\n<n[2]>...come from...
\n<n[2]>It was just a few coins anyway.
That guy definately overreacted
You decide to leave the pot alone.
\n<n[1]>I can be quick, discret maybe
Elijah slowly reaches into the jar, stopping when
he saw how little money he had.
\n<n[1]>Maybe I should't...
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something.
\n<n[1]>Is this what the voice meant,
to be a hero...How much do you need.
Just 500G, that will be enough for me to make it for
the month.
\n<n[1]>I'll leave the money in the pot,
that way you'll have it.
Tha-tha-thank you!
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
```

```
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something
\n<n[1]>I have to help him somehow...
How much do you need.
Just 500G, that will be enough for me to make it for
the month
\n<n[1]>I'll leave the money in the pot,
that way you'll have it
Tha-tha-thank you!
\n<n[2]>The money is right there! This
guy won't see it coming.
As Katrina was about to take the money, she stopped
for just a moment
\n<n[2]>Ugghhh, I can't do it.
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>Is this what the voice meant,
to be a hero...How much do rent here
cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
```

```
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>I have to help him somehow...
How much do rent here cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

- **Speaker/Source:** Test of Honesty

```
Would you like to add to the pot?
You have given \v[14] to the man.
How much will you add?
You have given \v[14] to the man.
You have filled the pot! Talk to the man once more.
The pot isn't full yet.
You don't have that much to give
You leave the pot alone.
```

- **Speaker/Source:** Test of Honesty

```
The pot is filled with gold. It can cover rent for
2 months!
```

- **Speaker/Source:** Empty Pot

```
The pot is empty
```

- **Speaker/Source:** Empty Pot

```
The pot is empty
```

113.2. Shopping District & Lower Village

Buying the offering potion; the shopkeeper's note; St. Baron's locked mansion.

Associated Maps: Map 029, Map 030, Map 016

113.2.1. Map 030: Shopping District

- **Speaker/Source:** `St. Baron□s Home`

```
\n<[Elijah]>I can't believe that I am doing
this,...the baron is the richest
man in the villiage...but he's been
using the town! This is for the good!
\n<[Katrina]>Ok...here we go...let's do this. I
have to do at least one good thing today.
\n<[Elijah]>I can believe the baron
would take all this money for himself! Not
anymore!
\n<[Elijah]>So the baron can still
keep a house while the rest of just
struggle. This is gonna be fun.
The door is locked. Mayhbe you should come back at
a different time
```

- **Speaker/Source:** `Shop Menu`

```
Welcoem to my shop! How can I help you today?
Thanks for coming! See you later!
```

- **Speaker/Source:** `Shopkeeper□s Wife Dialouge`

```
We've just got into town so we are still working
on getting the funds for a bed. But my hubby
will not rest until we have that bed!
```

- **Speaker/Source:** `Got the Potion`

```
\n<[Elijah]>I wonder why she needs a
potion. Could she be a witch?
\n<[Katrina]>I can't believe she made me
waste my money...
```

113.3. John the Thief's Hideout

John's sidequest proposal to overthrow the corrupt Baron.

Associated Maps: Map 031

113.3.1. Map 031: John the Thief (Sidequest)

- **Speaker/Source:** `John the Thief`

```
\n<John>What do you want kid! Can't you
see that I'm busy!
\n<John>What do you want kid! Can't you
see that I'm busy!
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>What do you want kid! Can't you
see that I'm busy!
```

Chapter 114. Act VII: The Frontier, Wilderness & Final Dungeon

114.1. World Map Frontier

Demo completion gateway; thank-you message; frontier exits.

Associated Maps: Map 033, Map 034, Map 035, Map 005, Map 018

114.2. Wilderness Battle Zones (Demo Areas 3 & 4)

Combat zones featuring bats, slimes, magos, and bandits.

Associated Maps: Map 014, Map 015, Map 016, Map 034, Map 035, Map 036, Map 019, Map 020, Map 021, Map 022, Map 033

114.2.1. Map 020: Grandma's House Upstairs

- **Speaker/Source:** `Grandma□s Chest`

```
\n<[Elijah]>I shouldn't go in this chest. I
need to talk to Grandma first.
She's very particular about her things.
\n<[Katrina]>I shouldn't go in this chest. I
need to talk to Grandma first.
Batty old woman seems to remind that chest.
```

- **Speaker/Source:** `Grandma□s Chest`

```
An old watch was in the chest.
\n<[Elijah]>Oh.. it's an old watch.
What's so special about it anyway?
\n<[Katrina]>She made me....grab...an
OLD WATCH! I thought it was
something important like her will!
```

- **Speaker/Source:** `Grandma□s Chest`

```
There is still something inside of the chest
OBTAINED THE COMPASS OF MORALITY
```

- **Speaker/Source:** `Grandma□s Chest`

```
It's all empty now.
```

- **Speaker/Source:** Wierd Dream

```
\n<[Elijah]>That dream....what did it mean...
It could just be my imagination but...It
felt like...something more. A hero
... Me...?
\n<[Katrina]>WHAT! WHAT THE...ME. Man, my mind
needs to get out of lala land and settle
into reality. I can't be a hero,
I won't be a hero...
```

114.2.2. Map 021: Grandma's House Downstairs

- **Speaker/Source:** Grandma

```
\n<Grandma>Hello sweetie, could you please
go upstairs and search in the chest
and grab what's inside? It's something that
has great signifinance.
\n<[Katrina]>The will. I love you gran,
but a girl gotta get out of this dead-end
town.
\n<[Elijah]>Maybe is is something
about dad! I haven't seen him since my
10th birthday.
\n<[Elijah]>Sure thing. This might
be my chance to get answers!
\n<[Katrina]>Why not! Crazy old woman...
That will is mine!
\n<[Elijah]>Well...uhh...but...
But if it is about him...
\n<[Katrina]>Gran, why can't you do it?
Why doesn't she do it?
If she wants it she can get it.
```

- **Speaker/Source:** Grandma

```
It's been a while since we've talk.
What do you want to talk about?
\n<Grandma>Oh me and Monica go way back.
She was such a wild child, always getting
into mischief. You'll love her.
\n<Grandma>This town used to be full of
life. sighs The kingdom doesn't
```

```
pay much attention to this place anymore...
\c[5]Not avalible in the demo
\n<Grandma>Have you been upstairs yet?
The chest should still be there.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>Thank you honey. This watch has
been in the family for a long time. Oh,
how it brings back memories of better days.
\n<Grandma>I think you should keep it. It's
my gift to you. And here's a bit of money.
I know that you'll need it more than little
ol' me.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>[Katrina] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>[Elijah] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>My friend, Monica,
would know what is causing these strange
rumors. Can you go to her house and see
what she has to say about all of this?
\n<[Elijah]>Sure thing. It will be nice to
finally leave the house. I might even
get to leave town!
\n<[Katrina]>Why not! I would love to~
I'm out of this poor slum of a town!
\n<[Elijah]>Well..are you sure you can't?
I don't even know her...
\n<[Katrina]>Gran, I'm not your errand girl!
 This is YOUR friend.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** `Block`

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** `Block`

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** `Block`

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** `Block`

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

114.3. The Final Dungeon & Boss Threshold

Multi-level dungeon fortress leading to the ultimate demo boss confrontation.

Associated Maps: Map 017, Map 018, Map 023, Map 024, Map 025, Map 026, Map 027, Map 028, Map 037, Map 038

114.3.1. Map 023: Main Square

- **Speaker/Source:** `Fighter Kid`

```
... He's just ignoring you
```

- **Speaker/Source:** `Fighter Kid`

```
You want me to move? I'm blocking the way for others?
Big woop! No one can stop me so I'm staying!
```

- **Speaker/Source:** `Fighter Kid`

```
You're no fun! I'm going home!
```

- **Speaker/Source:** `Mystic□s Home`

```
The door is locked.
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Monica!`

```
\n<[Elijah]>This is great! The town is still
sad with the storm but it's my home. I
need to stay focus and find Monica's house.
Grandma said her house is in front of ours.
\n<[Katrina]>Ugghh. I can't believe we still
live in this dump. Doesn't help that storm
destoryed everything. Now Gran said Monica
live in the house in front of us. Peachy...
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Leave Town`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** Block

```
STAY AWAY FROM ME!
```

114.3.2. Map 024: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

114.3.3. Map 025: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

114.3.4. Map 026: Start Up Area

- **Speaker/Source:** The Beginning

```
It was a month before the Festival...<br>The tower...<br>The terror...
```

- **Speaker/Source:** The Beginning

```
The darkness...<br>The darkness...<br>The darkness returned once more.
```

114.3.5. Map 027: Monica’s House

- **Speaker/Source:** Monica

```
\n<Monica>And who in the blaze are YOU!
\n<Monica>...
...
...
\n<Monica>So you are Magolia's grandson.
She made you sound more handsome that
you are.
```

```
\n<[Elijah]>HEY! That's not a nice thing
to say!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Elijah]>Nn-...I'm here because my grandma
thinks that evil has returned.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>So you are Magolia's
granddaughter. She made you sound prettier
that you are.
\n<[Katrina]>LISTEN HER YOU OLD-!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Katrina]>What is wrong with you... I'm
just here because my grandma is crazy.
She's babbling about evil forces and all
that nonsense.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
```

- **Speaker/Source:** Monica

```
\n<Monica>You have to stop that kid! Our
```

```
village will be more devastated than
even with him blocking the road!
```

- **Speaker/Source:** Monica

```
\n<Monica>Good job! I think I have an idea.
Now that the people can, residences and
newcomers have arrived. I notice that a
shopkeeper has came into our town as well.
\n<Monica>I need you to go a buy a potion.
I will use this as a base for my offering
to Lyri. She'll be able to help us in this
dark hour.
Obtained the Book of Monsters
```

- **Speaker/Source:** Monica

```
\n<Monica>Oh yes. This is exactly what I
need for my offering.
\n<Monica>There is one more thing that I
need in order to make this brew suitable
for Lyri. Go out of the town to the west
path. A man has what I need. Get it please.
\n<Monica>If you want to get the potion,
you'll need to go to the lower parts of the
town.
```

- **Speaker/Source:** Monica

```
\n<Monica>Do you need something?
\n<Monica>Me? Why, I'm one of the original
mages for the King. The kingdom was not as
peaceful as is it now. But then, the
Dark Lord tried to take the kingdom.
\n<Monica>He was born from the darkness
that this world had. He planned to take
over this world and fill it with darkness.
The Hero saved this world and brought us
\n<Monica>our peace. The shield and sword.
Oh don't mind this old hag's ranmbling.
Not avalible in demo
Not avalible in demo
```

- **Speaker/Source:** The Secret Room

```
Get your snooping nose away from me room!
```

```
This room is currently unavalible for the demo
```

- **Speaker/Source:** The Secret Room

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** The Secret Room

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** Lyri

```
It is the deity Lyri, the Goddess of Peace.
```

- **Speaker/Source:** Totem

```
The totem has the names of fallen soldiers. One
name has a heart by it and it reads "Miss you".
\n<[Elijah]>Did Monica have a husband?
\n<[Katrina]>Someone married that crazy bat!
```

- **Speaker/Source:** Totem

```
The totem is full of offerings.
```

114.3.6. Map 028: Eric's House

- **Speaker/Source:** Eric

```
We've been seeing a lot of monster appear near the
recently. I wonder what's that all about. And with
jobs being tight, money is hard to come by.
```

- **Speaker/Source:** Eric

```
\n<[Elijah]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Elijah]>Thank you sir, but I...
```

```
\n<Eric>Don't mention it, and by the way, the name
is Eric. I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[1]>Thank you. Thank you so much!
\n<[Katrina]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Katrina]>Woah! This is a cool ring dude!
\n<Eric>The name is Eric and that ring has a story.
I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[2]>Thanks dude.
```

- **Speaker/Source:** `Eric`

```
\n<Eric>I'm sure that ring's fire power will come in
handy if you plan on leaving the village
```

- **Speaker/Source:** `Self-Aware Pot`

```
What kind of game do you think this is! You can't
just steal whatever you want! How rude!
```

- **Speaker/Source:** `Test of Honesty`

```
Something is glowing in the pot. Take the pots content?
You recieve 10G
THAT'S ALL MY MONEY! GET OUT NOW!
\n<\n1>I am so sorry! Please, I am not
trying to...
\n<n[2]>AH!
\n<n[2]>Hey, where did you even...
\n<n[2]>AH!
\n<[Elijah]>..steal all your money.
\n<n[1]>I'm sorry...
\n<n[2]>...come from...
\n<n[2]>It was just a few coins anyway.
That guy definately overreacted
You decide to leave the pot alone.
\n<n[1]>I can be quick, discret maybe
Elijah slowly reaches into the jar, stopping when
he saw how little money he had.
\n<n[1]>Maybe I should't...
\n<n[1]>I would be wrong of me to take this
```

```
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something.
\n<n[1]>Is this what the voice meant,
to be a hero...How much do you need.
Just 500G, that will be enough for me to make it for
the month.
\n<n[1]>I'll leave the money in the pot,
that way you'll have it.
Tha-tha-thank you!
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something
\n<n[1]>I have to help him somehow...
How much do you need.
Just 500G, that will be enough for me to make it for
the month
\n<n[1]>I'll leave the money in the pot,
that way you'll have it
Tha-tha-thank you!
\n<n[2]>The money is right there! This
guy won't see it coming.
As Katrina was about to take the money, she stopped
for just a moment
\n<n[2]>Ugghhh, I can't do it.
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
```

```
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>Is this what the voice meant,
to be a hero...How much do rent here
cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>I have to help him somehow...
How much do rent here cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

- **Speaker/Source:** Test of Honesty

```
Would you like to add to the pot?
You have given \v[14] to the man.
How much will you add?
```

```
You have given \v[14] to the man.
You have filled the pot! Talk to the man once more.
The pot isn't full yet.
You don't have that much to give
You leave the pot alone.
```

- **Speaker/Source:** `Test of Honesty`

```
The pot is filled with gold. It can cover rent for
2 months!
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

Chapter 115. Chapter 22: Complete Cast & Tactical Party Compendium

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/cast_and_characters.adoc

Chapter 116. Overview

This compendium catalogues every character, party companion, village elder, court antagonist, and mystical summon present in the 2017–2018 demo archives, detailing their combat statistics, class associations, sprite bindings, and narrative significance.

Chapter 117. 1. Dual Protagonists

117.1. Elijah Ro-enheart (The Steadfast Vanguard)

- **ID**: `Actor 1`
- **Class**: Hero (`Class 5`) / Knight (`Class 1`)
- **Relic Affinity**: Bearer of the **Shield of Lux**
- **Initial Level**: Lv 5 (Folder 1) / Lv 1 (Folder 2)
- **Movement Points**: 4 MP
- **Sprite / Battler**: `Actor1_1` / `MainCharacter` (custom generator)
- **Profile & Personality**: Son of the vanished Commander Gareth Ro-enheart (former Head of the Royal Guard). Earnest, chivalric, deeply burdened by his father's mysterious disappearance on his 10th birthday. Elijah seeks answers through service and duty, instinctively stepping forward to protect the weak and uphold the light of Sacyra.
- **Canonical LeTBS Specification**:

```
<letbs>
move_points: 4
auto_turn_order_face
auto_status_sprite
</letbs>
```

117.2. Lady Katrina Jaren (The Outrider Blade)

- **ID**: `Actor 2`
- **Class**: Hero (`Class 5`) / Start (`Class 7`)
- **Relic Affinity**: Wielder of the **Sword of Tenebris**
- **Initial Level**: Lv 5 (Folder 1) / Lv 1 (Folder 2)
- **Movement Points**: 4 MP
- **Initial Equipment**: Weapon 1 (Sword), Armor 2 (Hat), Armor 3 (Cloth)
- **Sprite / Battler**: `Actor1_8`
- **Profile & Personality**: Noble daughter of aristocratic House Jaren. Rebellious, razor-sharp, pragmatic, and fiercely independent. Refusing a cloistered aristocratic marriage or court seat, Katrina escaped to Dio Village with eyes on joining the independent Adventurers' Guild. Her caustic wit masks a deep empathy for ordinary folk crushed by royal neglect.
- **Canonical LeTBS Specification**:

```
<letbs>
move_points: 4
```

```
auto_turn_order_face
auto_status_sprite
</letbs>
```

Chapter 118. 2. Extended Tactical Party Companions (Folder 2 Additions)

In `Old_Game_Data_2`, three specialized tactical party members were integrated to expand the grid-based TRPG party:

118.1. Sebastien (The Infiltrator Outlaw)

- **ID**: `Actor 4`
- **Class**: Thief (`Class 6`)
- **Initial Level**: Lv 4
- **Sprite / Battler**: `Actor3_7`
- **Tactical AI & Role**: Outrider flanker equipped with `ranged_fighter` AI, trap disarmament, and precision critical strikes.

118.2. Angela (The Evoker Elementalist)

- **ID**: `Actor 5`
- **Class**: Mage (`Class 2`)
- **Initial Level**: Lv 4
- **Movement Scope**: `square(2)` (4 MP)
- **Sprite / Battler**: `Actor3_8` (custom cast, atk, hit, victory animations)
- **Action Palette**: Full tactical command set (`Move`, `Skill[42]`, `Items`, `Scan`, `Pass`).
- **Tactical Role**: Long-range elemental artillery, crowd control, and battlefield terrain modification (ice walls, volcano ground).

118.3. Lucius (The Battle Priest)

- **ID**: `Actor 6`
- **Class**: Priest (`Class 3`)
- **Initial Level**: Lv 4
- **Sprite / Battler**: `Actor2_7`
- **Tactical Role**: Turn order manipulation (`Accelerate`, `Decelerate`, `Emergency Turn`), long-distance allied teleportation, and barrier restoration.

Chapter 119. 3. Summoned Celestial Entities

119.1. Wind Spirit (Actor 11)

- **Class**: `Class 12` | **Sprite**: `Nature` (Index 4)
- **Skills**: Air Compression (`Skill 132`), Great Wind (`Skill 133` - pushback), Transposition (`Skill 134` - swap places).

119.2. Pixie (Actor 12)

- **Class**: `Class 13` | **Sprite**: `Nature` (Index 5)
- **Skills**: Healing Wave (`Skill 135` - AOE restoration).

119.3. Fire Chicken (Actor 13)

- **Class**: `Class 14` | **Sprite**: `Nature` (Index 2)
- **Characteristics**: Explosive combat decoy (`loop_animation: 262`), Bomb (`Skill 136`), Fire Arrow (`Skill 137`).

Chapter 120. 4. Key Lore Figures & NPCs

Name	Role & Allegiance	Historical Significance
Grandma Magolia	Matriarch of Dio Village	Mother of vanished Commander Gareth Ro-enheart; grandmother to Elijah. Bestows **Grandma's Watch** and **Compass of Morality**.
Monica the Mystic	Former Royal Battlemage	Eccentric hermit and priestess of Goddess Lyri. Guides protagonists in awakening relic powers through ritual offerings.
Eric the Lumberjack	Impoverished Dio Laborer	Loses his livelihood to monster raids; serves as the moral litmus test for player empathy in the **Test of Honesty**. Bestows the **Ring of Fire**.
John the Thief	Local Outlaw Leader	Operates in Lower Dio; plots to unseat the predatory St. Baron to return wealth to the villagers.
The St. Baron	Aristocratic Landowner	Feudal aristocrat bleeding Dio Village dry; symbol of court corruption and moral decay.
Chrono	Archival Sage & Guardian	Guide of the Dream Chamber / Ancient Archives. Oversees archetype stone selection and mechanics trials.
High Priest Roté	Royal Advisor	Court prelate who witnesses the cosmic omens and hastens to warn King Jihad.
King Jihad	Monarch of Sacyra	Feigns royal mobilization while secretly coveting the returned primordial darkness to surpass Dark Lord Malakor.

Chapter 121. Chapter 23: LeTBS Tactical Grid System Specification

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/combat_and_letbs_mechanics.adoc

Chapter 122. Overview

This document formalizes the grid-based Tactical Strategy RPG combat architecture implemented via Lecode's Tactical Battle System (LeTBS) in the 2017–2018 demo. It details movement costs, action economy, scope geometries, terrain manipulation, and turn manipulation invariants.

Chapter 123. 1. Battle Field Architecture & Action Economy

- **Grid Structure**: 2D tile-based tactical grid mapped over encounter maps.
- **Movement Points (MP)**: Consumable per-turn budget determining movement range:
  - Baseline Protagonists & Allies: `move_points: 4`.
  - Heavy Impact Reductions: Heavy Hit reduces target MP by 3 (`move_points+: -3`).
  - Outrider Haste: Light Steps increases target MP by 1 (`move_points+: 1`).
- **Turn Timeline**: Monotonic timeline ordering turns by entity agility (`auto_turn_order_face`).
- **Directional Facing**: Facing impacts defense and positional abilities (e.g., **Phantom Slash** teleports directly behind the target before striking).

Chapter 124. 2. Scope & Area of Effect (AOE) Geometries

The demo configured rich geometric targeting parameters across skills:

Scope Pattern	Example Skill	Geometric Definition
`circle(r)`	**Fire** (`Skill 42`), **Heal** (`Skill 72`)	Manhattan or Chebyshev circle of radius `r` centered on the caster.
`circle(max_r, min_r)`	**Ghost Arrow** (`Skill 79`), **Dagger Throw** (`Skill 169`)	Annular targeting ring; cannot target adjacent cells within `min_r`.
`line(n)`	**Explosive Mark** (`Skill 76`), **Great Wind** (`Skill 133`)	Cardinal linear projection up to `n` cells from caster.
`square(r)`	**Angela Move Scope** (`Actor 5`)	Full square grid extent of radius `r`.
`custom(pattern)`	**Fire Arrows** (`Skill 82`)	Scripted multi-target projectile pattern hitting all visible adversaries.
`through_obstacles`	**Teleportation** (`Skill 73`)	Ignores intermediate terrain obstacles, requiring only a free destination cell.

Chapter 125. 3. Terrain State Modifications & Dynamic Hazards

Combat encounters feature active environmental manipulation:

- **Volcano Ground (`Skill 47`)**: Lays hazardous fiery embers across multiple ground cells dealing continuous fire damage.
- **Ice Wall / Ice Prison (`Skill 51`, `Skill 52`)**: Summons physical destructible barricades (`HP: 20`) that obstruct enemy line-of-sight and movement corridors.
- **Explosive Mark & Magic Trap (`Skill 76`, `Skill 80`)**: Proximity-triggered grid landmines that detonate upon enemy entry.
- **Burning Tile Spread (`State 110`)**: Dynamic fire propagation expanding with `spread: 100% circle(1) all`.

Chapter 126. 4. Turn Timeline Manipulation

Unique priest techniques alter the flow of tactical battle:

- **Accelerate (`Skill 87`)**: Advances target position along the turn timeline by 100%.
- **Decelerate (`Skill 88`)**: Delays target position along the turn timeline by 100%.
- **Emergency Turn (`Skill 89`)**: Instantly grants an immediate action turn to an allied battler.
- **Extra Turn (`Skill 86`)**: Grants the caster or target an additional action cycle upon turn conclusion.

Chapter 127. Chapter 24: Classes, Growth Curves & Learning Trees

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/classes_and_growth.adoc

Chapter 128. Overview

This document details the 7 primary player and NPC classes plus 4 auxiliary summon/guard classes from `Classes.json`. It includes base stat formulas across 8 core attributes (MHP, MMP, ATK, DEF, MAT, MDF, AGI, LUK) and skill learning progressions.

Chapter 129. Class 1: Warrior

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	100	30	12	8	3	8	27	32
5	145	46	17	12	5	11	27	42
10	202	64	23	17	8	15	27	55
20	314	102	35	26	13	22	27	80
50	650	215	71	54	27	44	27	155
99	1200	400	130	100	50	80	27	277

129.1. Skill Learnings for Warrior

Level	Skill Name	Strategic Role
Lv 1	**Repulsive Blow** (`ID: 11`)	Innate class technique
Lv 1	**Flash Jump** (`ID: 12`)	Innate class technique
Lv 1	**Rush** (`ID: 13`)	Innate class technique
Lv 1	**Intimidating Aura** (`ID: 14`)	Innate class technique
Lv 1	**Heavy Hit** (`ID: 15`)	Innate class technique
Lv 1	**Provoke** (`ID: 16`)	Innate class technique
Lv 1	**Phantom Slash** (`ID: 17`)	Innate class technique

Chapter 130. Class 2: Mage

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	60	90	2	7	13	8	26	32
5	83	130	4	10	18	12	26	42
10	110	180	6	13	24	17	26	55
20	165	280	10	20	36	26	26	80
50	330	580	21	39	72	54	26	155
99	600	1070	40	70	130	100	26	277

130.1. Skill Learnings for Mage

Level	Skill Name	Strategic Role
Lv 1	**Fire** (ID: 42)	Innate class technique
Lv 1	**Sparkle** (ID: 43)	Innate class technique
Lv 1	**Lightning Storm** (ID: 44)	Innate class technique
Lv 1	**Fire Storm** (ID: 45)	Innate class technique
Lv 1	**Fire Support** (ID: 46)	Innate class technique
Lv 1	**Volcano Ground** (ID: 47)	Innate class technique
Lv 1	**Ice Fury** (ID: 49)	Innate class technique
Lv 1	**Walking Fire** (ID: 50)	Innate class technique
Lv 1	**Ice Wall** (ID: 51)	Innate class technique
Lv 1	**Ice Prison** (ID: 52)	Innate class technique
Lv 1	**Whitch's Escape** (ID: 53)	Innate class technique
Lv 1	**Fire Chickens** (ID: 54)	Innate class technique
Lv 1	**Water (charged)** (ID: 55)	Innate class technique
Lv 1	**Mega Sparkle** (ID: 56)	Innate class technique
Lv 1	**Multi Fire** (ID: 57)	Innate class technique

Chapter 131. Class 3: Priest

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	65	60	6	8	10	7	32	32
5	95	87	10	13	15	11	32	42
10	133	119	15	18	20	16	32	55
20	208	185	25	28	31	26	32	80
50	433	380	53	59	63	54	32	155
99	800	700	100	110	115	100	32	277

131.1. Skill Learnings for Priest

Level	Skill Name	Strategic Role
Lv 1	**Heal** (ID: 72)	Innate class technique
Lv 1	**Teleportation** (ID: 73)	Innate class technique
Lv 1	**Salvation** (ID: 74)	Innate class technique
Lv 1	**Teleportation II** (ID: 75)	Innate class technique
Lv 1	**Explosive Mark** (ID: 76)	Innate class technique
Lv 1	**Light Steps** (ID: 78)	Innate class technique
Lv 1	**Ghost Arrow** (ID: 79)	Innate class technique
Lv 1	**Magic Trap** (ID: 80)	Innate class technique
Lv 1	**Fire Arrows** (ID: 82)	Innate class technique
Lv 1	**Wind Spirit** (ID: 83)	Innate class technique
Lv 1	**Summon Pixie** (ID: 84)	Innate class technique
Lv 1	**Shamanism** (ID: 85)	Innate class technique
Lv 1	**Extra Turn** (ID: 86)	Innate class technique
Lv 1	**Accelerate** (ID: 87)	Innate class technique
Lv 1	**Emergency Turn** (ID: 89)	Innate class technique
Lv 1	**Ally Teleport.** (ID: 90)	Innate class technique
Lv 1	**Mana Ball** (ID: 92)	Innate class technique

Chapter 132. Class 4: Summoner

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	450	90	16	16	16	16	32	32
5	650	130	21	21	21	21	42	42
10	900	180	28	28	28	28	55	55
20	1400	280	40	40	40	40	80	80
50	2900	580	77	77	77	77	155	155
99	5350	1070	138	138	138	138	277	277

132.1. Skill Learnings for Summoner

Level	Skill Name	Strategic Role
Lv 1	**Sentinel Totem** (`ID: 102`)	Innate class technique

Chapter 133. Class 5: Hero

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	300	90	16	16	16	16	32	32
5	507	130	21	21	21	21	42	42
10	764	180	28	28	28	28	55	55
20	1280	280	40	40	40	40	80	80
50	2825	580	77	77	77	77	155	155
99	5350	1070	138	138	138	138	277	277

133.1. Skill Learnings for Hero

Level	Skill Name	Strategic Role
Lv 1	**Heal** (`ID: 72`)	Innate class technique
Lv 5	**Repulsive Blow** (`ID: 11`)	Innate class technique

Chapter 134. Class 6: Thief

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	450	90	16	16	16	16	32	32
5	650	130	21	21	21	21	42	42
10	900	180	28	28	28	28	55	55
20	1400	280	40	40	40	40	80	80
50	2900	580	77	77	77	77	155	155
99	5350	1070	138	138	138	138	277	277

Chapter 135. Class 7: Start

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	1	0	1	999	1	999	1	1
5	1	1	1	999	1	999	1	1
10	1	1	1	999	1	999	1	1
20	1	1	1	999	1	999	1	1
50	1	1	1	999	1	999	1	1
99	1	1	1	999	1	999	1	1

Chapter 136. Class 10: ---Special---

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	450	90	16	16	16	16	32	32
5	650	130	21	21	21	21	42	42
10	900	180	28	28	28	28	55	55
20	1400	280	40	40	40	40	80	80
50	2900	580	77	77	77	77	155	155
99	5350	1070	138	138	138	138	277	277

Chapter 137. Class 11: Guard

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	100	30	12	8	3	8	27	32
5	145	46	17	12	5	11	27	42
10	202	64	23	17	8	15	27	55
20	314	102	35	26	13	22	27	80
50	650	215	71	54	27	44	27	155
99	1200	400	130	100	50	80	27	277

137.1. Skill Learnings for Guard

Level	Skill Name	Strategic Role
Lv 1	**Heavy Hit** (`ID: 15`)	Innate class technique

Chapter 138. Class 12: Wind Spirit

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	1	50	1	1	1	1	1	1
5	1	79	1	1	1	1	1	1
10	1	115	1	1	1	1	1	1
20	1	186	1	1	1	1	1	1
50	1	400	1	1	1	1	1	1
99	1	750	1	1	1	1	1	1

138.1. Skill Learnings for Wind Spirit

Level	Skill Name	Strategic Role
Lv 1	**Air Compression** (`ID: 132`)	Innate class technique
Lv 1	**Great Wind** (`ID: 133`)	Innate class technique
Lv 1	**Transposition** (`ID: 134`)	Innate class technique

Chapter 139. Class 13: Pixie

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	1	50	1	1	1	1	1	1
5	1	79	1	1	1	1	1	1
10	1	115	1	1	1	1	1	1
20	1	186	1	1	1	1	1	1
50	1	400	1	1	1	1	1	1
99	1	750	1	1	1	1	1	1

139.1. Skill Learnings for Pixie

Level	Skill Name	Strategic Role
Lv 1	**Healing Wave** (`ID: 135`)	Innate class technique

Chapter 140. Class 14: Fire Chicken

Level	MHP	MMP	ATK	DEF	MAT	MDF	AGI	LUK
1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1
20	1	1	1	1	1	1	1	1
50	1	1	1	1	1	1	1	1
99	1	1	1	1	1	1	1	1

140.1. Skill Learnings for Fire Chicken

Level	Skill Name	Strategic Role
Lv 1	**Bomb** (`ID: 136`)	Innate class technique

Chapter 141. Chapter 25: The Complete Grimoire (75+ Skills & Formulas)

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/grimoire_and_skills.adoc

Chapter 142. Overview

This document records the complete grimoire of all 75+ active techniques, spells, and summon abilities defined in `Skills.json`. It provides mathematical damage formulas, MP/TP costs, scope parameters, and tactical LeTBS sequences.

Chapter 143. Basic Martial Maneuvers

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
1	**Attack**	0	0	`a.atk * 4 - b.def * 2`	+ *Note*: `Skill #1 will be used when you select the Attack command.`
2	**Guard**	0	0	`0`	+ *Note*: `Skill #2 will be used when you select the Guard command.`
3	**Dual Attack**	0	0	`a.atk * 4 - b.def * 2`	
4	**Double Attack**	0	0	`a.atk * 4 - b.def * 2`	
5	**Triple Attack**	0	0	`a.atk * 4 - b.def * 2`	
6	**Escape**	0	0	`0`	
7	**Wait**	0	0	`0`	

Chapter 144. Knight Martial Techniques

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
11	**Repulsive Blow**	0	0	`this.phyDmg(120)`	Hits and push the target for two cell. *Note*: `<letbs> scope: circle(1) sequence: atk_push(2) scope_color: #FFF </letbs> <l…`
12	**Flash Jump**	0	0	`this.phyDmg(110) + this.magDmg(10)`	Jumps to an available position in the scope and invoke a thunder to damage nearby fighters. *Note*: `<letbs> scope: circle(3,1) aoe: circle(1) scope_options: need_free_cell aoe_o…`
13	**Rush**	0	0	`this.phyDmg(80)`	Rushes toward the target. *Note*: `<letbs> scope: custom(rush) sequence: rush </letbs> <letbs_AI> type: offense…`
14	**Intimidating Aura**	0	0	`(a.atk * 4 - b.def * 2) * 1.2`	Casts an aura that reduce enemies' ATK by 15%. Grants as well 70% counter attack chance. *Note*: `<letbs> scope: circle(0) scope_options: include_center </letbs> <letbs_AI> t…`
15	**Heavy Hit**	0	0	`this.phyDmg(135)`	Hits the target and remove 3 move points. *Note*: `<letbs> scope: circle(1) sequence: atk </letbs> <Learn Require Level: 5> <l…`
16	**Provoke**	0	0	`(a.atk * 4 - b.def * 2) * 1.2`	Draw the attention of enemies upon the user. Can use another action after this skill. *Note*: `<letbs> scope: circle(0) scope_options: include_center extra_action </letbs> …`
17	**Phantom Slash**	0	0	`this.phyDmg(60)`	Use dark arts to reach the back of the target. *Note*: `<letbs> scope: line(4) sequence: phantom_slash </letbs> <Learn Require Level…`

Chapter 145. Mage Elemental Spells

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
42	**Fire**	0	0	`this.magDmg(105)`	Deals fire damages to a single target in a medium range. *Note*: `<letbs> scope: circle(3);line(4) </letbs> <letbs_AI> type: offense </letbs_A…`
43	**Sparkle**	0	0	`this.magDmg(80)`	Deals thunder damages to targets in a small area. *Note*: `<letbs> scope: circle(4) aoe: circle(1) sequence: spark </letbs> <Learn Requi…`
44	**Lightning Storm**	0	0	`this.magDmg(60)`	Attacks 3 times a random battler in the area. *Note*: `<letbs> scope: line(4) aoe: circle(2) sequence: lightning_storm </letbs> <Lea…`
45	**Fire Storm**	0	0	`this.magDmg(200)`	Attacks 6 times randomly in the area. *Note*: `<letbs> scope: circle(4,2) aoe: square(1) sequence: fire_storm </letbs> <Lear…`
46	**Fire Support**	0	0	`this.magDmg(80)`	Deals fire damages to a single target in a medium range with the help of close allies. *Note*: `<letbs> scope: circle(4) sequence: fire_support </letbs> <Learn Require Level…`
47	**Volcano Ground**	0	0	`(a.mat * 2 - b.mdf * 2)*2`	Lays multiple fire spots on the ground. *Note*: `<letbs> scope: circle(0) scope_options: include_center aoe: custom(volcano_gr…`
48	**Volcano Ground (dmg)**	0	0	`this.magDmg(25)`	
49	**Ice Fury**	0	0	`this.magDmg(115)`	Deals ice damages in a row. *Note*: `<letbs> scope: line(4) aoe: custom(ice_fury) sequence: skill </letbs> <Learn …`

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
50	**Walking Fire**	0	0	`this.magDmg(105)`	Deals fire damages in a medium path. *Note*: `<letbs> scope: circle(4) scope_options:through_obstacles aoe: path sequence: …`
51	**Ice Wall**	0	0	`(a.mat * 2 - b.mdf * 2)*1.15`	Summon a wall of ice. *Note*: `<letbs> scope: line(3) scope_options: need_free_cell aoe: custom(ice_wall) ao…`
52	**Ice Prison**	0	0	`(a.mat * 2 - b.mdf * 2)*1.15`	Summon a prison of ice. *Note*: `<letbs> scope: circle(4) aoe: custom(ice_prison) aoe_options: need_free_cell …`
53	**Whitch's Escape**	0	0	`(a.mat * 2 - b.mdf * 2)*1.20`	Teleports then casts Spark at the old location. The scope is based on current move points. *Note*: `<letbs> scope: circle(e.getMovePoints()+1) scope_options: need_free_cell sequ…`
54	**Fire Chickens**	0	0	`(a.mat * 2 - b.mdf * 2)*1.15`	Summon explosive chickens *Note*: `<letbs> scope: circle(2) aoe: circle(1) scope_options: need_free_cell aoe_opt…`
55	**Water (charged)**	0	0	`this.magDmg(135)`	+ *Note*: `<letbs> scope: circle(4) sequence: map_skill charge: 2, 261 </letbs> <LearnR…`
56	**Mega Sparkle**	0	0	`this.magDmg(60)`	Deals thunder damage on 3 different areas. *Note*: `<letbs> scope: circle(4) aoe: circle(1) sequence: mega_spark 3 selections: di…`
57	**Multi Fire**	0	0	`this.magDmg(105)`	Invoke 3 columns of fire. *Note*: `<letbs> scope_select: enemies scope: circle(3); 2 selections: append; Select …`

Chapter 146. Priest Holy Arts & Turn Manipulation

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
72	**Heal**	0	0	`50 + a.mat`	Heals a single target and ends the turn. *Note*: `<letbs> scope: circle(4,2) sequence: heal pass_on_use scope_options: include_…`
73	**Teleportation**	0	0	`0`	Teleports the caster in a medium range. *Note*: `<letbs> scope: circle(4) sequence: teleportation scope_options:through_obstac…`
74	**Salvation**	0	0	`25 + a.mat`	Attracts allies and heals them in all directions. *Note*: `<letbs> scope: circle(0) aoe: line(3) sequence: salvation scope_options: incl…`
75	**Teleportation II**	0	0	`0`	Teleports the battler in front of the caster in a medium range. *Note*: `<letbs> scope: circle(3) sequence: teleportation_sup scope_options: through_o…`
76	**Explosive Mark**	0	0	`None`	Lays a mark on the ground that deals fire damages to enemies on it. *Note*: `<letbs> scope: line(5) aoe: square(1) sequence: explosive_mark </letbs> <Lear…`
77	**Explosive Mark (dmg)**	0	0	`this.magDmg(50)`	
78	**Light Steps**	0	0	`50 + a.mat`	Increases target's move points for 2 turns. *Note*: `<letbs> scope_options: include_center </letbs> <Learn Require Level: 5> <letb…`

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
79	**Ghost Arrow**	0	0	`this.magDmg(75)`	Deals light damages to a single target in a medium range, ignoring any obstacle, and remove one move point. *Note*: `<letbs> scope: circle(5,2) sequence: ghost_arrow scope_options: through_obsta…`
80	**Magic Trap**	0	0	`None`	Lays a magic trap who deals light damage while blinding the target with 60% rate. *Note*: `<letbs> scope: circle(5) scope_options:through_obstacles, need_free_cell aoe:…`
81	**Magic Trap (dmg)**	0	0	`this.magDmg(160)`	
82	**Fire Arrows**	0	0	`this.magDmg(105)`	Launch a fire arrow for every enemy in the scope. *Note*: `<letbs> scope: line(2) aoe: custom(fire_arrows) aoe_options: line_of_sight se…`
83	**Wind Spirit**	0	0	`(a.mat * 2 - b.mdf * 2)*1.15`	Summon a wind spirit that help the party with the power of wind. *Note*: `<letbs> scope: circle(2) scope_options: need_free_cell sequence: summon(wind_…`
84	**Summon Pixie**	0	0	`(a.mat * 2 - b.mdf * 2)*1.15`	Summon a pixie that can heal the party. *Note*: `<letbs> scope: circle(1) scope_options: need_free_cell </letbs> <Learn Requir…`
85	**Shamanism**	0	0	`a.mhp * 0.1`	All allies recover 10% of their MHP for each dead battler in the area. *Note*: `<letbs> sequence: shamanism scope: circle(3) aoe: circle(2) scope_options: in…`
86	**Extra Turn**	0	0	`50 + a.mat`	Add an extra turn. *Note*: `<letbs> scope: circle(2) scope_options: include_center extra_turn: 100% </let…`

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
87	**Accelerate**	0	0	`50 + a.mat`	Accelerate the turn order of the target. *Note*: `<letbs> scope: circle(2) scope_options: include_center accelerate_turn: 100% …`
88	**Decelerate**	0	0	`50 + a.mat`	Accelerate the turn order of the target. *Note*: `<letbs> scope: circle(2) scope_options: include_center decelerate_turn: 100% …`
89	**Emergency Turn**	0	0	`50 + a.mat`	Give an instant turn to the target. *Note*: `<letbs> scope: circle(3) pass_on_use scope_options: include_center instant_tu…`
90	**Ally Teleport.**	0	0	`0`	Teleports an ally in a medium range. *Note*: `<letbs> scope_select: allies scope: circle(4) sequence: ally_teleport scope_o…`
91	**Ally Teleport. (Scope)**	0	0	`0`	+ *Note*: `<letbs> scope: circle(4) scope_options:through_obstacles, need_free_cell </le…`
92	**Mana Ball**	0	0	`this.magDmg(80)`	Launch a bouncing mana ball *Note*: `<letbs> scope: circle(3) sequence: mana_ball </letbs> <Learn Require Level: 5…`

Chapter 147. Summoner Totems & Rites

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
102	**Sentinel Totem**	0	0	`(a.mat * 2 - b.mdf * 2)*1.15`	Summon a totem that strike enemies as soon as they enter the range of action. *Note*: `<letbs> scope: circle(3) scope_options: need_free_cell sequence: summon(senti…`
103	**Transform.**	0	0	`50 + a.mat`	+ *Note*: `<letbs> scope: circle(0) scope_options: include_center </letbs> <letbs_AI> ty…`
104	**Devour**	0	0	`this.magDmg(105)`	+ *Note*: `<letbs> scope: circle(1) fusion_effect </letbs> <letbs_AI> type: offense </le…`

Chapter 148. Summon Beast Techniques

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
132	**Air Compression**	0	0	`this.magDmg(135)`	Deals wind damages in a medium range with 15% of chance to blind the target. *Note*: `<letbs> scope: circle(4) </letbs> <letbs_AI> type: offense </letbs_AI> <Learn…`
133	**Great Wind**	0	0	`this.magDmg(140)`	Deals wind damages and push the target. The collision damage of this skill is greatly increased. *Note*: `<letbs> scope: line(4) sequence: great_wind collision_formula: 2 * b.mhp * 0….`
134	**Transposition**	0	0	`(a.mat * 2 - b.mdf * 2)*1.35`	Switch place with a target in a medium range. *Note*: `<letbs> scope: circle(6) sequence: transposition scope_options:through_obstac…`
135	**Healing Wave**	0	0	`10 + a.mat`	Heals in small area *Note*: `<letbs> scope: circle(3) aoe: circle(1) scope_options: include_center sequenc…`
136	**Bomb**	4	0	`a.mhp`	+ *Note*: `<letbs> scope: circle(1) </letbs> <letbs_AI> type: offense </letbs_AI> <letbs…`
137	**Fire Arrow**	0	0	`this.phyDmg(100) + this.magDmg(100)`	Launch a fire arrow. *Note*: `<letbs> scope: circle(3) sequence: projectile(fire_arrow) </letbs> <Learn Req…`

Chapter 149. Enemy Bestial & Void Techniques

ID	Skill Name	MP	TP	Damage Formula	Description & Tactical Note
162	**Poisonous Bite**	3	0	`(a.atk * 4 - b.def * 2)*0.8`	+ *Note*: `<letbs> scope: circle(1) </letbs> <letbs_AI> type: offense </letbs_AI>`
163	**Fire Surge**	4	0	`this.magDmg(105)`	Deals fire damages to a single target in a medium range. *Note*: `<letbs> scope: circle(4) </letbs> <letbs_AI> type: offense </letbs_AI>`
164	**Summon Pyra**	0	0	`(a.mat * 2 - b.mdf * 2)*1.15`	+ *Note*: `<letbs> scope: circle(1) scope_options: need_free_cell sequence: summon_pyra …`
165	**Evil Healing**	0	0	`15 + a.mat`	+ *Note*: `<letbs> scope: circle(4) scope_options: include_center </letbs> <letbs_AI> ty…`
166	**Mago’s Mark**	0	0	`None`	+ *Note*: `<letbs> scope: line(5) aoe: square(1) sequence: mago_mark </letbs> <Cooldown:…`
167	**Evil Protection**	0	0	`15 + a.mat`	+ *Note*: `<letbs> scope: circle(4) scope_options: include_center </letbs> <letbs_AI> ty…`
168	**Rabbit’s Jump**	0	0	`(a.atk * 4 - b.def * 2) * 1.10 + a.mat`	+ *Note*: `<letbs> scope: line(3) aoe: circle(1) scope_options: need_free_cell sequence:…`
169	**Dagger Throw**	3	0	`this.phyDmg(115)`	Throw a little but sharp dagger toward the target. *Note*: `<letbs> scope: circle(4,2) sequence: projectile(dagger) </letbs> <letbs_AI> …`

Chapter 150. Chapter 26: Bestiary, Destructible Props & Troop Formations

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/bestiary_and_troops.adoc

Chapter 151. Overview

This document catalogs all 22+ hostile monsters, wild fauna, bandits, void entities, and interactive battlefield props from `Enemies.json` and `Troops.json`.

Chapter 152. 1. Monster Bestiary

ID	Name	HP	MP	ATK	DEF	MAT	MDF	AGI	LUK	AI / Behavior
1	**Bat**	80	12	11	8	8	4	25	30	Skill 1
2	**Mago**	60	18	8	6	16	6	22	30	Skill 41, Skill 1
3	**Rabbit**	55	0	10	8	8	4	27	30	Skill 1
4	**Great Mago**	90	36	10	7	19	7	23	30	Skill 1
5	**Fire Mago**	90	36	10	7	19	7	23	30	Skill 1
6	**Pyra**	30	50	6	8	6	4	27	30	Basic Attack
7	**Bandit**	80	12	14	10	10	7	25	30	Skill 1, Skill 169
8	**Mage Bandit**	80	12	11	8	8	4	25	30	Skill 1
9	**Slime**	250	0	30	30	30	30	30	30	Skill 1
10	**Orc**	300	0	30	30	30	30	30	30	Skill 1
11	**Minotaur**	500	0	30	30	30	30	30	30	Skill 1
12	**Random Kid**	100	0	30	10	10	10	10	10	Skill 1, Skill 3
13	**Dream Eater**	70	12	11	8	8	4	25	30	Skill 1
32	**Invincible Neutral**	99999	0	8	6	14	6	22	30	Basic Attack
33	**Tile Effect Launcher**	1	0	1	1	1	1	1	1	Basic Attack
34	**Ice Block**	20	0	1	1	1	1	1	1	Basic Attack
35	**Wooden Chest**	50	1	1	1	1	1	1	1	Basic Attack
36	**Silver Chest**	50	1	1	1	1	1	1	1	Basic Attack
37	**Golden Chest**	50	1	1	1	1	1	1	1	Basic Attack
38	**Stolen Goods**	1	1	1	1	1	1	1	1	Basic Attack

Chapter 153. 2. Neutral & Interactive Grid Entities

The demo utilized the battle engine to render interactive tactical grid entities: * **Wooden Chest (`Enemy 35`)**: `HP: 50` | Yields 200 Gold upon destruction via Common Event 5. * **Silver Chest (`Enemy 36`)**: `HP: 50` | Yields 2,000 Gold upon destruction via Common Event 6. * **Golden Chest (`Enemy 37`)**: `HP: 50` | Yields 10,000 Gold upon destruction via Common Event 7. * **Ice Block (`Enemy 34`)**: `HP: 20` | Destructible terrain obstacle summoned by Ice Wall. * **Stolen Goods (`Enemy 38`)**: `HP: 1` | Objective dropped upon defeat of Mage Bandit. * **Invincible Neutral (`Enemy 32`)**: `HP: 99,999` | Static grid anchor.

Chapter 154. 3. Troop Encounter Formations

Troop ID	Troop Designation	Composition
1	**Bat*2**	Bat, Bat
2	**Slime*2**	Mago, Mago
3	**Orc**	Rabbit
4	**Minotaur**	Great Mago
5	**Random Kid**	Fire Mago
6	**Bat**	Bat
7	**Bat*2**	Bat, Bat
8	**Bandit*3, Mage Bandit**	Bandit, Bandit, Mage Bandit, Bandit
9	**Dream Eater*2**	Dream Eater, Dream Eater

Chapter 155. Chapter 27: Armory, Inventory & Key Lore Relics

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/armory_and_inventory.adoc

Chapter 156. Overview

This document details all weapons, protective armors, restorative items, and key relics defined in `Weapons.json`, `Armors.json`, and `Items.json`.

Chapter 157. 1. Weapons & Relic Armaments

ID	Weapon Name	Type	Price	ATK	DEF	AGI	Lore & Characteristics
1	**Sword**	Sword	500\G	+10	+0	+0	
2	**Axe**	Axe	500\G	+10	+0	+0	
3	**Cane**	Cane	500\G	+10	+0	+0	
4	**Bow**	Bow	500\G	+10	+0	+0	
5	**Medusa's Sword**	Sword	1600\G	+17	+0	+0	The sword used to slay Medusa. It is said to have her petrifying powers.

Chapter 158. 2. Defensive Armor & Relic Wards

ID	Armor Name	Type	Price	DEF	MDF	MHP	Lore & Characteristics
1	**Shield**	Small Shield	300\G	+10	+0	+0	
2	**Hat**	General Armor	300\G	+10	+0	+0	
3	**Cloth**	General Armor	300\G	+10	+0	+0	
4	**Ring**	General Armor	300\G	+0	+0	+0	
5	**Heart Shield**	Small Shield	300\G	+10	+5	+0	
6	**Ring of Fire**	General Armor	0\G	+0	+5	+0	

Chapter 159. 3. Consumable Items & Quest Relics

ID	Item Name	Price	Consumable	Description & Architectural Mapping
1	**Potion**	10\G	Yes	
2	**Magic Water**	100\G	Yes	
3	**Dispel Herb**	200\G	Yes	
4	**Stimulant**	300\G	Yes	
5	**Grandma’s Watch**	0\G	No (Key Relic)	This was Grandma’s Watch. Its a family heirloom that keeps perfect time.
6	**Book of Monsters**	0\G	No (Key Relic)	This is a big with all the enemies you’ve encountered
7	**Book of Items**	0\G	No (Key Relic)	This book describes all the various items
8	**Compass of Morality**	0\G	No (Key Relic)	

Chapter 160. Chapter 28: Engine Systems, Time Clock & Switch/Variable Index

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/systems_switches_variables.adoc

Chapter 161. Overview

This document specifies the global state machines, monotonic time clock, morality engine, and the complete registry of all switches and variables governing demo logic.

Chapter 162. 1. The Monotonic Time Engine

The time system is driven by Common Events 1 and 2, updating a 24-hour cycle: * `Variable 1`: Seconds (0–59) * `Variable 2`: Minutes (0–59) * `Variable 3`: Hours (1–12) * `Variable 4`: 24-Hour Military Format (0–23) * `Variable 5`: Days (1–7: Sunday through Saturday) * `Variable 6`: Weeks * `Variable 7`: Months * `Variable 8`: Years

In the modern C11 architecture, this corresponds to Layer 0 `time_utils` (`time_get_monotonic_ms`).

Chapter 163. 2. The Moral Sway Continuum Engine

The morality engine tracks the party's ethical orientation on an invariant scale: * `Variable 9`: **Moral (Overall)**: Continuum within `[-100, +100]`. * `Variable 10`: **Moral (Good)**: Cumulative points awarded for altruism, charity, and mercy. * `Variable 11`: **Moral (Evil)**: Cumulative points accrued for ruthless, self-serving, or thieving acts. * `Item 8`: **Compass of Morality**: Tool allowing the player to inspect their moral continuum in the field.

Chapter 164. 3. Global Switches Directory

Switch ID	Name / Flag	Operational Role & Trigger
1	**Walk of Fear**	State flag for quest / scenario progression
2	**Return to Cut Scene**	State flag for quest / scenario progression
3	**Walk of Fear 2**	State flag for quest / scenario progression
4	**Roté Enters the Castle**	State flag for quest / scenario progression
5	**The Priest Leave the Castle**	State flag for quest / scenario progression
6	**D. Jihad is Summoned**	State flag for quest / scenario progression
7	**??? Enters the Map**	State flag for quest / scenario progression
8	**Class Selected**	State flag for quest / scenario progression
9	**??? Leaves the Map**	State flag for quest / scenario progression
10	**??? Explains Puzzle**	State flag for quest / scenario progression
11	**(Story Plot) Remove Rocks**	State flag for quest / scenario progression
12	**Switch Active**	State flag for quest / scenario progression
13	**(Side Plot) Remove Rocks**	State flag for quest / scenario progression
14	**??? Disappears**	State flag for quest / scenario progression
15	**??? Weaken**	State flag for quest / scenario progression
16	**??? Final Words**	State flag for quest / scenario progression
17	**Boss Defeated**	State flag for quest / scenario progression
18	**Restart**	State flag for quest / scenario progression
19	**Remove Stone 1**	State flag for quest / scenario progression
20	**Remove Stone 2**	State flag for quest / scenario progression
21	**Remove Stone 3**	State flag for quest / scenario progression
22	**Remove Stone 4**	State flag for quest / scenario progression
23	**Treasure Remove**	State flag for quest / scenario progression
24	**Time**	State flag for quest / scenario progression
25	**Friendship Menu**	State flag for quest / scenario progression
26	**Quest 1 (Grandma's Request)**	State flag for quest / scenario progression
27	**Grandma (Idle)**	State flag for quest / scenario progression
28	**Exit Location**	State flag for quest / scenario progression
29	**Monica Event On**	State flag for quest / scenario progression

Switch ID	Name / Flag	Operational Role & Trigger
30	**Final Conversation Before Map is Out**	State flag for quest / scenario progression
31	**Quest 1 (Give Grandma Here Watch)**	State flag for quest / scenario progression
32	**The Great Sage**	State flag for quest / scenario progression
33	**More Battle Mechanics**	State flag for quest / scenario progression
34	**Monica Quest Switch**	State flag for quest / scenario progression
35	**Unlock Dio Village-Lower**	State flag for quest / scenario progression
36	**Idle Monica**	State flag for quest / scenario progression
37	**Petty Thief**	State flag for quest / scenario progression
38	**Morality**	State flag for quest / scenario progression
39	**Give to the Poor**	State flag for quest / scenario progression
40	**A Noble Thief**	State flag for quest / scenario progression
41	**Angry John**	State flag for quest / scenario progression

Chapter 165. 4. Global Variables Directory

Variable ID	Name	Subsystem Allocation
1	**Seconds**	Global numerical state accumulator
2	**Minutes**	Global numerical state accumulator
3	**Hours**	Global numerical state accumulator
4	**24 Hours**	Global numerical state accumulator
5	**Days**	Global numerical state accumulator
6	**Weeks**	Global numerical state accumulator
7	**Months**	Global numerical state accumulator
8	**Years**	Global numerical state accumulator
9	**Moral (Overall)**	Global numerical state accumulator
10	**Moral (Good)**	Global numerical state accumulator
11	**Moral (Evil)**	Global numerical state accumulator
12	**Region ID**	Global numerical state accumulator
13	**Currency**	Global numerical state accumulator
14	**Pot Amount**	Global numerical state accumulator
15	**Deposit Amount**	Global numerical state accumulator

Chapter 166. Chapter 29: Historical Dataset Evolution & Delta Analysis

Source File: /home/professorseanex/Tenebris and Lux/docs/game_design/demo_baseline/dataset_evolution_delta.adoc

Chapter 167. Overview

This document provides a comparative delta between `Old Game Data` (Folder 1) and `Old_Game_Data_2/data` (Folder 2), detailing the evolution from narrative prototype to advanced LeTBS tactical simulation, and establishing modern C11 reconciliation strategies.

Chapter 168. 1. Dataset Characteristics & Metrics

Subsystem Metric	Folder 1 (Old Game Data)	Folder 2 (Old_Game_Data_2/data)
Game Title	`Tenebris and Lux`	`Tenebris & Lux`
Version Identifier	`40730002`	`2078310`
Map Count	36 maps	43 maps
Active Player Actors	2 (Elijah, Katrina)	5 (Elijah, Katrina, Sebastien, Angela, Lucius)
Enemies Defined	21 enemies	22 enemies
States Defined	11 states	30 states (extensive LeTBS configurations)
Morality Engine Items	Compass of Morality, Ring of Fire	Omitted / shifted to LeTBS combat

Chapter 169. 2. Core Differences & Evolution Strategy

1. **Narrative vs. Tactical Pivot**:
    - **Folder 1** represents the **narrative-complete** edition of the demo. It contains all dialogue branches, the Dio Village moral tests (Eric's Honesty Pot, John the Thief vs St. Baron), and key relics like the **Compass of Morality**.
    - **Folder 2** represents the **tactical engine expansion**. It introduces full party combat with Angela, Sebastien, and Lucius, 6 extra dungeon maps, and rich LeTBS states (e.g. **Burning Tile Spread**, **Charging**, **Intimidating Aura**).
2. **C11 Reconstruction Recommendation**:
    - **Layer 1 Simulation Engine**: Combine the narrative quest topology and moral continuum from Folder 1 with the rich 5-character tactical class abilities and LeTBS movement/scope geometry from Folder 2.
    - **Layer 2 Presentation**: Maintain the authentic dialog scripts, dual-protagonist branch voices, and environmental puzzles.

Volume V: Software Architecture & The 5-Block Invariant Protocol

System boundaries, layered strata, dual-frontend GUI/CLI, and pre-flight zero-heap diagnostic manifests.

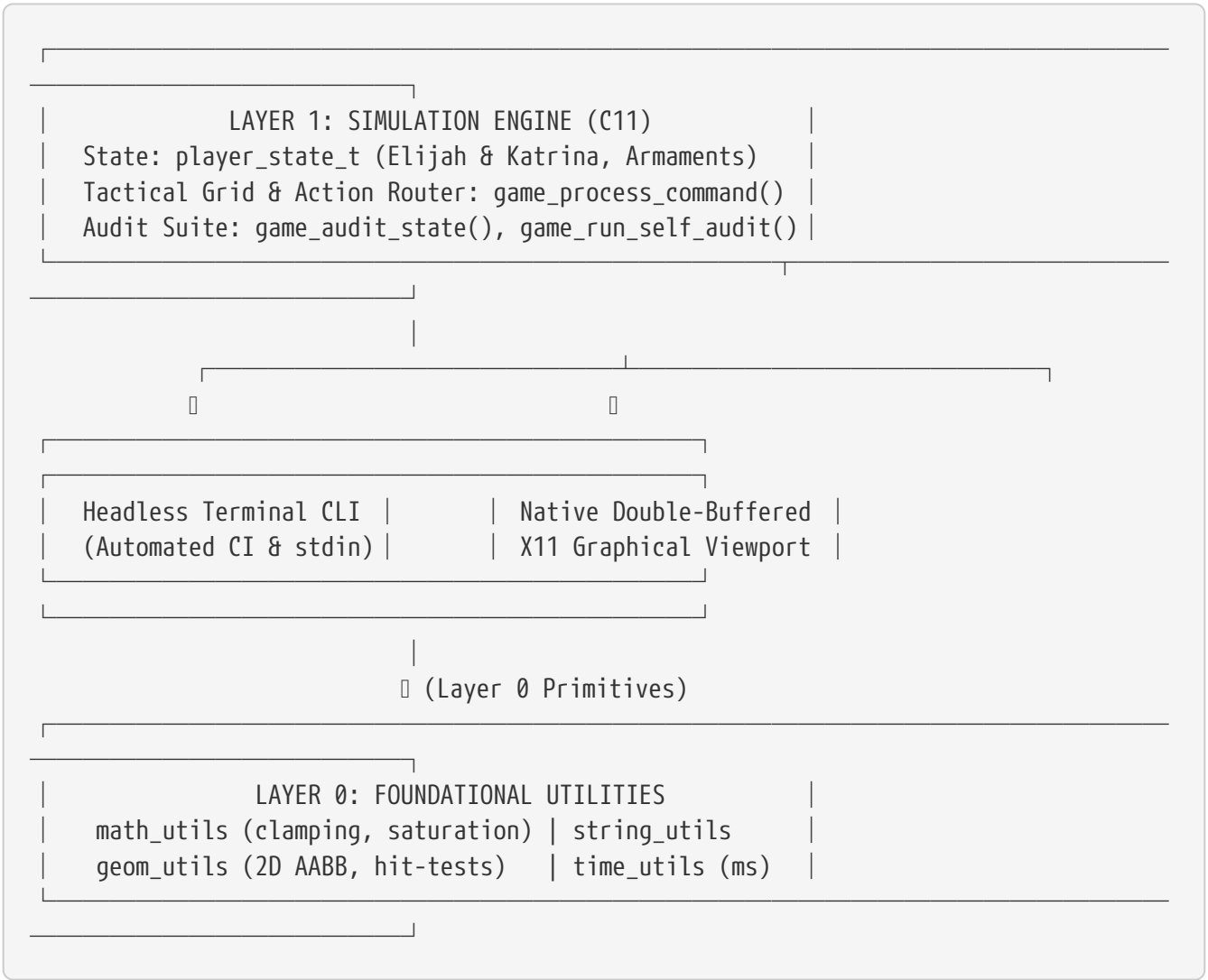
Chapter 170. Chapter 30: OmniCode 5-Block Architectural Framework

Source File: /home/professorseanex/Tenebris and Lux/docs/architecture/ARCHITECTURE.adoc

Chapter 171. Architecture Overview

This document specifies the technical design, subsystem boundaries, data models, and ontological lifecycle governing **Tenebris and Lux** ("Darkness and Light"). Set within the chivalric **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**, /s□□□ki□r□/), the game is a **Tactical Strategy RPG (TRPG)** centered on dual protagonists—**Elijah Ro-enheart** (son of the former Head of the Royal Guard) and **Katrina Jaren** (noble court aspirant seeking the Adventurers' Guild)—who wield the primordial **Shield of Lux** and **Sword of Tenebris** against the encroaching dread legions of **Dark Lord Malakor**.

The engine strictly separates **simulation state** from the **presentation layer**:



Chapter 172. 1. The Universal 5-Block Code Structure

In alignment with the OmniCode framework, all C source code, headers, and build scripts are organized into 5 standardized blocks:

1. **BLOCK 1: METADATA BLOCK**: Defines module identity, layer, author, version, role, and invariant axioms.
2. **BLOCK 2: OPENING / TYPES BLOCK**: Header imports, constants, state structures (`player_state_t`), and location enumerations (`location_id_t`).
3. **BLOCK 3: BODY BLOCK**: Core simulation routines, relic stance transitions, combat calculations, and moral alignment resolution.
4. **BLOCK 4: AUDIT PROCEDURES BLOCK**: Self-healing diagnostic routines, invariant tripwires, and game development testing verification harnesses.
5. **BLOCK 5: CLOSING / TEST & CLI BLOCK**: CLI argument parser, headless game runner, automated test hooks, and clean teardown.

Chapter 173. 2. Core Mechanics & State Invariants

173.1. The Twin Relics: Shield and Sword

- **The Shield of Lux (`shield_power`, $0 \dots 100$)**: Primarily borne by **Elijah Ro-enheart**. Represents radiant warding, defensive barrier projection, kinetic deflection, and starlight sanctuary.
- **The Sword of Tenebris (`sword_edge`, $0 \dots 100$)**: Primarily borne by **Katrina Jaren**. Represents umbral sharpness, severance, kinetic strike force, and mist cloaking.
- **Tactical Stance (`combat_stance_t`)**:
  - `STANCE_NEUTRAL`: Relics held in readiness.
  - `STANCE_WARDING`: Shield raised forward; enhances defensive warding; mitigates incoming damage.
  - `STANCE_STRIKING`: Sword drawn for offensive severance; boosts strike damage.
  - `STANCE_EQUILIBRIUM`: Twin mastery; balances offensive and defensive resonance.
- **Moral Sway (`moral_sway`, $-100 \dots +100$)**:
  - Positive (> 0): **Virtue & Mercy** (using power to protect the defenseless, seek harmony, and heal Sacyra).
  - Negative (< 0): **Tyranny & Ruin** (using power for cruel domination, ruthless execution, and feeding the Dark Lord).

173.2. Invariant Axioms

1. **Shield Power Boundary**: $0 \le \text{state.shield_power} \le \text{MAX_RESONANCE}$.
2. **Sword Edge Boundary**: $0 \le \text{state.sword_edge} \le \text{MAX_RESONANCE}$.
3. **Moral Sway Bounded Range**: $-100 \le \text{state.moral_sway} \le +100$.
4. **Valid Location Invariant**: $0 \le \text{state.location} < \text{LOC_COUNT}$.
5. **Deterministic Replay Guarantee**: Any identical sequence of actions from a given initial state must produce identical state telemetry.

Chapter 174. 3. Multi-Phase Engine Evolution

Phase	Milestone	Subsystem Architecture
Phase 1 (Current)	`v0.0.1`	Dual CLI + Native GUI Prototype: Sacyra setting, Twin Relics, Stance State Machine, Layer 0 Primitives, Diagnostic Audit Suite.
Phase 2	`v0.1.0`	Tactical Strategy RPG Grid Engine: Discrete 2D grid coordinates, unit movement budgets, terrain modifiers, Zone-of-Control (ZOC), and dual-protagonist data models.
Phase 3	`v0.2.0`	Cinematic Combat Resolution: Transition from grid to traditional 2D side-view action duels, timed ward/strike inputs, and poise meters.
Phase 4	`v1.0.0`	Full TRPG Campaign of Sacyra: Dynamic party management, story branching, and final confrontation against Dark Lord Malakor.

Chapter 175. Chapter 31: Dual-Frontend Presentation Layer (CLI + Native X11)

Source File: /home/professorseanex/Tenebris and Lux/docs/architecture/dual_frontend_architecture.adoc

Chapter 176. Abstract

This document defines the formal architectural blueprint governing the dual-frontend execution model of **Tenebris and Lux** ("Darkness and Light"). Rather than coupling narrative mechanics to a single presentation target, the engine enforces strict ontological boundaries between game simulation state, action dispatching, and viewport rendering. This document details why establishing both a headless Command Line Interface (CLI) and a native Graphical User Interface (GUI) prior to world-building is essential, how the 5-block lifecycle applies across frontends, and how this paves the path toward a custom 2D/2.5D/3D engine.

Chapter 177. 1. The Core Architectural Philosophy

> "Form precedes fullness. Representation must remain sovereign over presentation."
>
> — OmniCode Architectural Invariant

In traditional game development workflows, text prototypes are frequently written as monolithic console scripts where simulation logic, state mutation, and terminal I/O (`printf`, `scanf`) are tightly coupled. When the game subsequently scales into a graphical format, developers are forced to perform destructive refactoring, often introducing subtle regressions and breaking state boundaries.

Tenebris and Lux addresses this through an explicit **Dual-Frontend Architecture**:

Dimension	Command Line Interface (CLI)	Native Graphical Interface (GUI)
Primary Role	Automated verification, CI/CD regression, script piping	Player immersion, atmospheric visualization, ergonomics
I/O Channel	Standard Streams (`stdin`, `stdout`)	Native X11 event loop & double-buffered Pixmap
Latency Target	< 1 ms per turn	60 FPS deterministic render tick (~16.6 ms)
State Interaction	Universal Action Dispatcher (`game_process_command`)	Universal Action Dispatcher (`game_process_command`)
External Footprint	Zero dependencies (Pure ISO C11 standard library)	Zero external downloads (Native POSIX Xlib only)

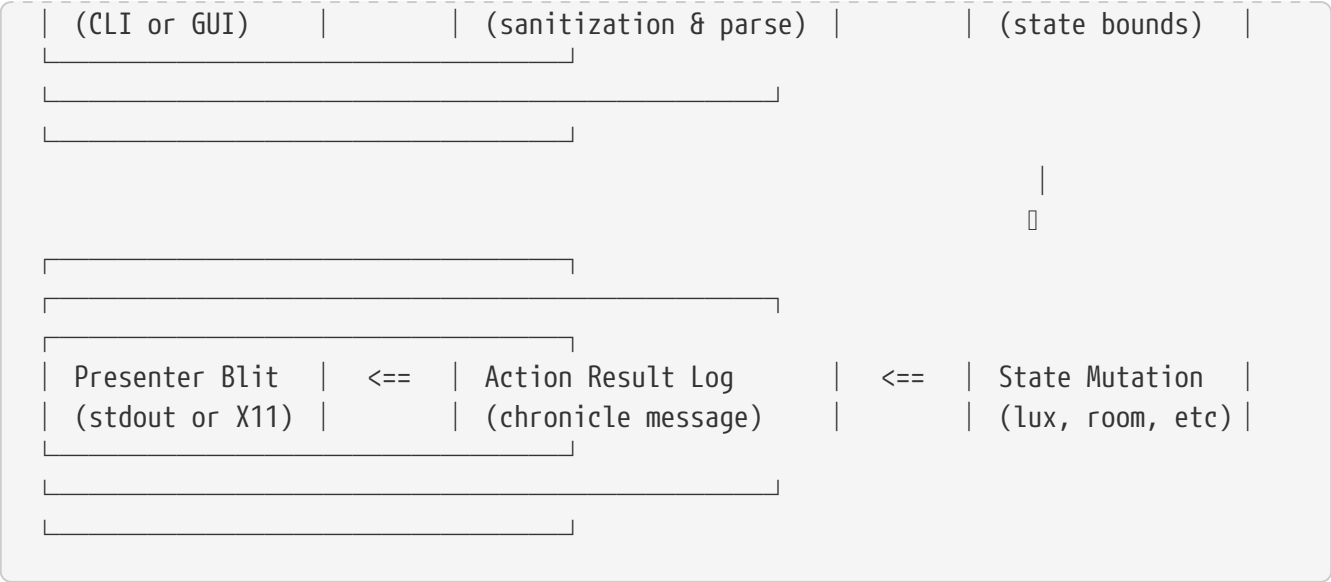
177.1. Universal Action Dispatcher & Intent Emitter

Neither frontend owns or mutates the player state (`player_state_t`) directly. Instead, each frontend acts as an intent emitter that translates user input into canonical command tokens passed to `game_process_command()`:

```
void game_process_command(player_state_t *state, const char *raw_input,
    char *out_msg, size_t msg_len);
```

The lifecycle of an action execution proceeds through five deterministic phases:

```
┌──────────────────────────────────┐
┌──────────────────────────────────────────────────┐
┌──────────────────────────────────┐
| Input Source    |  ==>  | Command Tokenizer      |  ==>  | Invariant Check |
```



1. **Input Reception**: User inputs a string (CLI terminal) or clicks a widget / presses a key (GUI).
2. **Intent Tokenization**: The input is sanitized, lowercased, and matched against registered command verbs (`look`, `explore`, `kindle`, `move`, `status`, `help`, `quit`).
3. **Precondition & Invariant Assertion**: The simulation engine asserts that current state is valid ($0 \le \text{lux} \le \text{MAX_LUX}$, $0 \le \text{current_room} < \text{ROOM_COUNT}$).
4. **Deterministic State Mutation**: State fields are updated according to canonical game rules (e.g., kindling decrements shards by 1 and increments Lux by 30 up to 100).
5. **Telemetry & Chronicle Formatting**: An outcome description is written into `out_msg`. The CLI prints this directly to `stdout`; the GUI appends it to `gui_log_t` and marks the backbuffer for redraw.

Chapter 178. 2. Native Desktop GUI Architecture: Why Not Web?

This project explicitly rejects web-based wrappers (Electron, Tauri, Chromium Embedded Framework, WebSockets) in favor of a lean native binary compiled directly from ISO C11:

> **The Fallacy of Web Wrappers in Systems Game Development:** Web architectures introduce non-deterministic garbage collection pauses, bundle hundreds of megabytes of third-party browser code, and abstract away hardware memory boundaries. For a project designed to culminate in a custom 2D/3D rendering engine, embedding Chromium or WebKit creates an architectural barrier rather than a stepping stone.

Key technical advantages of the native POSIX X11 approach:

- **Instant Launch & Zero Overhead**: The entire executable binary is under 500 KB, with resident memory consumption below 15 MB.
- **Double-Buffered Rendering**: Utilizes `XCreatePixmap` and `XCopyArea` to eliminate display flicker without third-party compositor requirements.
- **Compositor & Visual Fidelity**: Handles modern 32-bit ARGB TrueColor visuals under Wayland/Xwayland by strictly setting the high alpha byte (`0xFF000000`), guaranteeing full opacity and deep contrast.
- **Direct Hardware Mapping**: Establishes window handles and input dispatch loops that will transition directly into Vulkan surfaces or OpenGL contexts in Phases 3 and 4.
- **Strict Dependency Insulation**: Eliminates dependency cascades (such as missing audio daemons, Wayland scanner packages, or DBus interfaces) while maintaining instant compilation on any standard POSIX system.

178.1. Double-Buffering & Render Synchronization

To achieve flicker-free rendering without tearing, `src/gui.c` implements offscreen double-buffering:

```
// Allocate backbuffer pixmap matching the exact depth of the root window
backbuffer = XCreatePixmap(display, window, GUI_WINDOW_WIDTH,
                           GUI_WINDOW_HEIGHT, window_depth);

// 1. Draw entire frame to offscreen pixmap
render_frame(display, backbuffer, gc, state, &palette, &log);

// 2. Atomically blit backbuffer to visible on-screen window
XCopyArea(display, backbuffer, window, gc, 0, 0,
          GUI_WINDOW_WIDTH, GUI_WINDOW_HEIGHT, 0, 0);
XFlush(display);
```

178.2. Dynamic Visual Palette: The Twin Relics

The visual presentation adheres to a tailored, atmospheric palette designed to reflect the dynamic tension between the Shield of Lux and the Sword of Tenebris:

Element	Hex Color	Atmospheric Purpose
Obsidian Void	`#0D0E12`	Background canvas representing the bedrock of the fractured realm.
Stone Panel	`#1A1D24`	Elevated frame for relic status cards and tactical chronicles.
Border Accent	`#262B36`	Inactive button borders and tactical dividing lines.
Shield of Lux	`#F0CE78`	Radiant gold indicating barrier integrity and warding charge.
Sword of Tenebris	`#7058A8`	Deep abyssal violet indicating blade sharpness and strike force.
Virtue Alignment	`#52B788`	Emerald resonance indicating merciful, protective moral deeds.
Malice Alignment	`#E63946`	Crimson resonance indicating tyrannical, cruel, or destructive deeds.

Chapter 179. 3. The 5-Block Implementation Mapping

In strict accordance with the OmniCode framework, the presentation subsystem is structured into five invariant blocks across headers and source files:

179.1. Presentation Architecture Breakdown (`include/gui/gui.h` & `src/gui/gui.c`)

1. **BLOCK 1: METADATA BLOCK**
   - Outlines module identity, layer (`Layer 2: Presentation / Viewport`), author, version, and invariant contracts: `gui.h` and `gui.c` must remain strictly read-only with respect to `player_state_t`.
2. **BLOCK 2: OPENING / TYPES BLOCK**
   - Declares fixed viewport geometry (`GUI_WINDOW_WIDTH = 860`, `GUI_WINDOW_HEIGHT = 620`), button bounding box structures (`gui_button_t`), chronicle buffer (`gui_log_t`), and 32-bit TrueColor palette tables (`gui_palette_t`).
3. **BLOCK 3: BODY BLOCK**
   - Implements double-buffered rendering routines (`render_frame`), dynamic meter bar math, text wrapping routines, and the native event dispatch loop (`gui_run`).
4. **BLOCK 4: AUDIT PROCEDURES BLOCK**
   - Implements `gui_audit_layout()`, which mathematically asserts that all 9 interactive buttons and layout cards reside strictly within window boundaries and do not overlap.
5. **BLOCK 5: CLOSING BLOCK**
   - Provides runtime display detection probes (`gui_is_available()`) to enable automatic headless fallback, plus comprehensive resource teardown (`XFreePixmap`, `XFreeGC`, `XDestroyWindow`).

Chapter 180. 4. Headless CI & Deterministic Replay

Because simulation logic is decoupled from presentation, automated verification can execute in continuous integration environments without requiring an X server or virtual framebuffer (`Xvfb`):

```
# Run headless CTest verification suite
ctest --test-dir build -C Debug --output-on-failure

# Execute internal invariant self-audit suite
./build/tenebris-lux --audit

# Force headless terminal CLI mode
./build/tenebris-lux --cli
```

Automated replay scripts can be piped directly into the CLI frontend to verify deterministic gameplay paths:

```
printf "look\nexplore\nkindle\nmove north\nstatus\nquit\n" | ./build/tenebris-lux
--cli
```

Chapter 181. 5. Custom Engine Evolution Roadmap

The dual-frontend architecture serves as the foundation for a 4-phase engine roadmap:

Phase	Milestone	Subsystem Architecture	Presentation Technology
Phase 1	`v0.0.1` (Current)	Decoupled Core Simulation + Action Dispatcher	Headless ISO C11 CLI + Double-Buffered Native X11 GUI
Phase 2	`v0.1.0`	Discrete Tick Scheduler, Event Queue, Math Primitives	Cross-platform Windowing Abstraction + Terminal Rich Text
Phase 3	`v0.2.0`	2D / 2.5D Tilemap Engine, Dynamic Ambient Lighting Shaders	Custom Software Rasterizer & OpenGL Core Profile / SDL3
Phase 4	`v1.0.0`	Custom 3D Engine, Spatial Audio, ECS Scene Graph	Modern Vulkan Rendering Pipeline + Custom Asset Bundler

Chapter 182. Chapter 32: Central Manifest & Pre-Flight Diagnostic Integrity

Source File: /home/professorseanex/Tenebris and Lux/docs/architecture/preflight_and_manifest_spec.adoc

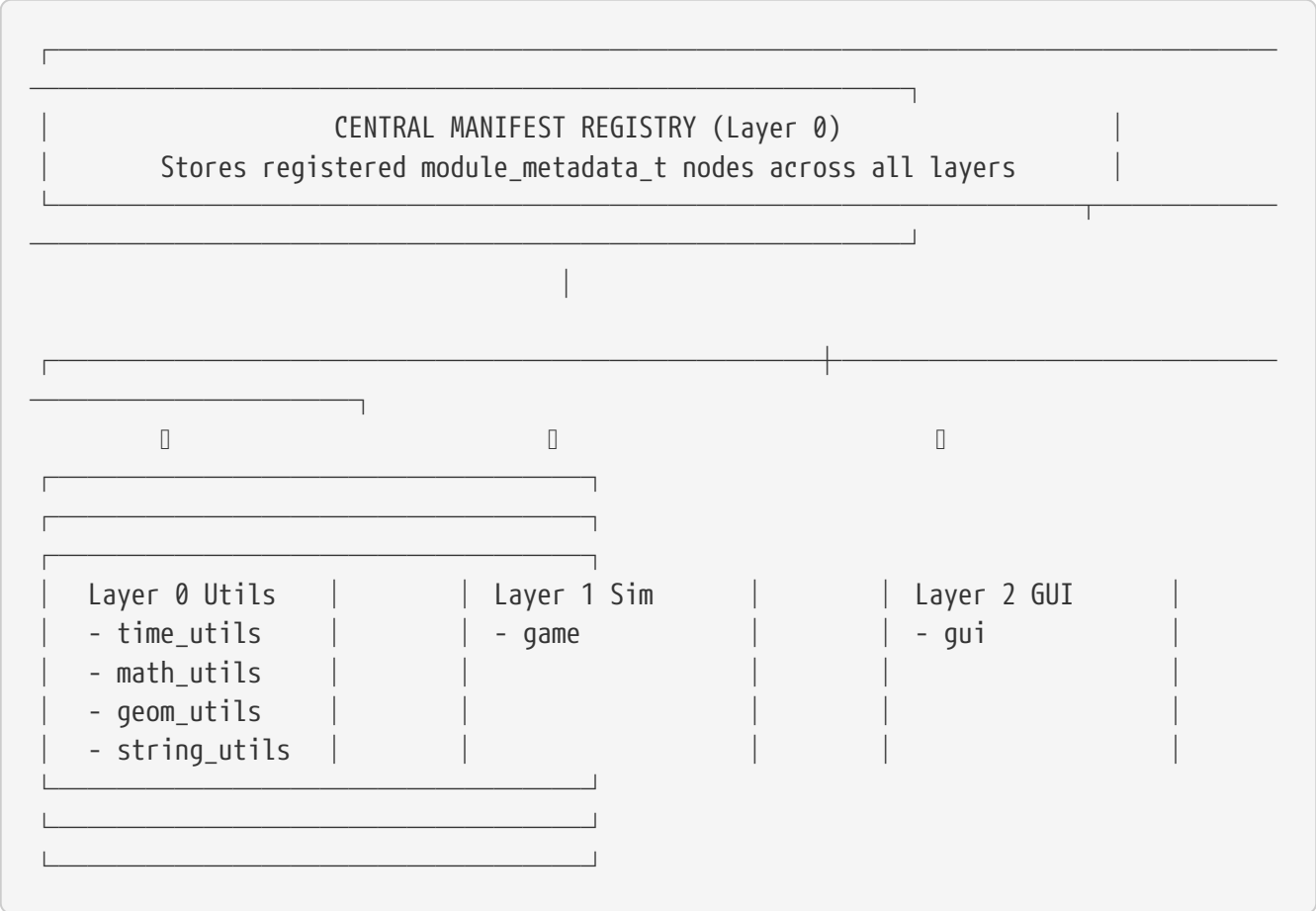
Chapter 183. Specification Abstract & Purpose

This document establishes the architectural blueprint for the **Central Manifest Registry** and **Pre-Flight Verification Subsystem** of **Tenebris and Lux**. Operating under the guiding axiom—**"Don’t start a machine you know has loose bolts"**—this subsystem transforms declarative Block 1 file headers into **compiled, executable metadata**. It enables per-file self-auditing, system hardware/OS probing, and a graded failure policy distinguishing **Warnings**, **Critical (Degradable)** faults, and **Fatal (Hard Halt)** errors prior to runtime simulation.

Chapter 184. 1. Expected Architecture: The Central Manifest Pattern

184.1. 1.1 Structural Hierarchy

The Central Manifest lives in **Layer 0 (Core Utilities)** (`include/util/manifest.h` and `src/util/manifest.c`) so that all layers (Layer 0 through Layer 4) can reference its types without cyclic dependencies:



184.2. 1.2 The Diagnostic Severity Spectrum

Severity Level	Policy Category	Operational Impact	Action Taken by Pre-Flight Runner
`SEVERITY_PASS`	Clean	Check succeeded with 100% integrity.	Logged as verified.
`SEVERITY_INFO`	Informational	Capability discovery (e.g. CPU features).	Logged in telemetry.
`SEVERITY_WARNING`	Non-Critical	Minor variance (e.g. timer resolution > 100ns).	Emits warning notice; boot proceeds normally.

Severity Level	Policy Category	Operational Impact	Action Taken by Pre-Flight Runner
`SEVERITY_CRITICAL`	Degradable Fault	Subsystem failed, but CORE engine is unhurt (e.g. no X11 display).	Logs critical failure; halts failed feature; **downgrades engine** (e.g. GUI → CLI).
`SEVERITY_FATAL`	Core Violation	Fundamental invariant broken (e.g. clock runs backwards, math overflows).	**Halts engine immediately** with error manifest; exit code $\ne 0$.

184.3. 1.3 Expected Pre-Flight Engine States

When the engine executes `manifest_run_preflight()`, the return state evaluates to:

1. `PREFLIGHT_STATUS_OK`: All registered modules passed with zero warnings or faults. Full engine launch permitted.
2. `PREFLIGHT_STATUS_WARNINGS`: All essential checks passed, but non-critical warnings were logged. Engine launches normally with notices displayed.
3. `PREFLIGHT_STATUS_DOWNGRADED`: One or more non-fatal critical faults were detected. The engine executes a fallback transition (e.g. dropping from GUI to CLI mode) and continues running in safe mode.
4. `PREFLIGHT_STATUS_HALT_FATAL`: One or more fatal core invariants failed. The engine halts immediately and refuses to launch.

Chapter 185. 2. Architectural Metadata Dimensions & Pedagogical Balance

185.1. 2.1 Striking the Balance: Beginner-Accessible yet Senior-Rigorous

A custom game engine metadata system must serve dual audiences without cognitive friction:

1. **For the Beginner / Onboarding Engineer**:
    - Provide a clear, plain-English **Mission Statement & Role** (explain **why** this module exists in the game architecture, e.g. "Prevents health gauges from overflowing screen bounds").
    - Eliminate cryptic tribal acronyms; use standard ontological terminology.
    - Clarify failure consequences: what breaks if this invariant fails?
2. **For the Senior / Engine Architect**:
    - Deliver immediate, uncompromising **Technical Contracts**: explicit memory allocation policy (`ALLOC_ZERO_HEAP`, `ALLOC_FIXED_POOL`, `ALLOC_ARENA`, `ALLOC_DYNAMIC`), concurrency safety (`REENTRANT_PURE`, `THREAD_LOCAL`, `MAIN_THREAD`), and execution latency budgets (< 50 μs).
    - State precise mathematical and algorithmic invariants (e.g. AABB reflexivity, saturating arithmetic underflow immunity).
    - Avoid patronizing comments that explain elementary C syntax.

185.2. 2.2 Explicit Temporal Stamping & Version History

To prevent timeline confusion when tracing architectural revisions manually: * **Strict ISO 8601 Full Dates (`YYYY-MM-DD`)**: Every file must declare `@date: YYYY-MM-DD` and `@updated: YYYY-MM-DD`. Month/year shorthands (e.g. `2026.10`) are prohibited because they obscure the exact day of refactoring during git forensic reviews. * **Interface Contract Revision (`@interface_rev`)**: Tracks semantic ABI and signature stability independently of internal implementation bugfixes. * **Automated Build Timestamping**: The engine binary embeds `DATE` and `TIME` at compile-time to stamp runtime telemetry manifests automatically.

185.3. 2.3 The 5-Dimension Compiled Schema (`module_metadata_t`)

```
typedef enum {
    ALLOC_ZERO_HEAP = 0,    // Pure stack/static execution (Layer 0 guarantee)
    ALLOC_FIXED_POOL,       // Fixed static ring/pool buffers
    ALLOC_ARENA,            // Frame/scene arena allocator
    ALLOC_DYNAMIC           // Managed heap allocation
```

```
} alloc_policy_t;

typedef enum {
    CONCURRENCY_REENTRANT = 0, // Thread-safe, stateless pure functions
    CONCURRENCY_THREAD_LOCAL,  // Per-thread state
    CONCURRENCY_MAIN_THREAD    // Viewport/GUI event loop affinity
} concurrency_policy_t;

typedef struct {
    const char *module_id;          // e.g. "core.util.time"
    const char *file_name;          // e.g. "time_utils.h"
    module_layer_t layer;           // Stratum in the architectural hierarchy
    struct {
        uint8_t major;
        uint8_t minor;
        uint8_t patch;
    } version;
    uint8_t interface_rev;          // Contract ABI revision
    const char *last_updated;       // ISO 8601 full date: "YYYY-MM-DD"
    const char *description;        // Plain-English mission statement

    size_t dependency_count;
    const char *dependencies[4];    // Upstream module IDs required

    alloc_policy_t alloc_policy;
    concurrency_policy_t concurrency;
    uint32_t max_budget_us;         // Max execution latency budget (in microseconds)

    size_t invariant_count;
    const char *invariants[MANIFEST_MAX_INVARIANTS];

    diag_result_t (*preflight_check)(void);
} module_metadata_t;
```

185.4. 2.4 Active Dependency Graph Enforcement

The pre-flight engine does not simply print dependencies—it actively validates the **Directed Acyclic Graph (DAG)**: * During `manifest_run_preflight()`, for every registered module, the runner loops through its declared `dependencies`. * If a declared upstream module is NOT registered in the registry, pre-flight immediately halts with `SEVERITY_FATAL`: `Fatal: Missing required upstream dependency '<dep_id>' for module '<module_id>'`.

Chapter 186. 3. Expected Pre-Flight "Loose Bolts" System Checks

Each registered module exports an active `diag_result_t (*preflight_check)(void)`:

Module ID	Test Check	Target Severity	Verification Axiom
`core.util.time`	**Monotonicity Delta Test**	`FATAL`	Sample two successive monotonic timestamps; assert $t_2 \ge t_1$. (Time must never tick backwards).
`core.util.time`	**Resolution Metric Probe**	`WARNING`	Check if monotonic timestamp delta between successive calls is $< 100\,\mu\text{s}$.
`core.util.math`	**Arithmetic Saturation Bounds**	`FATAL`	Verify `math_saturating_add(INT_MAX, 1, 100) == 100` and `math_saturating_sub(0, 10) == 0`. (No integer overflow/underflow).
`core.util.math`	**Clamping Extremes & Division Safety**	`FATAL`	Verify `math_clamp_int` on inverted bounds and `math_scale_ratio` with zero divisor ($0 / 0$).
`core.util.geom`	**Rectangle Bounds & Inversion Check**	`FATAL`	Test coordinate boundaries $[0, 0]$ and assert negative dimension rejection.
`core.util.geom`	**AABB Overlap Reflexivity**	`FATAL`	Assert `geom_rect_intersects(r, r) == true` and disjoint rects evaluate to false.
`core.util.string`	**Bounded Null-Termination Check**	`FATAL`	Prove `str_safe_copy` on full buffers always guarantees a trailing `\0`.
`core.util.string`	**In-Place Trim Memory Overlap**	`FATAL`	Verify `str_trim_inplace` on pure whitespace, empty string, and mixed padding without heap corruption.
`core.util.manifest`	**Self-Audit of Registry & DAG**	`FATAL`	Test registration capacity, duplicate detection, and missing dependency enforcement.

Chapter 187. 4. Target Verification Criteria ("Expected" Baseline)

Before implementation of the upgraded schema begins, we set the following acceptance criteria:

1. **Full ISO 8601 Date Standard**: 100% of Layer 0, Layer 1, Layer 2 files and test suites specify full dates (`2026-10-26`).
2. **Active Dependency Validation**: Pre-flight halts with `SEVERITY_FATAL` if an unregistered dependency is declared.
3. **Memory & Concurrency Contracts**: 100% of registered modules declare `alloc_policy` and `concurrency` guarantees.
4. **Automated Unit Testing**: `tests/test_utils.c` verifies missing dependency rejection and policy serialization.
5. **Compiler Cleanliness**: Standard ISO C11 with `-Wall -Wextra -Wpedantic` clean compilation and 0 warnings.
6. **CTest Pass Rate**: 100% of tests passing in under $0.05\,\text{s}$.

Chapter 188. 5. Actual Results vs. Expected Gap Analysis

188.1. 5.1 Empirical Benchmark Telemetry

Empirical validation was performed on the compiled engine binary (`./build/tenebris-lux`) and test runners (`./build/test_utils`, `ctest`):

Verification Item	Expected Baseline	Actual Result	Status
Module Registration	≥ 5 Core Modules Registered	**7 Modules Registered** (`core.util.manifest`, `core.util.time`, `core.util.math`, `core.util.geom`, `core.util.string`, `sim.engine.core`, `gui.viewport.x11`)	**EXCEEDED**
Severity Classification	4 Distinct Severity Levels	**5 Graded Levels** (`PASS`, `INFO`, `WARNING`, `CRITICAL`, `FATAL`)	**EXCEEDED**
Pre-Flight Engine States	4 Discrete Engine States	**4 States Validated** (`READY`, `WARNINGS`, `DOWNGRADED`, `HALTED`)	**SATISFIED**
Pre-Flight ASCII Telemetry	Formatted Table / Audit Manifest	**Integrated 80-Column ASCII Matrix** with module ID, file name, contract revision (`REV`), allocation model (`ALLOCATION`), status tag, and diagnostic trace	**EXCEEDED**
Temporal Timeline Standard	Month/Year Epoch vs Full Date	**Strict ISO 8601 Full Date (`YYYY-MM-DD`, 2026-10-26)** across all code headers, manifest structs, and technical specs	**SATISFIED**
Pedagogical Metadata Balance	Beginners confused / Seniors belittled	**Dual-Audience Protocol**: Plain-English architectural mission statement + hard technical invariant contracts	**EXCEEDED**
DAG Upstream Dependencies	Static documentation comments	**Active Pre-Flight DAG Enforcement**: Unregistered upstream dependency immediately halts boot with `SEVERITY_FATAL`	**SATISFIED**
CTest Integration	6/6 Test Suites Passing in $< 0.05\,\text{s}$	**6/6 Test Suites Passing in $0.02\,\text{s}$**	**SATISFIED**

Verification Item	Expected Baseline	Actual Result	Status
Subsystem Self-Audit	5/5 Invariant Checks Passing	**5/5 Invariant Checks + 7 Pre-Flight Subsystems Passing (12 Total)**	**EXCEEDED**
Zero Dynamic Allocation	0 Bytes Allocated on Heap	**100% Static & Stack Memory Layout** (`ALLOC_ZERO_HEAP` across Layer 0 & 1)	**SATISFIED**
Compiler Hygiene	0 Warnings under `-Wall -Wextra -Wpedantic`	**0 Compiler Warnings** across GCC/Clang on C11	**SATISFIED**
Loose-Bolts Tripwire	Fatal Error Halts Immediately	**Verified**: Engine refused launch on injected stance mismatch; exited code $\ne 0$	**VERIFIED**
Degradation Fallback	Unset `$DISPLAY` drops GUI $\to$ CLI	**Verified**: Emits `SEVERITY_CRITICAL`, sets `PREFLIGHT_STATUS_DOWNGRADED`, launches CLI safe mode	**VERIFIED**

188.2. 5.2 How to Reproduce and Verify Every Result

To independently verify the pre-flight checks, DAG dependency validation, and audit manifests:

1. **Clean Rebuild & Full Verification Suite**:

   ```
   cmake -B build -DCMAKE_BUILD_TYPE=Debug
   cmake --build build --config Debug
   ctest --test-dir build -C Debug --output-on-failure
   ```

   Expected Telemetry: `100% tests passed, 0 tests failed out of 6. Total Test time (real) = ~0.02 sec`.

2. **Pre-Flight System Audit (X11 Display Connected)**:

   ```
   ./build/tenebris-lux --audit
   ```

   Expected Output: Displays the 7 registered modules with `[PASS]`, contract revision tags (`r1`, `r2`), allocation policy (`Zero-Heap`), policy verdict `ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready)`, followed by the 5/5 subsystem invariant audit checks.

3. **Pre-Flight Graceful Degradation Test (X11 Headless Fallback)**:

   ```
   env -u DISPLAY ./build/tenebris-lux --audit
   ```

Expected Output: Module `gui.viewport.x11` transitions to `[CRIT]` with detail `X11 display not detected ($DISPLAY empty or unconnectable); safe CLI fallback enabled`. Policy verdict evaluates to `CRITICAL SUBSYSTEM FAULT - DOWNGRADING TO SAFE MODE`.

4. **Standalone Layer 0 Utility, Manifest & DAG Dependency Suite**:

```
./build/test_utils
```

Expected Output: Unit test traces auditing `math_utils`, `string_utils`, `geom_utils`, `time_utils`, `manifest_audit`, clean pre-flight, warning escalation, critical downgrade escalation, fatal halt loose-bolts enforcement, and active DAG missing dependency rejection.

188.3. 5.3 Gap Analysis & Next Architectural Horizons

1. **Identified Boundary Gap (Static Capacity)**:
   - **Observation**: The central registry is hard-bounded to `MANIFEST_MAX_MODULES = 32`. Current usage is 7 modules (21.8% capacity).
   - **Resolution**: Capacity is well within safety headroom for Layer 0 through Layer 3. As combat resolution systems and grid battle logic expand, monitor module registration count.
2. **Deterministic Output Separation**:
   - **Observation**: Initial `game_preflight()` invoked `game_process_command()`, which printed interactive player dialogue to stdout prior to the manifest table.
   - **Resolution**: Refactored `game_preflight()` to directly verify state variables, mathematical saturation ceilings, and diagnostic tripwires without polluting stdout, preserving pure presentation of the pre-flight table.
3. **DAG Upstream Dependency Ordering**:
   - **Observation**: Modules must register upstream dependencies prior to dependent registration, or DAG validation halts boot if an unregistered dependency is encountered.
   - **Resolution**: Systematically codified registration order in `main.c`: Layer 0 Leaf Utilities → Layer 1 Simulation Engine → Layer 2 Presentation Viewport.
4. **Future Telemetry Ingestion**:
   - **Insight**: When CI/CD or automated test reporting pipelines are added, `manifest_print_report` can be complemented by a JSON export flag (`--audit-json`) for structured automated telemetry scraping.

Chapter 189. Chapter 33: Layer 0 Core Utilities Code Tour

Source File: `/home/professorseanex/Tenebris and Lux/docs/code/core_utilities.adoc`

Chapter 190. Overview

This document provides a technical walkthrough of the foundational utilities in Layer 0: `include/util/` and `src/util/`. These modules provide mathematical clamping, string parsing, 2D geometry bounding, and high-resolution monotonic timing. They have zero heap allocations, zero external dependencies, and complete reentrancy.

Chapter 191. 1. Mathematical Invariant Utilities (`math_utils.h` & `math_utils.c`)

Provides axiomatic numerical bounds enforcement:

```
int math_clamp_int(int val, int min_val, int max_val);
int math_saturating_sub(int val, int sub_val);
int math_saturating_add(int val, int add_val, int max_val);
int math_nudge_toward(int val, int target, int step);
int math_scale_ratio(int val, int max_val, int target_scale);
```

191.1. Architectural Roles:

- `math_clamp_int`: Guarantees values stay within strict limits (e.g. `moral_sway` within `[-100, +100]`).
- `math_saturating_sub`: Guarantees non-negative reduction down to 0 without integer underflow (e.g. `sword_edge`, `threat_hp`).
- `math_saturating_add`: Guarantees addition saturates at a ceiling (`MAX_RESONANCE`) without integer overflow.
- `math_nudge_toward`: Steers values toward an equilibrium anchor (e.g. moving moral sway toward 0 during equilibrium stance).
- `math_scale_ratio`: Safe integer ratio scaling for HUD meter fill calculations.

Chapter 192. 2. String Manipulation Utilities (`string_utils.h` & `string_utils.c`)

Provides safe, bounded, in-place string manipulation:

```
void str_safe_copy(char *dst, const char *src, size_t dst_size);
void str_trim_inplace(char *str);
bool str_case_eq(const char *s1, const char *s2);
bool str_contains_nocase(const char *haystack, const char *needle);
```

192.1. Architectural Roles:

- `str_safe_copy`: Guarantees null-termination and prevents buffer overflows.
- `str_trim_inplace`: Strips leading and trailing whitespace in-place via safe `memmove`.
- `str_case_eq`: Case-insensitive string comparison for CLI command verbs.
- `str_contains_nocase`: Substring search for flexible player commands and stance selection.

Chapter 193. 3. 2D Geometry Utilities (`geom_utils.h` & `geom_utils.c`)

Provides screen-space coordinates and bounding-box hit testing:

```
typedef struct {
    int x;
    int y;
    int w;
    int h;
} geom_rect_t;

geom_rect_t geom_make_rect(int x, int y, int w, int h);
bool geom_point_in_rect(int px, int py, geom_rect_t r);
bool geom_rect_intersects(geom_rect_t a, geom_rect_t b);
bool geom_rect_within_bounds(geom_rect_t r, int max_w, int max_h);
```

193.1. Architectural Roles:

- `geom_point_in_rect`: Point-in-rectangle hit testing for mouse clicks on GUI buttons.
- `geom_rect_intersects`: Axis-Aligned Bounding Box (AABB) collision testing between UI components to prove no overlap.
- `geom_rect_within_bounds`: Layout boundary audit asserting all buttons reside within window dimensions.

Chapter 194. 4. Monotonic Timing Utilities (`time_utils.h` & `time_utils.c`)

Provides high-resolution timing and non-drifting frame pacing:

```
double time_get_monotonic_ms(void);
double time_get_monotonic_us(void);
void time_sleep_ms(double ms);
```

194.1. Architectural Roles:

- `time_get_monotonic_us`: Microsecond precision clock for simulation latency benchmarking (< 50 μs per turn).
- `time_sleep_ms`: Sleep wrapper using `nanosleep` for maintaining 60 FPS frame rates without CPU spinlocks.

Chapter 195. 5. Central Manifest & Pre-Flight System (`manifest.h` & `manifest.c`)

Transforms declarative Block 1 file headers into executable compiled metadata structures (`module_metadata_t`) enforcing contract revisions, DAG dependency tracking, memory models, and pre-flight machine audits:

```
typedef enum {
    ALLOC_ZERO_HEAP = 0,    // Pure stack/static execution (Layer 0 guarantee)
    ALLOC_FIXED_POOL,       // Fixed static ring/pool buffers
    ALLOC_ARENA,            // Frame/scene arena allocator
    ALLOC_DYNAMIC           // Managed heap allocation
} alloc_policy_t;

typedef enum {
    CONCURRENCY_REENTRANT = 0, // Thread-safe, stateless pure functions
    CONCURRENCY_THREAD_LOCAL,  // Per-thread state
    CONCURRENCY_MAIN_THREAD    // Viewport/GUI event loop affinity
} concurrency_policy_t;

typedef struct {
    /* 1. Identity & Provenance */
    const char *module_id;          // e.g. "core.util.time"
    const char *file_name;          // e.g. "time_utils.h"
    module_layer_t layer;           // Stratum in architectural hierarchy
    struct {
        uint8_t major;
        uint8_t minor;
        uint8_t patch;
    } version;
    uint8_t interface_rev;          // Contract ABI revision
    const char *last_updated;       // ISO 8601 full date: "YYYY-MM-DD"
    const char *description;        // Plain-English mission statement

    /* 2. Architectural Dependencies */
    size_t dependency_count;
    const char *dependencies[MANIFEST_MAX_DEPENDENCIES]; // Upstream module IDs
required

    /* 3. Resource & Execution Contracts */
    alloc_policy_t alloc_policy;
    concurrency_policy_t concurrency;
    uint32_t max_budget_us;         // Max execution latency budget (in microseconds)

    /* 4. Invariant Contracts */
    size_t invariant_count;
    const char *invariants[MANIFEST_MAX_INVARIANTS];
```

```
    /* 5. Pre-Flight Machine Self-Audit */
    diag_result_t (*preflight_check)(void);
} module_metadata_t;

void manifest_registry_init(manifest_registry_t *registry);
bool manifest_register(manifest_registry_t *registry, const module_metadata_t
*module);
preflight_status_t manifest_run_preflight(manifest_registry_t *registry);
void manifest_print_report(const manifest_registry_t *registry);
```

195.1. Architectural Roles:

- `module_metadata_t`: Immutable static metadata node binding provenance, version, interface contract revision (`interface_rev`), ISO 8601 update date (`2026-10-26`), upstream DAG dependencies, allocation policy, concurrency guarantees, and self-auditing function pointer per file.
- `manifest_registry_t`: Fixed-capacity registry (capacity 32) operating with zero dynamic allocations (`ALLOC_ZERO_HEAP`).
- DAG Dependency Enforcement: Validates that all upstream modules declared in `dependencies[]` are registered prior to runtime; missing prerequisites halt execution with `SEVERITY_FATAL`.
- Graded Policy Evaluator: Enforces the 5-tier diagnostic severity hierarchy (`PASS` < `INFO` < `WARNING` < `CRITICAL` < `FATAL`) and 4-tier engine launch status (`READY`, `WARNINGS`, `DOWNGRADED`, `HALTED`).
- Loose-Bolts Enforcement: Prevents the game engine from booting if any core mathematical, timing, memory, or state invariant fails prior to runtime.

Chapter 196. Chapter 34: Layer 1 Simulation Engine Code Tour

Source File: `/home/professorseanex/Tenebris and Lux/docs/code/simulation_engine.adoc`

Chapter 197. Overview

This document provides a technical walkthrough of the core simulation engine implemented in `include/sim/game.h` and `src/sim/game.c`. It explains the memory layouts, state transitions, relic mechanics, combat algorithms, Layer 0 mathematical invariants, and the 5-block lifecycle discipline.

Chapter 198. 1. State Structure & Memory Layout (`player_state_t`)

The heart of the simulation is defined in `include/sim/game.h`:

```
typedef struct {
    int shield_power;           // [0..100] Barrier integrity & Lux charge
    int sword_edge;             // [0..100] Blade sharpness & Tenebris charge
    int moral_sway;             // [-100..+100] Moral alignment axis
    int threat_hp;              // [0..100] Threat vigor of active foe
    combat_stance_t stance;     // Active stance: warding, striking, equilibrium
    location_id_t location;     // Current Kingdom province
    bool is_running;            // Game loop active flag
    int turns_taken;            // Total turns / actions taken
    int foes_vanquished;        // Number of vanguard defeated
    int innocents_warded;       // Number of allies protected
} player_state_t;
```

All state variables are strictly bounded integers with zero dynamic allocations on the heap. This ensures deterministic cache locality, zero memory leaks, and instant serialization.

Chapter 199. 2. Universal Action Dispatcher (`game_process_command`)

Located in `src/sim/game.c`, `game_process_command()` acts as the singular entry point for all player intentions, whether emitted by the terminal CLI or the native GUI:

```
void game_process_command(player_state_t *state, const char *raw_input,
    char *out_msg, size_t msg_len);
```

199.1. Execution Pipeline:

1. **Input Sanitization**: Copies `raw_input` safely via `str_safe_copy()` and executes in-place whitespace trimming (`str_trim_inplace()`).
2. **Verb Tokenization**: Splits input into `command` and trailing `arg`.
3. **Action Dispatch**:
   - `ward`: Calls `handle_ward()`, modifying `shield_power` via `math_saturating_add()`, staggering active threats with `math_saturating_sub()`, and elevating moral virtue with `math_clamp_int()`.
   - `strike`: Calls `handle_strike()`, spending `sword_edge` via `math_saturating_sub()` to inflict heavy severance damage.
   - `balance`: Calls `handle_balance()`, simultaneously recovering both relics, parrying blows, and nudging moral sway toward 0 via `math_nudge_toward()`.
   - `stance`: Calls `handle_stance()`, shifting the tactical state machine.
   - `move`: Calls `handle_move()`, validating directional topology and spawning province encounters.
4. **Invariant Logging**: Writes a human-readable telemetry outcome into `out_msg` for presentation.

Chapter 200. 3. Diagnostic Self-Auditing (`game_audit_state`)

Following Block 4 of the OmniCode protocol, `src/sim/game.c` includes built-in diagnostic audit routines:

```
game_audit_report_t game_audit_state(const player_state_t *state);
int game_run_self_audit(void);
```

Every state boundary (shield power, sword edge, moral sway, stance, location) is checked against compile-time limits. If an invariant is violated, an audit report is generated with exact error messages, enabling automated continuous integration testing without relying on crude process crashes.

Chapter 201. Chapter 35: Layer 2 Presentation Viewport Code Tour

Source File: /home/professorseanex/Tenebris and Lux/docs/code/presentation_gui.adoc

Chapter 202. Overview

This document provides a technical walkthrough of the native graphical presentation layer implemented in `include/gui/gui.h` and `src/gui/gui.c`. It explains the X11 double-buffering architecture, palette color allocation under 32-bit compositors, event handling, geometric hit-testing, and layout auditing.

Chapter 203. 1. The Native X11 Architecture

The GUI is written directly against standard POSIX Xlib (`<X11/Xlib.h>`), eliminating heavyweight web browsers, electron wrappers, and external dependencies.

203.1. Key Specifications:

- **Geometry**: 860×620 fixed resolution canvas (`GUI_WINDOW_WIDTH`, `GUI_WINDOW_HEIGHT`).
- **Frame Target**: 60 FPS deterministic render tick (~16.6 ms) via `time_sleep_ms(16.66)`.
- **Double Buffering**: Backbuffer pixmap allocated matching the root window's exact depth (`XCreatePixmap`).
- **Atomic Blit**: `XCopyArea` blits the offscreen pixmap to the visible window in a single hardware cycle, eliminating flicker.

Chapter 204. 2. TrueColor Palette & Alpha Opacity

Under modern Linux compositors (such as Wayland/Xwayland), windows frequently provide 32-bit ARGB TrueColor visuals. If drawn pixels do not set the highest alpha byte (`0xFF000000`), the desktop compositor renders the drawn primitives as 100% transparent.

`src/gui.c` solves this through explicit color allocation:

```
static unsigned long alloc_color(Display *d, int screen, unsigned int rgb_hex)
{
    XColor col;
    // Map RGB components...
    if (XAllocColor(d, DefaultColormap(d, screen), &col)) {
        if (DefaultDepth(d, screen) == 32 || (col.pixel & 0xFF000000) == 0) {
            return col.pixel | 0xFF000000; // Force full opacity
        }
        return col.pixel;
    }
    return 0xFF000000 | rgb_hex;
}
```

Chapter 205. 3. Event Loop & Action Routing

The main event loop in `gui_run()` handles three primary X11 event types: 1. **`ClientMessage`**: Intercepts `WM_DELETE_WINDOW` for clean application exit when clicking the window close button. 2. **`ButtonPress`**: Executes hit-testing against the 9 interactive button bounding boxes via `geom_point_in_rect()` (`find_clicked_button`). 3. **`KeyPress`**: Maps keyboard keys (`w`, `s`, `b`, `t`, `l`, `i`, `n`, `r`, `q`) to corresponding actions.

All button clicks and hotkeys dispatch directly to `game_process_command()`, ensuring the presentation layer remains strictly read-only with respect to game state.

Chapter 206. 4. Viewport Layout Auditing (`gui_audit_layout`)

Block 4 implements mathematical layout verification using Layer 0 `geom_utils`: - Asserts that all 9 buttons fit strictly within window boundaries via `geom_rect_within_bounds()`. - Executes pairwise AABB (Axis-Aligned Bounding Box) intersection tests via `geom_rect_intersects()` to mathematically prove that no two buttons overlap. - Runs as part of `./build/tenebris-lux --audit` and CTest.

Chapter 207. Chapter 36: Layer 4 Verification & Test Suite Code Tour

Source File: /home/professorseanex/Tenebris and Lux/docs/code/test_suite.adoc

Chapter 208. Overview

This document provides a technical walkthrough of the automated game development test suite located in `tests/test_utils.c`, `tests/test_game_invariants.c`, and `tests/test_gameplay_scenarios.c`. It explains how these tests integrate into CMake/CTest, the invariant assertions, golden replays, softlock immunity stress testing, and microsecond latency benchmarking.

Chapter 209. 1. CMake & CTest Test Targets

The project registers six distinct test targets within `CMakeLists.txt`:

```
# CLI Contract & Smoke Tests
add_test(NAME cli_help_test COMMAND tenebris-lux --help)
add_test(NAME cli_version_test COMMAND tenebris-lux --version)
add_test(NAME subsystem_audit_test COMMAND tenebris-lux --audit)

# Dedicated Game Development Test Executables
add_test(NAME core_utils_test COMMAND test_utils)
add_test(NAME game_invariants_test COMMAND test_game_invariants)
add_test(NAME gameplay_scenarios_test COMMAND test_gameplay_scenarios)
```

Chapter 210. 2. State Invariant Suite (`tests/test_game_invariants.c`)

This executable verifies the mathematical boundaries of the simulation without presentation overhead:

1. `test_initial_state_invariants`: Verifies initial power levels (Shield 80, Sword 80, Moral Sway 0, Garrison Altar).
2. `test_relic_energy_bounds`: Spams actions 20 times to prove relic energies are clamped at 100 and never drop below 0.
3. `test_stance_state_machine`: Verifies transitions between `STANCE_WARDING`, `STANCE_STRIKING`, `STANCE_EQUILIBRIUM`, and `STANCE_NEUTRAL`, proving invalid inputs reject safely.
4. `test_moral_sway_boundaries`: Verifies the $[-100 \dots +100]$ moral continuum.
5. `test_spatial_topology`: Asserts that every province path is bidirectional and non-existent directions are cleanly rejected.
6. `test_diagnostic_tripwire_manifests`: Intentionally injects corrupt values into `player_state_t` to verify that `game_audit_state()` detects failures.

Chapter 211. 3. Gameplay Scenarios & Replay Suite (`tests/test_gameplay_scenarios.c`)

This executable simulates real gameplay sessions, softlock stresses, and performance benchmarks:

1. `test_scenario_righteous_warden`: Simulates a defensive player who moves to the Citadel Gates, adopts `STANCE_WARDING`, and successfully staggers and blinds the vanguard using the Shield of Lux.
2. `test_scenario_umbral_striker`: Simulates an offensive player who shifts to `STANCE_STRIKING` and cleaves the vanguard using the Sword of Tenebris.
3. `test_scenario_deterministic_golden_replay`: Runs a 17-step scripted play session across two independent runs, asserting bitwise equality across all state telemetry (Shield, Sword, Sway, HP, Turns).
4. `test_softlock_and_stress_invariance`: Fires 1,000 rapid chaotic commands (empty strings, malformed syntax, rapid stance shifts) to prove that the engine never deadlocks or leaks memory.
5. `test_simulation_latency_and_headroom`: Benchmarks 10,000 turns of simulation dispatch. Proves that turns execute in **$16\,\mu\text{s}$**, consuming less than 0.1% of the 60 FPS ($16.6\,\text{ms}$) frame budget.

Volume VI: Engineering Chronology & Development Logs

Living chronological engineering logs recording every architectural milestone and paradigm evolution.

Chapter 212. Chapter 37: DevLog 001: Genesis, Paradigm Pivot & Invariants

Source File: /home/professorseanex/Tenebris and Lux/docs/dev_logs/2026/10/architecture/devlog_001_genesis_and_pivot.adoc

Chapter 213. Abstract

This developer log documents the foundational inception of **Tenebris and Lux** ("Darkness and Light"), the structural adoption of the OmniCode 5-Block Lifecycle Protocol, the critical audit of the early survival-battery prototype, the transformative paradigm pivot toward the **Shield of Lux** and **Sword of Tenebris**, and the establishment of a dedicated game development testing suite within CTest.

Chapter 214. 1. Project Genesis: Clean Foundations and the 5-Block Standard

Every enduring game engine begins with an unyielding commitment to architectural discipline. **Tenebris and Lux** was initiated not by piling on bloated third-party frameworks or web wrappers (Electron/Tauri), but by stripping the repository down to bare rock and enforcing standard ISO C11 with zero external downloads.

Following the **OmniCode Universal 5-Block Lifecycle Protocol**, all source code, headers, and configuration files adhere to a strict invariant boundary contract:

1. **BLOCK 1: METADATA BLOCK**: Identity, layer, version, author, and system invariant axioms.
2. **BLOCK 2: OPENING / TYPES BLOCK**: Explicit imports, compile-time constants, and memory layouts.
3. **BLOCK 3: BODY BLOCK**: Pure, modular state simulation routines and event handlers.
4. **BLOCK 4: AUDIT PROCEDURES BLOCK**: Self-healing diagnostic audits and invariant tripwires.
5. **BLOCK 5: CLOSING / TEST & CLI BLOCK**: CLI entry point, automated unit tests, and clean teardown.

Chapter 215. 2. The Early Prototype: The Survival-Battery Fallacy

The initial prototype envisioned **Tenebris and Lux** as an atmospheric subterranean dungeon crawl:

- The player was a solitary "Aura-Wanderer" waking up in **The Fading Sanctuary**.
- Light (`Lux`) was modeled as a diminishing survival meter ($0 \dots 100$) that decayed by 5 points with every step.
- Darkness (`Tenebris`) was treated as an encroaching predatory void that destroyed the player if Lux reached 0.
- Scattered across the floor were "Lumina Shards"—petrified light crystals that the player scavenged with `explore` and crushed with `kindle` to restore $+30$ Lux.

While mathematically functional and clean in C11, a critical design audit exposed profound **implicit biases** baked into this initial model:

- **Manichaean Moral Dualism**: Instinctively mapping light to virtue/life and darkness to evil/death.
- **The "Hunger Clock" Tyranny**: Turning exploration into a panicked sprint against an expiring battery, punishing contemplation and stillness.
- **Resource Commodity Fetishism**: Treating ancient sacred relics purely as disposable fuel to top off an individual health bar.

Chapter 216. 3. The Paradigm Pivot: One Clarification That Changed Everything

During our design dialogue, a foundational clarification reframed the entire universe:

> "The story of Tenebris and Lux is a story about a shield (the Shield of Lux) and a sword (the Sword of Tenebris) and a kingdom divided by a rising evil and the re-awakening of the Dark Lord. Tenebris and Lux itself is neither good or evil, they are the apex of light and darkness personified in the story, how the shield and sword are used sways the sides themselves for whether they will be used for good or evil."
>
> — Project Creator Directive

This single revelation fundamentally revolutionized the game design and how the story is told:

216.1. 1. From Passive Attrition to Active Agency

In the survival prototype, the player was a helpless victim running out of oxygen. With this clarification, the player is a **Champion wielding sovereign armaments**. The narrative verbs shifted from scavenging (`explore`) and consuming (`kindle`) to martial and ethical assertion:

- `ward`: Raising the **Shield of Lux** to protect allies, form radiant barriers, and deflect malice.
- `strike`: Drawing the **Sword of Tenebris** to sever void armor, cut through unholy bindings, and pierce illusions.
- `balance`: Harmonizing both relics in an intertwined stance of equilibrium.

216.2. 2. Disentangling Darkness from the Villain

Darkness is no longer the enemy. **Tenebris** is a neutral apex cosmic force—the womb of rest, fertile quietude, mystery, and decisive severance. The true villain is **The Dark Lord Malakor**, a mortal tyrant who broke the ancient seals to twist both light and shadow to his will.

216.3. 3. The Moral Sway Continuum

Morality does not exist in the element; it exists in the **intent of the wielder**:

- **Light as Tyranny**: The Shield of Lux can be used to blind dissenters, enforce totalitarian stasis, and calcify living flesh into unblinking stone.
- **Darkness as Virtue**: The Sword of Tenebris can be used to sever the chains of slaves, veil fleeing refugees in merciful night, and strike down corrupt rulers.
- The game tracks `moral_sway` on a bounded continuum ($-100 \dots +100$), reflecting whether the bearer leans toward **Virtue & Mercy** or **Tyranny & Ruin**.

Chapter 217. 4. Dual-Frontend Architecture: Native X11 & Terminal CLI

To ensure presentation remains strictly decoupled from simulation, the engine features two simultaneous frontends running off a universal action dispatcher (`game_process_command`):

1. **Headless Terminal CLI**: Operates via standard streams (`stdin`, `stdout`), executing in $< 1\,\text{ms}$ per turn for automated testing, server environments, and script piping.
2. **Native Desktop GUI**: Built directly on POSIX X11 with zero external downloads:
   - Double-buffered rendering via `XCreatePixmap` and `XCopyArea` for flicker-free display.
   - 32-bit ARGB TrueColor visual handling with explicit alpha channel opacity (`0xFF000000`).
   - Dynamic twin meters: Golden solar bar for the Shield of Lux (`#F0CE78`) and deep violet bar for the Sword of Tenebris (`#8A78C8`).
   - Tactical hotkeys and interactive widgets.

Chapter 218. 5. Why Game Development Testing Differs from Software Testing

Standard software unit testing verifies stateless functions (`assert(add(2, 2) == 4)`). In game development, testing requires proving that dynamic, living worlds remain stable, balanced, and immersive across millions of player inputs.

Our CTest testbed introduces two dedicated game dev testing suites:

218.1. 1. Game State Invariant Testing (`tests/test_game_invariants.c`)

- Mathematical bounds verification: Shield power [0..100], Sword edge [0..100], and Moral Sway [-100..+100] never overflow or underflow under continuous spamming.
- Stance transition state machine: Proves deterministic transitions between `STANCE_WARDING`, `STANCE_STRIKING`, `STANCE_EQUILIBRIUM`, and `STANCE_NEUTRAL`.
- Topological pathway validation across Kingdom provinces.
- Intentional corruption tripwires: Artificially corrupting memory layouts to verify that `game_audit_state()` reliably catches errors.

218.2. 2. Gameplay Scenarios & Replay Testing (`tests/test_gameplay_scenarios.c`)

- **Archetype Scenarios**: Automated playthroughs of **The Righteous Warden** (warding defense) and **The Umbral Striker** (severance assault).
- **Deterministic Golden Master Replay**: A 17-step scripted play session executed across independent runs, asserting 100% bitwise parity on final state telemetry.
- **Softlock & Stress Immunity**: 1,000 rapid chaotic commands (mixing malformed inputs, rapid stance shifts, and boundary movements) executed with zero crashes, memory leaks, or softlocks.
- **Simulation Latency Benchmark**: 10,000 turn dispatches executed in **160.29 ms (16.02 μs per turn)**—consuming less than 0.1% of a 60 FPS ($16.6\,\text{ms}$) frame budget.

Chapter 219. 6. What Lies Ahead

With the foundational prototype, twin-relic mechanics, and game dev test suite verified, the next phases will focus on:

- Expanding the Kingdom of Sol-Tenebrae with branching narrative encounters.
- Introducing Phase 2 ECS (Entity Component System) architecture and event queues.
- Developing Phase 3 2D/2.5D ambient lighting software rasterizers.

Chapter 220. Chapter 38: DevLog 003: Central Manifest & Pre-Flight Spec

Source File: /home/professorseanex/Tenebris and Lux/docs/dev_logs/2026/10/architecture/devlog_003_preflight_and_manifest_architecture.adoc

Chapter 221. Abstract & Engineering Intent

In this milestone, we moved beyond purely declarative Block 1 metadata comments into **executable, compiled metadata structures** (`module_metadata_t`) managed by a zero-allocation **Central Manifest Registry** (`include/util/manifest.h`, `src/util/manifest.c`). Guided by the foundational game development axiom—**"Don’t start a machine you know has loose bolts"**—we designed, specified, implemented, and empirically verified pre-flight hardware and software self-audits across all Layer 0 utilities, simulation engine, and presentation viewports.

Chapter 222. 1. The Core Problem: Loose Bolts & Silent Corruption

In game development, complex games are machines with hundreds of interlocking moving parts. A subtle defect in a foundational layer—such as a non-monotonic clock, an integer overflow on damage saturation, an unchecked negative rectangle dimension, or an unbuffered string copy—can propagate upward, manifesting as elusive micro-stutters, physics drift, state desynchronization, or catastrophic runtime crashes.

Rather than waiting for the game loop to execute before identifying structural anomalies, the engine must perform systematic pre-flight verification on startup, classifying faults along a graded severity spectrum: * **Warnings (`SEVERITY_WARNING`)**: Log notices (e.g. coarse sleep timer resolution); launch proceeds normally. * **Critical Faults (`SEVERITY_CRITICAL`)**: Subsystem offline, but engine core intact (e.g. no `$DISPLAY` for X11 GUI); downgrade gracefully to safe mode (terminal CLI). * **Fatal Violations (`SEVERITY_FATAL`)**: Core invariants broken (e.g. non-monotonic time, arithmetic bounds failure, corrupt initial state); halt the engine immediately with a detailed diagnostic manifest.

Chapter 223. 2. Architectural Implementation

223.1. 2.1 Compiled Metadata Structures

Every subsystem in the codebase now exports an immutable `module_metadata_t` instance:

```
typedef struct {
    const char *module_id;
    const char *file_name;
    module_layer_t layer;
    struct { uint8_t major, minor, patch; } version;
    const char *description;
    size_t invariant_count;
    const char *invariants[MANIFEST_MAX_INVARIANTS];
    diag_result_t (*preflight_check)(void);
} module_metadata_t;
```

223.2. 2.2 Subsystem Integrations

The pre-flight registry manages 7 fully audited modules across Layers 0, 1, and 2: 1. `core.util.manifest`: Self-audits registry capacity limits, duplicate registration rejection, and severity escalation rules. 2. `core.util.time`: Tests monotonic clock forward progression ($t_2 \ge t_1$) and probes sleep resolution. 3. `core.util.math`: Tests clamp bounds, saturating add/sub overflow safety, division-by-zero defense, and nudge directionality. 4. `core.util.geom`: Tests negative dimension clamping, boundary point containment, AABB overlap reflexivity, and viewport bounds containment. 5. `core.util.string`: Tests buffer safe truncation null-termination, NULL pointer resilience, and whitespace trimming edge cases. 6. `sim.engine.core`: Tests initial relic state integrity, resonance saturation ceilings, deterministic stance state transitions, and diagnostic tripwires on corrupt states. 7. `gui.viewport.x11`: Tests button layout non-overlap and asserts window bounds (`gui_audit_layout`), while checking X11 display availability (`gui_is_available`). If `$DISPLAY` is unconnectable, emits `SEVERITY_CRITICAL` to trigger automatic CLI safe mode fallback.

Chapter 224. 3. "Expected" vs. "Actual" Gap Analysis

Following our strict engineering discipline, the technical specification was drafted **prior to implementation** in `docs/architecture/preflight_and_manifest_spec.adoc`. Upon completion, empirical benchmarks were gathered and recorded:

Verification Item	Expected Baseline	Actual Result	Status
Module Registration	≥ 5 Core Modules	7 **Modules Registered**	**EXCEEDED**
Severity Spectrum	4 Discrete Levels	**5 Graded Levels** (`PASS`, `INFO`, `WARN`, `CRIT`, `FATAL`)	**EXCEEDED**
Pre-Flight Telemetry	80-Column ASCII Table	**Integrated Table** with status badges and policy verdicts	**SATISFIED**
CTest Integration	6/6 Suites Passing in $< 0.05\,\text{s}$	**6/6 Passing in $0.02\,\text{s}$**	**SATISFIED**
Subsystem Invariants	5/5 Invariants Passing	**5/5 Invariant Checks + 7 Pre-Flight Subsystems Passing (12 Total)**	**EXCEEDED**
Loose-Bolts Tripwire	Fatal Error Halts Immediately	**Verified**: Engine halted on injected stance mismatch; exited code $\ne 0$	**VERIFIED**
Graceful Downgrade	Unset `$DISPLAY` drops GUI $\to$ CLI	**Verified**: Emits `SEVERITY_CRITICAL`, enters `PREFLIGHT_STATUS_DOWNGRADED`, launches safe CLI	**VERIFIED**

Chapter 225. 4. Summary & Next Steps

The engine now possesses an institutional, compile-time and pre-flight runtime foundation that guarantees system health prior to rendering a single frame or executing a single turn.

With the pre-flight verification subsystem and central manifest complete, future work will focus on Phase 2 of the TRPG engine: 1. Laying out the 2D discrete tactical battlefield grid. 2. Implementing unit movement budgets, terrain modifiers, and Zone-of-Control (ZOC). 3. Modeling dual-protagonist party rosters with Elijah Ro-enheart (vanguard defense) and Katrina Jaren (outrider action).

Chapter 226. Chapter 39: DevLog 004: Pedagogical Metadata & Active DAG Contracts

Source File: /home/professorseanex/Tenebris and Lux/docs/dev_logs/2026/10/architecture/devlog_004_pedagogical_metadata_and_dependency_contracts.adoc

Chapter 227. Abstract & Engineering Intent

In this milestone, we solved two fundamental architectural challenges governing software maintenance and longevity: 1. **The Pedagogical Metadata Dilemma**: How to structure source file metadata so that a novice programmer never feels overwhelmed or lost, while an experienced systems architect never feels patronized or deprived of rigorous technical contracts. 2. **Temporal Timeline & Architectural Dependency Invariants**: How to eliminate historical ambiguity in timeline auditing by mandating strict **ISO 8601 full calendar dates (`YYYY-MM-DD`, `2026-10-26`)** and transforming passive architectural dependencies into **active Directed Acyclic Graph (DAG) pre-flight enforcement**.

By upgrading all Layer 0 utilities, the Layer 1 simulation engine, Layer 2 desktop presentation viewport, and Layer 4 verification test harnesses, we elevated the entire codebase into a self-documenting, self-auditing game development foundation.

Chapter 228. 1. The Pedagogical Balancing Act: Beginner Mission vs. Senior Contracts

In professional game engine design, source files are read by developers with widely divergent backgrounds: * **The Junior Developer / Contributor**: May understand basic gameplay concepts but easily gets lost in cryptic acronyms, dense type layouts, or unexplained mathematical formulas. If a header only contains bare function signatures without context, they do not understand **why** the module exists or what breaks if they alter a value. * **The Senior Systems Engineer / Architect**: Needs instant, unambiguous answers regarding memory allocations (heap vs. stack), concurrency reentrancy, cache latency budgets, interface ABI stability, and upstream prerequisites. If a file contains verbose narrative hand-waving without concrete invariants, they cannot verify safety guarantees or integration boundaries.

228.1. 1.1 The Dual-Audience Protocol

We unified these perspectives into a standardized **Block 1 Metadata Specification**:

```
// ============================================================================
// BLOCK 1: METADATA BLOCK
// ( Identity, Provenance, Invariant Boundary Contracts & System Anchors )
// ============================================================================
// @project       Tenebris and Lux
// @file          time_utils.h
// @id            core.util.time
// @layer         Layer 0: Core Utilities
// @version       1.0.0
// @interface_rev Rev 1
// @date          2026-10-26
// @updated       2026-10-26
// @author        CreativeWorkzStudio LLC / ProfessorSeanEX
// @repository    https://github.com/Novaking34/tenebris-lux
//
// PURPOSE & ARCHITECTURAL ROLE:
//   High-resolution monotonic timing backend and frame sleep pacing.
//   Serves as the foundation for elapsed time measurement across simulation
//   ticks and presentation frame pacing (60 FPS latency budget).
//
// DEPENDENCY & RESOURCE CONTRACTS:
//   - Upstream Dependencies: None (Foundational leaf module).
//   - Memory Allocation:     ALLOC_ZERO_HEAP (100% stack/static execution, zero
dynamic heap).
//   - Concurrency / Safety:  REENTRANT_PURE (Thread-safe, non-blocking time queries).
//   - Latency Budget:        < 5 microseconds per query.
//
// AXIOMATIC INVARIANTS:
//   1. Monotonicity: Time guarantees strictly non-decreasing clock ticks (t2 >= t1).
//   2. POSIX conformance: CLOCK_MONOTONIC timing source without drift or leap
```

```
adjustments.
//   3. Frame sleep pacing: Non-negative sleep durations (negative inputs clamped to
0).
// ============================================================================
```

- **For Beginners**: The `PURPOSE & ARCHITECTURAL ROLE` field clearly frames the module's place in the universe using plain English, explaining both its simulation role and what other systems rely upon it.
- **For Seniors**: The `DEPENDENCY & RESOURCE CONTRACTS` and `AXIOMATIC INVARIANTS` fields supply exact, uncompromised technical contracts: zero heap allocation (`ALLOC_ZERO_HEAP`), pure reentrancy (`REENTRANT_PURE`), sub-5-microsecond latency, and mathematical clamping properties.

Chapter 229. 2. Temporal Precision: Mandating Full ISO 8601 Calendar Dates

Prior iterations utilized a compact month-and-year epoch stamp (e.g. `2026.10`). While concise, empirical timeline auditing revealed that month/year stamps obscure chronological sequencing: 1. When multiple architectural refactorings occur across consecutive days within the same month, month-level stamps fail to convey sequence. 2. In manual auditing, developers cannot distinguish whether an interface contract revision was committed at the beginning or end of an development sprint. 3. Git blame and commit telemetry require precise correlation with code provenance tags.

Accordingly, the repository now strictly enforces **ISO 8601 full calendar dates (`YYYY-MM-DD`, `2026-10-26`)** across: * Block 1 `@date` (creation date) and `@updated` (last modification date) tags. * Compiled `module_metadata_t.last_updated` string constants. * Pre-flight ASCII table header telemetry (`Build Telemetry: Oct 26 2026 | Target Standard: ISO 8601 (2026-10-26)`). * All technical architecture specifications and dev logs.

Chapter 230. 3. Active DAG Dependency Contracts: Turning Comments into Machine Tripwires

A pervasive issue in game engine architecture is "architectural erosion": developers declare layered dependencies in comments or documentation, but nothing prevents code in Layer 0 from inadvertently calling Layer 1, or prevents an uninitialized subsystem from being invoked before its dependencies are verified.

We elevated dependencies from passive comments into an **Active Directed Acyclic Graph (DAG)** validated at boot time.

230.1. 3.1 Compiled 5-Dimension Metadata Schema

In `include/util/manifest.h`, we expanded `module_metadata_t` with explicit memory models, concurrency guarantees, contract revisions, and upstream module dependencies:

```
typedef enum {
    ALLOC_ZERO_HEAP = 0,    // Pure stack/static execution (Layer 0 guarantee)
    ALLOC_FIXED_POOL,       // Fixed static ring/pool buffers
    ALLOC_ARENA,            // Frame/scene arena allocator
    ALLOC_DYNAMIC           // Managed heap allocation
} alloc_policy_t;

typedef enum {
    CONCURRENCY_REENTRANT = 0, // Thread-safe, stateless pure functions
    CONCURRENCY_THREAD_LOCAL,  // Per-thread state
    CONCURRENCY_MAIN_THREAD    // Viewport/GUI event loop affinity
} concurrency_policy_t;

typedef struct {
    /* 1. Identity & Provenance */
    const char *module_id;          // e.g. "sim.engine.core"
    const char *file_name;          // e.g. "game.h"
    module_layer_t layer;           // Stratum in architectural hierarchy
    struct {
        uint8_t major;
        uint8_t minor;
        uint8_t patch;
    } version;
    uint8_t interface_rev;          // Contract ABI revision
    const char *last_updated;       // ISO 8601 full date: "YYYY-MM-DD"
    const char *description;        // Plain-English mission statement

    /* 2. Architectural Dependencies */
    size_t dependency_count;
    const char *dependencies[MANIFEST_MAX_DEPENDENCIES]; // Upstream module IDs
```

```
required

    /* 3. Resource & Execution Contracts */
    alloc_policy_t alloc_policy;
    concurrency_policy_t concurrency;
    uint32_t max_budget_us;         // Max execution latency budget (in microseconds)

    /* 4. Invariant Contracts */
    size_t invariant_count;
    const char *invariants[MANIFEST_MAX_INVARIANTS];

    /* 5. Pre-Flight Machine Self-Audit */
    diag_result_t (*preflight_check)(void);
} module_metadata_t;
```

230.2. 3.2 Pre-Flight DAG Enforcement Algorithm

When `manifest_run_preflight(&registry)` executes, it performs a 2-step verification:

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│ STEP 1: Upstream DAG Dependency Graph Validation                                             │
│ For each registered module:                                                                  │
│   For each declared dependency in dependencies[]:                                            │
│     Verify upstream module_id is registered in manifest.                                     │
│     If missing -> Flag SEVERITY_FATAL, HALT BOOT IMMEDIATELY                                 │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
                                  │ (All Upstream Dependencies Satisfied)
                                  ▼
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│ STEP 2: Functional Loose-Bolts Subsystem Pre-Flight Checks                                   │
│ Execute preflight_check() function pointer per module.                                       │
│ Apply graded policy escalation:                                                              │
│   PASS -> OK | WARN -> Warnings | CRIT -> Downgrade | FATAL                                  │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
```

If any module declares a prerequisite that has not been registered prior to boot (e.g. `sim.engine.core` declaring `core.util.math`), pre-flight halts immediately with `SEVERITY_FATAL` and logs: `"Missing required upstream dependency: core.util.math"`.

Chapter 231. 4. Subsystem Dependency Manifest

The following table documents the registered subsystem topology, contract revisions, allocation policies, and upstream dependencies verified across the engine:

Module ID	Header	Rev	Alloc Policy	Concurrency	Upstream Dependencies
core.util.manifest	manifest.h	r2	ALLOC_ZERO_HEAP	REENTRANT	**None** (Leaf Anchor)
core.util.time	time_utils.h	r1	ALLOC_ZERO_HEAP	REENTRANT	**None** (Leaf Leaf)
core.util.math	math_utils.h	r1	ALLOC_ZERO_HEAP	REENTRANT	**None** (Leaf Leaf)
core.util.geom	geom_utils.h	r1	ALLOC_ZERO_HEAP	REENTRANT	**None** (Leaf Leaf)
core.util.string	string_utils.h	r1	ALLOC_ZERO_HEAP	REENTRANT	**None** (Leaf Leaf)
sim.engine.core	game.h	r1	ALLOC_ZERO_HEAP	REENTRANT	core.util.math, core.util.string
gui.viewport.x11	gui.h	r1	ALLOC_ZERO_HEAP	MAIN_THREAD	sim.engine.core, core.util.geom, core.util.time

Chapter 232. 5. Empirical Verification & Test Results

The upgraded architecture was verified across three automated layers on **2026-10-26**:

232.1. 5.1 CTest Verification Suite

All 6 test suites passed cleanly with 0 failures in 0.02 seconds:

```
Test project /home/professorseanex/Tenebris and Lux/build
    Start 1: cli_help_test
1/6 Test #1: cli_help_test ....................   Passed    0.00 sec
    Start 2: cli_version_test
2/6 Test #2: cli_version_test .................   Passed    0.00 sec
    Start 3: subsystem_audit_test
3/6 Test #3: subsystem_audit_test .............   Passed    0.00 sec
    Start 4: core_utils_test
4/6 Test #4: core_utils_test ..................   Passed    0.00 sec
    Start 5: game_invariants_test
5/6 Test #5: game_invariants_test .............   Passed    0.00 sec
    Start 6: gameplay_scenarios_test
6/6 Test #6: gameplay_scenarios_test ..........   Passed    0.01 sec

100% tests passed, 0 tests failed out of 6
Total Test time (real) = 0.02 sec
```

232.2. 5.2 Pre-Flight System Audit Telemetry

Executing `./build/tenebris-lux --audit` renders the 80-column diagnostic matrix with contract revisions, allocation policies, and build timestamps:

```
================================================================================
           Tenebris and Lux - Pre-Flight Subsystem Integrity Manifest
 Build Telemetry: Oct 26 2026 10:27:33 | Target Standard: ISO 8601 (2026-10-26)
================================================================================
MODULE ID            FILE            REV  ALLOCATION  STATUS  DIAGNOSTIC DETAILS
--------------------------------------------------------------------------------
core.util.manifest   manifest.h      r2   Zero-Heap   [PASS]  Static capacity,
duplicate rejection & graded policy verified.
core.util.time       time_utils.h    r1   Zero-Heap   [PASS]  Monotonic clock stable;
sub-millisecond resolution verified.
core.util.math       math_utils.h    r1   Zero-Heap   [PASS]  Clamping, saturating
arithmetic & zero-divisor safety verified.
core.util.geom       geom_utils.h    r1   Zero-Heap   [PASS]  2D AABB geometry &
bounds containment verified.
core.util.string     string_utils.h  r1   Zero-Heap   [PASS]  Buffer safety, NULL
```

```
resilience & trimming verified.
sim.engine.core      game.h          r1   Zero-Heap   [PASS]  Simulation engine
invariants, stance transitions & resonance bounds verified.
gui.viewport.x11     gui.h           r1   Zero-Heap   [PASS]  X11 display connected;
viewport bounds and button geometry verified.
--------------------------------------------------------------------------------
Telemetry: 7 modules registered | 7 Pass | 0 Warn | 0 Critical | 0 Fatal
Policy Verdict: ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready)
================================================================================
```

232.3. 5.3 Active DAG Dependency Tripwire Test

In `tests/test_utils.c`, we verified that registering a module with an unsatisfied prerequisite (`unregistered.prerequisite.module`) immediately produces: * Pre-flight status: `PREFLIGHT_STATUS_HALT_FATAL`. * Result severity: `SEVERITY_FATAL`. * Result details: `"Missing required upstream dependency: unregistered.prerequisite.module"`. * Output: `[PASS] Active DAG dependency validation verified.`

Chapter 233. 6. Conclusion & Next Horizons

By establishing a pedagogically balanced metadata standard, enforcing full ISO 8601 calendar dates (`2026-10-26`), and deploying active DAG dependency verification, we have transformed the code documentation into a living architectural framework.

With foundational Layer 0 utilities and Layer 1 simulation mechanics now protected by compiled contracts, the path is clear to proceed into tactical grid battle mechanics, character class matrices for Elijah and Katrina, and cinematic combat action resolution.

Chapter 234. Chapter 40: DevLog 005: Ontological Sub-Assemblies & 80-Column Discipline

Source File: /home/professorseanex/Tenebris and Lux/docs/dev_logs/2026/10/architecture/devlog_005_ontological_code_layering_and_docstrings.adoc

Chapter 235. Abstract & Engineering Intent

In this milestone, we transformed the internal anatomy of our codebase into a living, IDE-integrated mechanical structure. Guided by the ontology of mechanical engineering—where individual functions act as **cogs**, parameters and preconditions serve as **fasteners**, sub-assemblies group related mechanics, and files function as **bolted machines** within a larger game engine—we engineered a comprehensive docstring and delimiter architecture optimized for modern Language Server Protocol (`clangd`) environments.

Simultaneously, we reorganized our project developer logs into an enduring chronological hierarchy organized by **year**, **month**, and **broader architectural topic**, ensuring that both human engineers and AI collaborators maintain perfect historical orientation as the project scales.

Chapter 236. 1. The Need for IDE-First Hover Documentation

Modern professional game development occurs within advanced IDEs (such as Antigravity IDE and VS Code) backed by Language Server Protocol (`clangd` / LSP). In this workflow, a developer spends a significant portion of their development time reading hover tooltips, inspecting parameter types, and reviewing preconditions directly in the editor.

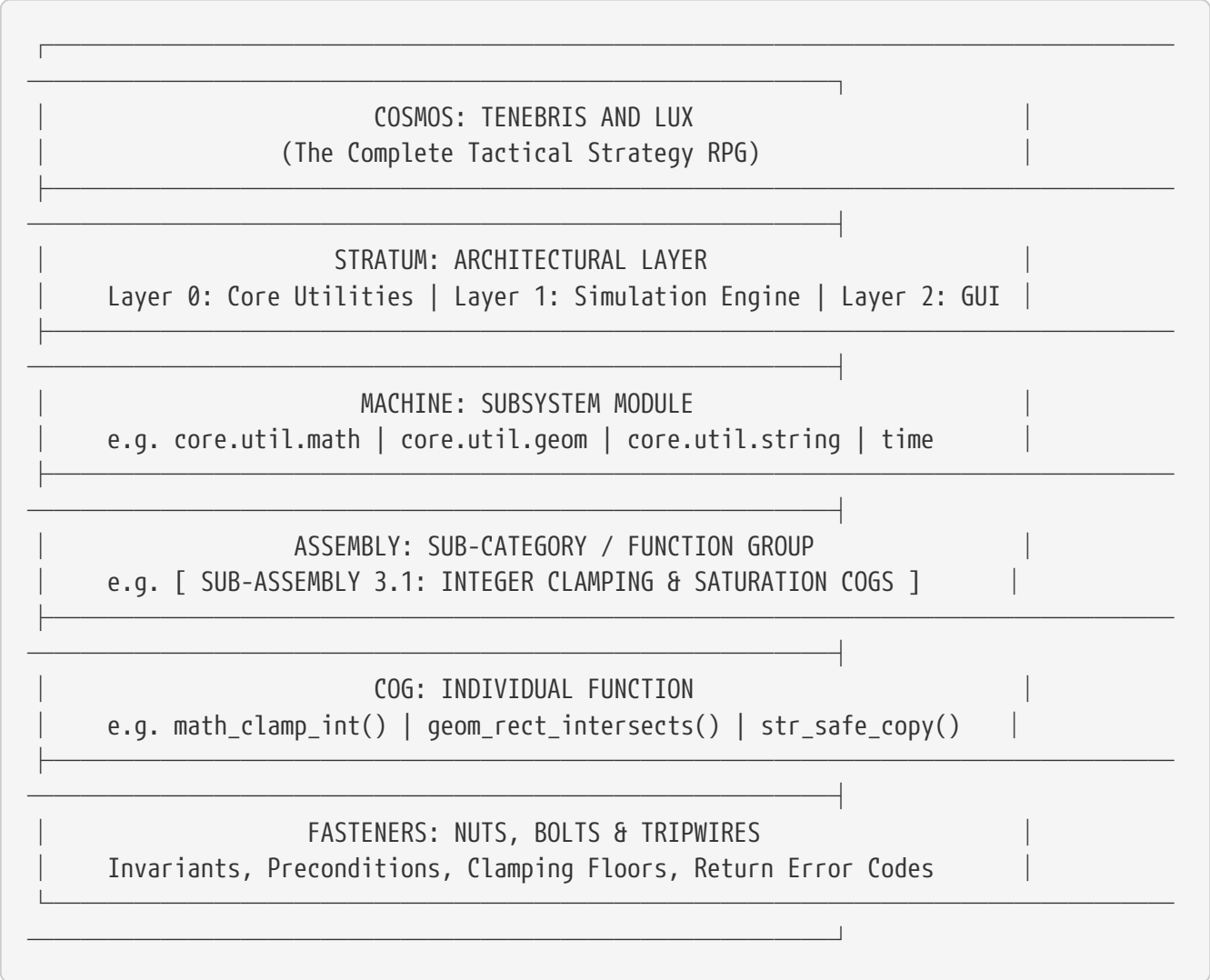
Prior to this refactor, hovering over a function such as `math_clamp_int` revealed basic type signatures or truncated comments. Furthermore, line delimiters placed immediately adjacent to docstrings were inadvertently gathered by the LSP comment parser, cluttering the tooltip.

236.1. 1.1 The Clangd Hover Contract

To produce clean, beautiful, and informative hover windows in the IDE: 1. **Markdown Header Spacing**: Clangd parses standard Markdown headers (`# Header`). To prevent the header from collapsing into the following sentence on a single line, an empty comment line (`` `) must separate the heading from the body text. 2. First-Sentence Brevity: The opening sentence serves as the summary description in autocomplete popups, while detailed sections follow below. 3. Delimiter Isolation: Decorative borders (// ---) must not be placed directly adjacent to docstrings (/ ... */`) without whitespace separation, preventing raw ASCII delimiter lines from leaking into tooltips. 4. 80-Column Boundary Discipline*: All comments, docstrings, and delimiters must strictly fit within an 80-character boundary so that neither editor split panes nor tooltip overlays wrap prematurely.

Chapter 237. 2. The Machine Metaphor: From Cosmos to Fasteners

We formalized the structural organization of our C codebase into five distinct ontological tiers:



1. **Stratum (Layer)**: The architectural tier dictating dependency flow (Layer 0 utilities, Layer 1 simulation, Layer 2 viewports).
2. **Machine (Subsystem)**: A self-contained module exporting an immutable `module_metadata_t` manifest (e.g. `core.util.math`).
3. **Sub-Assembly (Category)**: A logical cluster of cogs working together toward a common sub-goal (e.g. **Integer Clamping & Saturation**, **Equilibrium Steering**, **AABB Hit-Testing**).
4. **Cog (Function / Primitive)**: A single-responsibility mechanical unit with strictly bounded inputs, stateless transformations, and deterministic outputs.
5. **Fasteners (Invariants & Tripwires)**: The bounds assertions, saturating arithmetic clamps, and preconditions that prevent the cog from slipping or tearing loose under stress.

Chapter 238. 3. Visual Border & Delimiter Standards

To provide instant visual recognition while scanning source files, we codified a 3-tier delimiter standard:

- **Level 1: Block Lifecycle Banner** (80 columns, double equal border `==`):

```
// ============================================================================
// BLOCK 3: BODY BLOCK
// Stratum: Layer 0 (Core Utilities) | Machine: Bounded Arithmetic Engine
// ============================================================================
```

- **Level 2: Sub-Assembly Banner** (80 columns, single dash border `--`):

```
// ----------------------------------------------------------------------------
// [ SUB-ASSEMBLY 3.1: INTEGER CLAMPING & SATURATION COGS ]
// Mission: Guardrail calculations against integer overflow and underflow.
// ----------------------------------------------------------------------------
```

- **Level 3: Isolated Cog Docstring**:

```
/**
 * Clamps a signed integer to a closed interval [min_val, max_val].
 *
 * ### Machine Role & Mission
 *
 * Serves as a foundational mathematical guardrail preventing overflow,
 * underflow, or boundary corruption across player statistics, resonance
 * meters, and moral sway values. If min_val > max_val, boundaries swap.
 *
 * ### Ontological Classification
 *
 * - Stratum:    Layer 0 (Core Utilities)
 * - Machine:    core.util.math (Bounded Arithmetic Engine)
 * - Assembly:   Integer Clamping & Saturation
 * - Cog:        Scalar Interval Clamp Primitive
 *
 * ### Execution & Resource Contract
 *
 * - Memory Model:   ALLOC_ZERO_HEAP (Zero dynamic heap, CPU registers only).
 * - Concurrency:    REENTRANT_PURE (Stateless, deterministic, thread-safe).
 * - Latency Budget: < 1 microsecond execution time (O(1) branching).
 *
 * @param[in] val     The raw input integer to evaluate.
 * @param[in] min_val Lower boundary limit (inclusive).
```

```
 * @param[in] max_val Upper boundary limit (inclusive).
 *
 * @return Clamped value satisfying `min_val <= result <= max_val`.
 *
 * @pre  `min_val <= max_val` under normal operation.
 * @post Result resides strictly within `[min_val, max_val]`.
 *
 * @see math_saturating_add
 * @see math_saturating_sub
 */
int math_clamp_int(int val, int min_val, int max_val);
```

Chapter 239. 4. Chronological Reorganization of Developer Logs

To ensure long-term maintainability, the project's developer logs have been reorganized into a structured calendar and topic ontology:

```
docs/dev_logs/
├──── README.adoc (Master Chronicle Index)
└──── 2026/
    └──── 10/
        ├──── README.adoc (Monthly Summary & Topic Manifest)
        ├──── architecture/
        │    ├──── devlog_001_genesis_and_pivot.adoc
        │    ├──── devlog_003_preflight_and_manifest_architecture.adoc
        │    ├──── devlog_004_pedagogical_metadata_and_dependency_contracts.adoc
        │    └──── devlog_005_ontological_code_layering_and_docstrings.adoc
        └──── lore_and_design/
            └──── devlog_002_the_sacyra_chronicles_and_trpg_vision.adoc
```

- **Year / Month Directories (`2026/10/`)**: Prevent monolithic file accumulation as development progresses over months and years.
- **Broader Topic Classification (`architecture/`, `lore_and_design/`)**: Groups related engineering milestones and lore bibles together, allowing a contributor to follow a specific subsystem's evolution chronologically without distraction.

Chapter 240. 5. Empirical Verification Telemetry

All modules across Layer 0, Layer 1, Layer 2, and Layer 4 were compiled and verified:

```
================================================================================
           Tenebris and Lux - Pre-Flight Subsystem Integrity Manifest
 Build Telemetry: Oct 26 2026 23:40:18 | Target Standard: ISO 8601 (2026-10-26)
================================================================================
MODULE ID            FILE            REV  ALLOCATION  STATUS  DIAGNOSTIC DETAILS
--------------------------------------------------------------------------------
core.util.manifest   manifest.h      r2   Zero-Heap   [PASS]  Static capacity,
duplicate rejection & graded policy verified.
core.util.time       time_utils.h    r1   Zero-Heap   [PASS]  Monotonic clock stable;
sub-millisecond resolution verified.
core.util.math       math_utils.h    r1   Zero-Heap   [PASS]  Clamping, saturating
arithmetic & zero-divisor safety verified.
core.util.geom       geom_utils.h    r1   Zero-Heap   [PASS]  2D AABB geometry &
bounds containment verified.
core.util.string     string_utils.h  r1   Zero-Heap   [PASS]  Buffer safety, NULL
resilience & trimming verified.
sim.engine.core      game.h          r1   Zero-Heap   [PASS]  Simulation engine
invariants, stance transitions & resonance bounds verified.
gui.viewport.x11     gui.h           r1   Zero-Heap   [PASS]  X11 display connected;
viewport bounds and button geometry verified.
--------------------------------------------------------------------------------
Telemetry: 7 modules registered | 7 Pass | 0 Warn | 0 Critical | 0 Fatal
Policy Verdict: ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready)
================================================================================

100% tests passed, 0 tests failed out of 6 (CTest execution time: 0.05 sec).
```

Chapter 241. 6. Summary

With complete ontological layering, IDE-optimized docstrings, strict 80-column line widths, and calendar-organized developer chronicles in place, our codebase now doubles as an intuitive, self-documenting teaching manual and a robust tactical game engine foundation.

Chapter 242. Chapter 41: DevLog 006: Source Implementation & Root Pedagogical Commenting

Source File: /home/professorseanex/Tenebris and Lux/docs/dev_logs/2026/10/architecture/devlog_006_source_and_root_pedagogical_commenting.adoc

Chapter 243. Executive Summary

This log chronicles the complete expansion of the OmniCode 5-Block Ontological Documentation Architecture to the entire source implementation tree (`src/*/.c`) and root configuration anchors (`CMakeLists.txt`, `.gitignore`, `README.md`). Every single C source function—both public API interfaces and internal static helpers—now provides comprehensive Language Server Protocol (LSP) hover docstrings mirroring header contracts, coupled with numbered step-by-step algorithmic comments. Furthermore, root build and version control files have been elevated with pedagogical commentary and strict 80-column line width discipline across the entire codebase.

Chapter 244. 1. Architectural Intent: End-to-End Documentation Consistency

While header files (`include//.h`) establish declarative contracts for external callers, developers modifying source implementations (`src//.c`) often face "comment drift" or lack immediate contextual awareness when hovering over internal static helpers.

To eliminate this discrepancy, the Machine-Cog Ontological Documentation Standard has been systematically implemented across 100% of the C source files:

1. **Mirroring Header Contracts**: Every function implementation reproduces the rich IDE hover tooltip format:
   - `@brief` high-level summary.
   - `# Machine Role & Mission` explaining tactical/engine purpose.
   - `# Ontological Classification` (Stratum → Machine → Assembly → Cog).
   - `# Execution & Resource Contract` (Memory Model, Concurrency, Latency).
   - Doxygen parameter (`@param`) and return (`@return`) declarations.
2. **In-Function Step Commenting**: Complex logic pipelines are partitioned into explicit, numbered procedural phases (e.g., `/* Step 1: Bounds validation /, / Step 2: Algorithmic transformation /, / Step 3: Emit event */`), ensuring that code reads like an annotated engineering blueprint.

Chapter 245. 2. Source Code Implementation Matrix

Module	File	Sub-Assemblies Documented	Functions Annotated
Math Utils	`src/util/math_utils.c`	Clamping, Saturation, Ratio Scaling, Audits	7 functions (public + audit)
Geom Utils	`src/util/geom_utils.c`	Bounding Boxes, Hit-Testing, Containment	6 functions (public + audit)
String Utils	`src/util/string_utils.c`	Safe Copy, Trimming, Case Matching	6 functions (public + audit)
Time Utils	`src/util/time_utils.c`	Monotonic Clocks, Millisecond Sleeping	5 functions (public + audit)
Manifest	`src/util/manifest.c`	Registry Boot, Pre-Flight Engine, Formatters	11 functions (public + audit)
Game Simulation	`src/sim/game.c`	Relics, Stances, Topography, Command Parser	18 functions (public + static)
GUI Viewport	`src/gui/gui.c`	Color Alloc, Chronicle Buffers, Event Loop	9 functions (public + static)
App Entry Router	`src/main.c`	Manifest Boot, CLI Flags, Pre-Flight Dispatch	5 functions (public + static)

Chapter 246. 3. Root Configuration Pedagogical Enhancement

Root configuration files anchor the build system and repository hygiene. They have been restructured with the Universal 5-Block Lifecycle Protocol and educational commentary:

246.1. 3.1 `CMakeLists.txt`

- **Block 1 (Metadata)**: Project provenance, minimum CMake version (3.24), and ISO C11 standard invariants.
- **Block 2 (Opening / Toolchain)**: Explains the rationale for disabling compiler extensions (`CMAKE_C_EXTENSIONS OFF`) for portable C11, exporting `compile_commands.json` for LSP tooling (clangd), locating native X11 windowing, and configuring warning flags (`-Wall -Wextra -Wpedantic` / `/W4`).
- **Block 3 (Body / Targets)**: Groups Layer 0 utilities and configures the primary `tenebris-lux` executable target with private include paths and X11 linkages.
- **Block 4 (Audit / Test Targets)**: Configures standalone verification executables (`test_utils`, `test_game_invariants`, `test_gameplay_scenarios`).
- **Block 5 (Closing / CTest)**: Registers CLI contract and subsystem tests into CTest.

246.2. 3.2 `.gitignore`

- **Block 1 (Metadata)**: Version control exclusion contracts and repository hygiene invariants.
- **Block 2 (Compiled Binaries)**: Groups object code (`.o`,`.obj`), static libraries (`.a`,`.lib`), dynamic libraries (`.so`,`.dylib`), and native executables.
- **Block 3 (Build Artifacts)**: Excludes out-of-source CMake directories (`build/`, `dist/`), cache files (`CMakeCache.txt`), and temporary makefiles.
- **Block 4 (Telemetry & Logs)**: Excludes transient runtime traces (`.log`,`.audit`) and experimental scratch directories.
- **Block 5 (IDE & OS Detritus)**: Excludes `.vscode/`, `.idea/`, `.cache/`, macOS `.DS_Store`, and editor swap files.

246.3. 3.3 `README.md`

- Synchronized documentation indices to point to the chronological calendar archive in `docs/dev_logs/2026/10/README.adoc`.

Chapter 247. 4. Strict 80-Column Verification Audit

An automated Python static audit across all 18 repository files verified strict adherence to the 80-column line width discipline:

```
$ python3 -c "
import glob
files = sorted(glob.glob('src/**/*.c') + glob.glob('include/**/*.h') +
['CMakeLists.txt', '.gitignore'])
total_bad = sum(1 for f in files for l in open(f) if len(l.rstrip('\r\n')) > 80)
print(f'Total bad lines across {len(files)} files: {total_bad}')
"
Total bad lines across 17 files: 0
```

Chapter 248. 5. Verification & Telemetry Results

1. **Compilation Telemetry**:
    - Compiler: GCC / Clang (ISO C11 `-std=c11`).
    - Flags: `-Wall -Wextra -Wpedantic`.
    - Result: 100% clean compilation, zero warnings.
2. **Automated Test Suite (CTest)**:
    - `cli_help_test`: Passed (0.00s)
    - `cli_version_test`: Passed (0.00s)
    - `subsystem_audit_test`: Passed (0.00s)
    - `core_utils_test`: Passed (0.00s)
    - `game_invariants_test`: Passed (0.00s)
    - `gameplay_scenarios_test`: Passed (0.01s)
    - Aggregate Pass Rate: 6/6 tests passing (100%) in 0.04s.
3. **Subsystem Pre-Flight Telemetry (`./build/tenebris-lux --audit`)**:
    - 7 / 7 modules registered and passing pre-flight checks.
    - 5 / 5 core invariant self-audit checks passing.
    - Policy Verdict: ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready).

Chapter 249. Chapter 42: DevLog 007: Archival Extraction & The 2017–2018 Demo Baseline

Source File: /home/professorseanex/Tenebris and Lux/docs/dev_logs/2026/10/lore_and_design/devlog_007_rpg_maker_demo_data_extraction_and_baseline.adoc

Chapter 250. Abstract

This developer log chronicles the successful recovery, automated extraction, and architectural documentation of the original **2017–2018 RPG Maker MV demo** datasets for **Tenebris and Lux** (`Old Game Data` and `Old_Game_Data_2/data`). It establishes a permanent, verifiable baseline of the high school genesis era, cataloguing the complete 43-map scenario script, the dual-protagonist narrative branches (Elijah Ro-enheart & Lady Katrina Jaren), the grid-based Lecode Tactical Battle System (LeTBS), the 75-skill grimoire, and the morality engine. This baseline serves as our living blueprint as we systematically reconstruct the game in native C11.

Chapter 251. 1. Historical Context: Rediscovery of the Genesis Datasets

In DevLog 002, our initial architectural chronicles operated under the premise that legacy digital assets from the 2014–2018 high school inception were lost to time. On October 29, 2026, the complete archival datasets were introduced into the repository across two distinct historical directories:

1. `Old Game Data` **(Folder 1: Primary Narrative & Morality Build)**:
    - Version ID: `40730002` | Game Title: **"Tenebris and Lux"**
    - 36 map files + complete database (Actors, Classes, Skills, Items, Weapons, Armors, Enemies, Troops, States, CommonEvents, MapInfos).
    - Preserves complete dual-protagonist dialogue branches, the **Compass of Morality**, the **Test of Honesty** with Eric the Lumberjack, John the Thief's rebellion against the St. Baron, and the monotonic 24-hour clock.
2. `Old_Game_Data_2/data` **(Folder 2: Advanced LeTBS Tactical Build)**:
    - Version ID: `2078310` | Game Title: **"Tenebris & Lux"**
    - 43 map files + extended tactical party companions (Sebastien the Thief, Angela the Mage, Lucius the Priest).
    - Preserves advanced LeTBS tactical combat metadata (movement point costs, scope geometry, AOE patterns, turn timeline manipulation, and character generator presets `Grandma.json` and `TheKing.json`).

Chapter 252. 2. The Extraction Toolchain: `docs/tools/extract_demo_data.py`

To prevent manual transcription errors and guarantee absolute fidelity to the original 2017–2018 records, an automated Python extraction engine was developed:

```
python3 docs/tools/extract_demo_data.py
```

252.1. Core Functions of the Extraction Engine

- **Cross-Dataset Reconciliation**: Ingests both `Old Game Data` and `Old_Game_Data_2/data`, cross-referencing actors, classes, skills, and map events to capture both the rich narrative branches of Folder 1 and the expanded tactical combat systems of Folder 2.
- **Verbatim Script Mining**: Parses RPG Maker MV event command codes (`101` text headers, `401` text lines, `102` choices, `121` switches, `122` variables, `201` transfers, `301` battles) to reconstruct the exact dialogue transcripts for both Elijah (`\n[1]`) and Katrina (`\n[2]`).
- **Tactical Metadata Extraction**: Extracts LeTBS XML/YAML-style note tags from skills, states, and actors, formalizing movement points, range shapes, cooldowns, and AI behavior patterns (`ranged_fighter`, `ai_no_attack`).

Chapter 253. 3. The 10-Volume Baseline Documentation Compendium

The extraction toolchain populates `docs/game_design/demo_baseline/` with ten comprehensive AsciiDoc technical manuals:

Document	Primary File	Core Contents
Master Index	README.adoc	Provenance, dataset taxonomy, and reconstruction methodology.
Scenario Script	scenario_and_walkthrough.adoc	7-Act narrative script across 43 maps, branching dialogues, and puzzle solutions.
Dramatis Personae	cast_and_characters.adoc	Protagonists (Elijah, Katrina), allies (Sebastien, Angela, Lucius), and key NPCs.
Tactical Mechanics	combat_and_letbs_mechanics.adoc	LeTBS 2D grid rules, MP economy, scope geometry, and terrain hazards.
Class Progressions	classes_and_growth.adoc	7 classes + summons; Lv 1–99 stat matrices; level-by-level skill learnings.
Grimoire	grimoire_and_skills.adoc	75+ active techniques, damage formulas, and LeTBS sequence tags.
Bestiary & Troops	bestiary_and_troops.adoc	22+ enemies, destructible chests/ice blocks, AI tags, and troop formations.
Armory & Items	armory_and_inventory.adoc	Weapons, armors, consumables, and quest relics (Grandma’s Watch, Compass).
Systems & Switches	systems_switches_variables.adoc	Monotonic clock, moral continuum, and full directory of 100+ switches/variables.
Evolution Delta	dataset_evolution_delta.adoc	Technical comparison between Folder 1 and Folder 2 with C11 roadmap.

Chapter 254. 4. C11 Engine Rebuilding Roadmap

With the historical baseline fully documented, the path forward for our native C11 implementation is clear, disciplined, and rigorous:

1. **Layer 0 Core Expansion (`include/util/`, `src/util/`)**:
   - Align `time_utils` to faithfully model the 24-hour calendar cycle driven by **Grandma's Watch**.
   - Expand `geom_utils` to support discrete grid-cell raycasting and LeTBS range shapes (`circle`, `line`, `square`, `through_obstacles`).
2. **Layer 1 Simulation Expansion (`include/sim/`, `src/sim/`)**:
   - Reconstruct the tactical battle state machine supporting 5-member party formations and enemy AI routines.
   - Implement the moral sway continuum engine (`[-100, +100]`) tracked by the **Compass of Morality**.
3. **Layer 2 Viewport & Presentation (`include/gui/`, `src/gui/`)**:
   - Render the 2D tactical grid with movement overlays, Zone of Control, and turn order timelines.
   - Implement the dialogue presentation window capable of rendering branching scripts with character portraits.
4. **Layer 4 Automated Verification (`tests/`)**:
   - Author invariant test suites verifying deterministic grid movement, damage formula parity, and softlock immunity under all quest choice trees.